

CHAPTER 11

BASICS OF AUTOCAD

11.1 *Introducing AutoCAD*

AutoCAD, created by Autodesk Corporation, is the most widely used technical drawing program. All over the world people are using this AutoCAD to generate different kinds of drawings. The major disciplines that use AutoCAD are:

- Mechanical
- Architecture
- GIS (Geographic Information Systems)
- Facilities Management
- Electrical/Electronic
- Multimedia etc.

AutoCAD creates and saves file in 'vector' format. Vector files store an image by mathematically defining an object that makes up the image. Opening AutoCAD and the different basic components of AutoCAD will be introduced in short in this section.

11.1.1 *Opening AutoCAD*

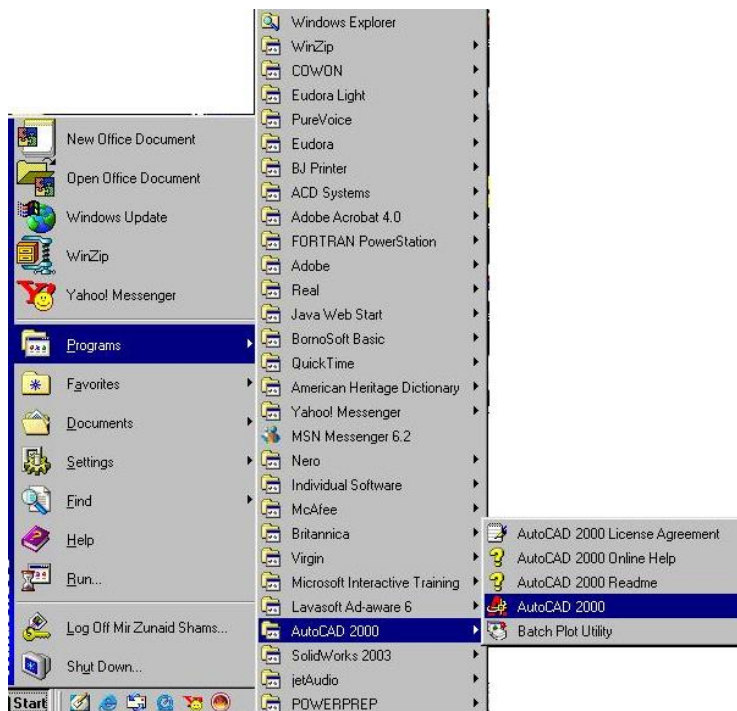


Figure 11.1: Steps to Open Auto CAD

Having AutoCAD installed in computer, one can easily open AutoCAD by clicking on the AutoCAD icon, which looks like  for AutoCAD 2000 version. It can also be opened selecting the following steps as mentioned below (Figure 11.1).

Start → Programs → AutoCAD 2000 → AutoCAD 2000

Opening AutoCAD will display AutoCAD window showing Startup dialogue box (Figure 11.2). Now selecting the **Start from Scratch** button and the radio button for the desired unit (English or Metric) followed by the **OK** button in the Startup dialog box, the Startup dialog box disappears from the window and the window becomes ready for the desired drawing.

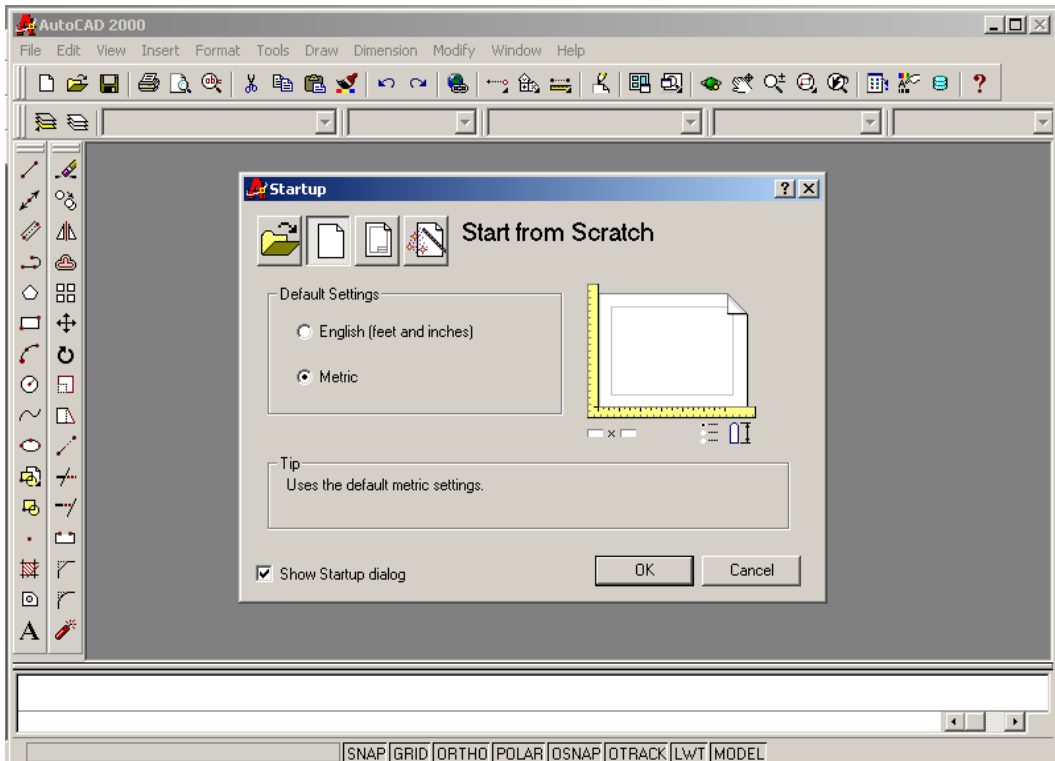


Figure 11.2: AutoCAD Window Showing Startup Dialogue Box

A portion of the Startup dialog box is shown in Figure 11.3. Of the four buttons located at the top left corner of the Startup dialog box, the first one from the left is for **Open a Drawing**, the second one is for **Start from Scratch**, the third one is for **Use a Template** and the fourth one is for **Use a Wizard**. By clearing check box of **Show Startup dialog** of the Startup dialog box (Figure 11.2), it will disappear from the window. However, it may be displayed again by selecting the following steps as mentioned below.

Tools → Options → System tab → Show Startup dialog check box

AutoCAD window showing **Open a Drawing** is presented in Figure 11.4. When the Open a Drawing button is selected as shown in this figure the four recent files will be displayed. Now one can select the desired one and click the **OK** button, the file will be ready for work. If the desired

file is not available here one has to click on the **Browse** button, a Select File dialog box will appear where from the required file has to be selected followed by **Open** button.

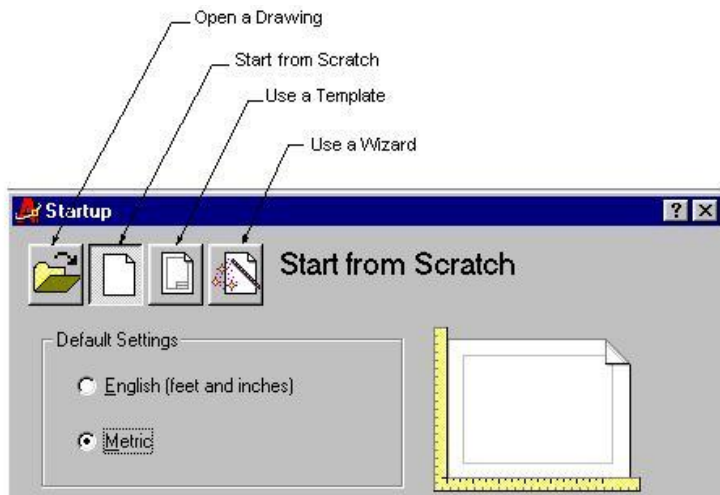


Figure 11.3: A Portion of Startup Dialog Box

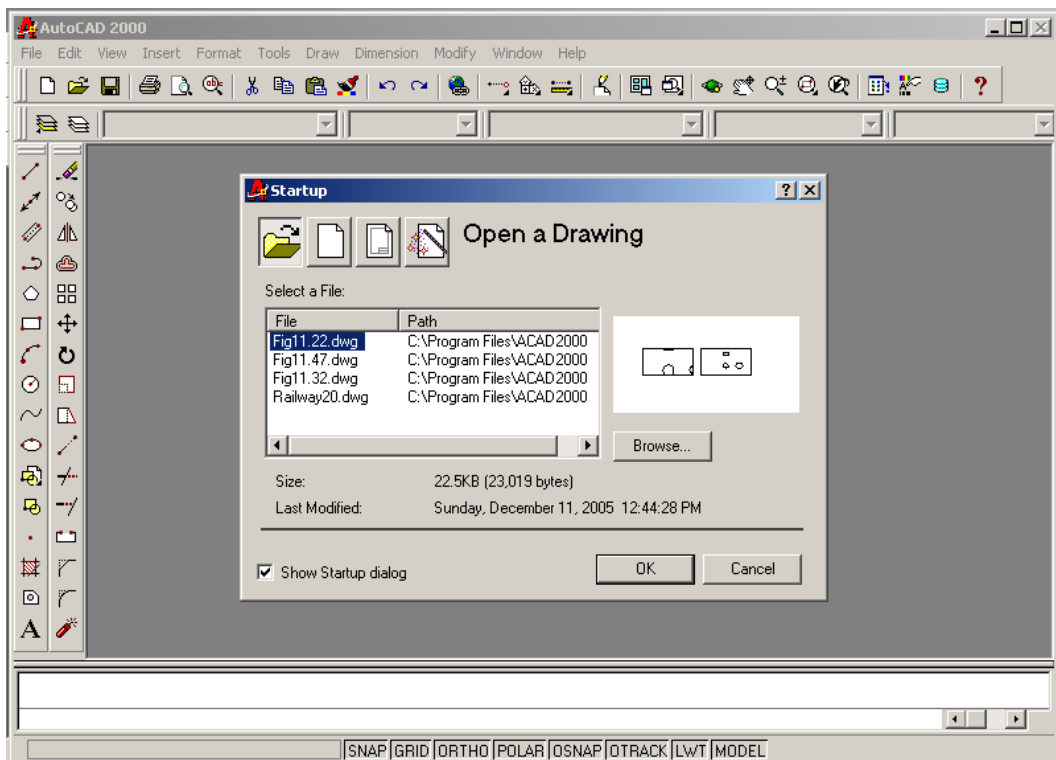


Figure 11.4: AutoCAD Window Showing Open a Drawing

11.1.2 Components of AutoCAD Window

In Figure 11.5 the various components of the AutoCAD window have been introduced. It is the basic view of AutoCAD window. However, depending on the user's preference there may be slight difference in orientation and arrangement of the toolbars. It contains title bar, menu bar, standard toolbar, object properties toolbar, draw toolbar, modify toolbar, command area (command prompts area) and status bar etc. as shown. The area at the middle of the window is known as the drawing area.

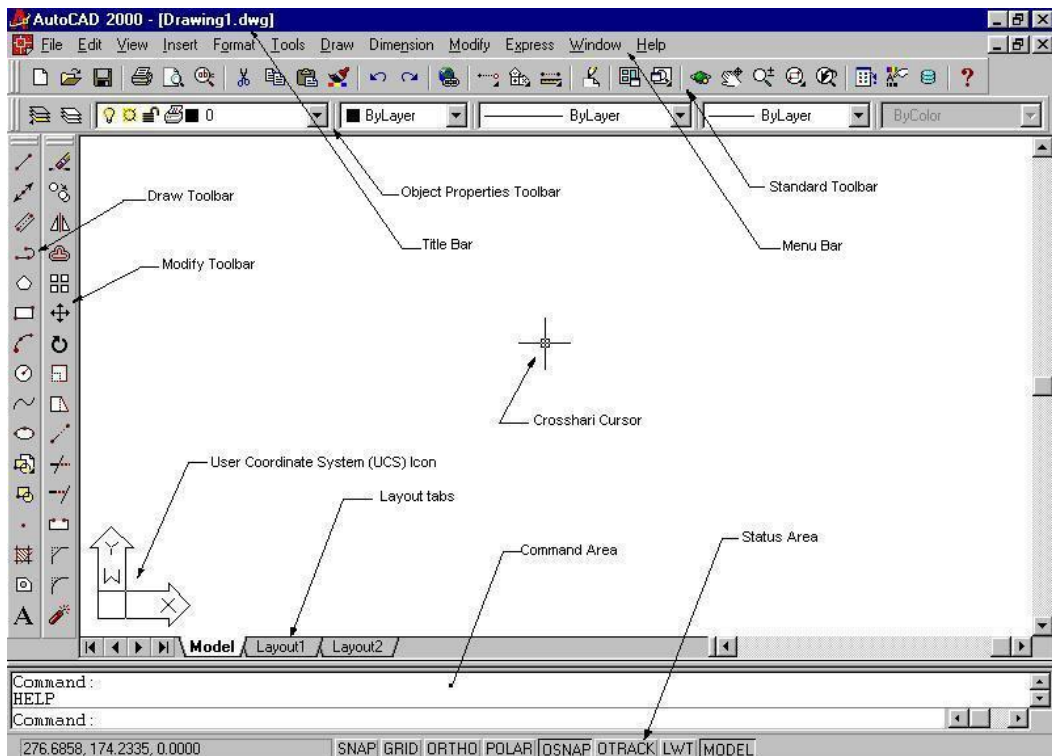


Figure 11.5: Components of AutoCAD Window

11.1.3 Crosshair Cursor

In AutoCAD drawing area, the crosshair looks like two intersecting lines with a small square box at their intersection. At the bottom of the AutoCAD screen, at the left end of the status bar, the X, Y coordinates are seen, the values of them change as the cross hair moves within the drawing area indicating its location as shown in Figure 11.6. As the cross hair cursor is moved to select any command, the cursor turns into the shape of an arrow.

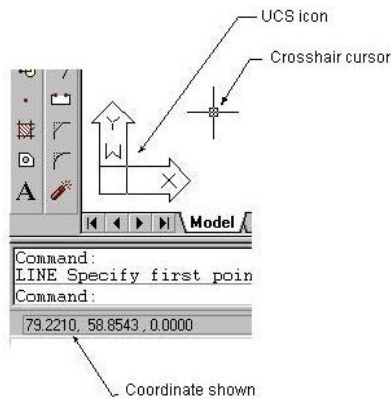


Figure 11.6: Crosshair Cursor

11.1.4 Menu Bar

The menu bar of AutoCAD window contains a number of titles such as, File, Edit, View, Insert, Format, Tools, Draw, Dimension, Modify, Window and Help. If any one of the menu bar titles is selected, it is highlighted and a menu is displayed just below the title consisting of a number of commands, which is called the **pulled-down menu**. In Figure 11.7 AutoCAD window displaying **Tools menu** is shown, which is obtained selecting Tools title from the menu bar.

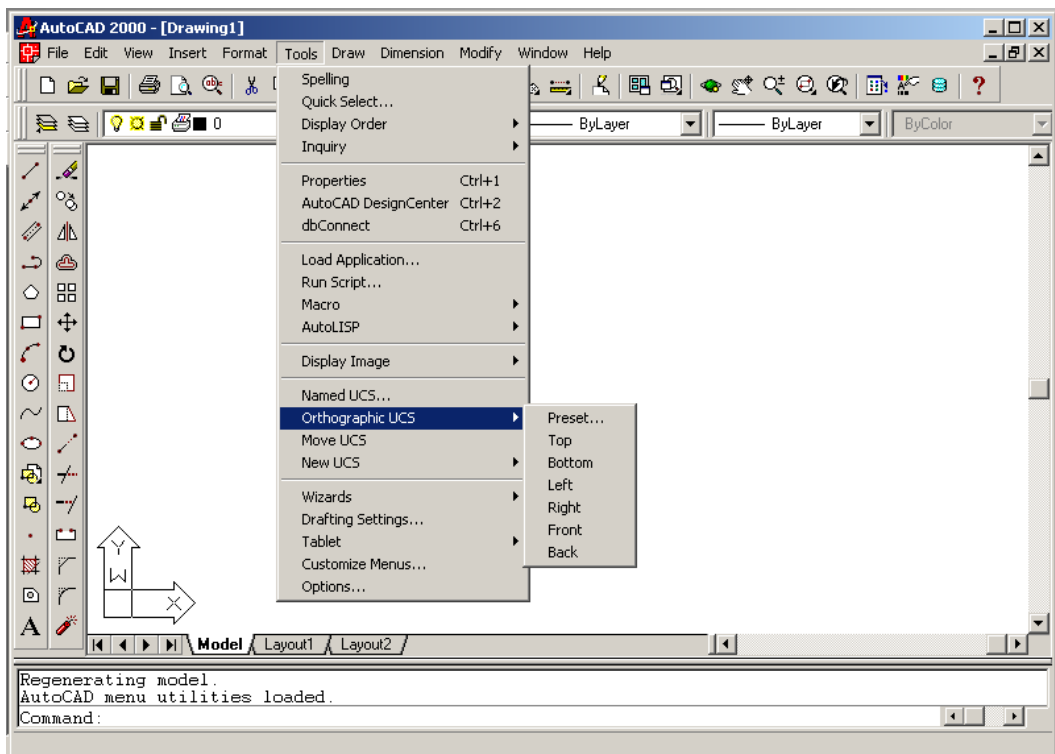


Figure 11.7: AutoCAD window Displaying Tools Menu

If one selects Draw title, draw menu will be displayed just below the draw title and so on. At the right side of some of the commands in the pull-down menu there are right arrows. If one takes the cursor on the command having an arrow at its right side, a parallel menu is displayed (Figure 11.7). It is called the **cascading menu**, which provides some command options.

11.1.5 Short Cut Menu

When right mouse button is clicked in the drawing area a menu is displayed, which is called the **short cut menu** (Figure 11.8). One can invoke some useful commands from the short cut menu. Once any command is initiated from the toolbar or menu or inserted in the command area, next the same command becomes available in the short cut menu at the top of the list of commands. Thus the available command in the short cut menu is changed according to the requirement of the user. It saves time to use the command from the shortcut menu directly.

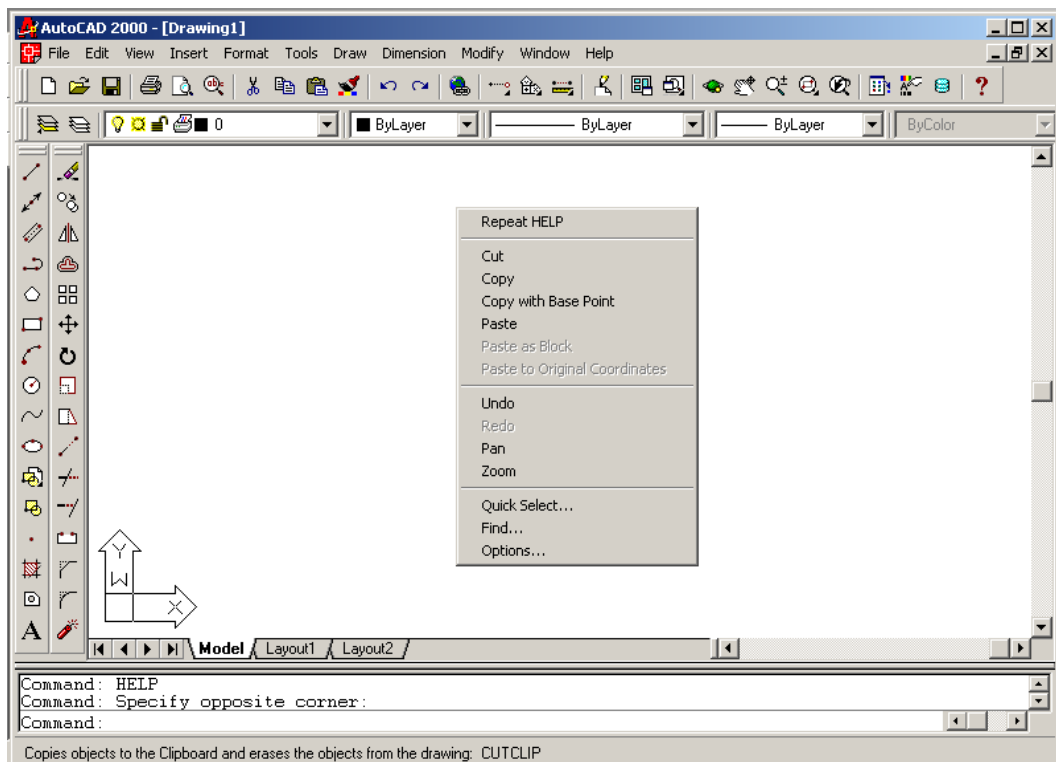


Figure 11.8: AutoCAD Window Showing Shortcut Menu

11.1.6 Toolbar

AutoCAD window consists of a number of toolbars. It becomes easier and convenient to invoke command from the toolbar directly. Each toolbar contains a number of icons for various commands. When the cursor is brought on any icon for selection, the command prompts are displayed in the command area. There is a standard toolbar at the top of the window. It contains a number of very useful commands, which are introduced in Figure 11.9.

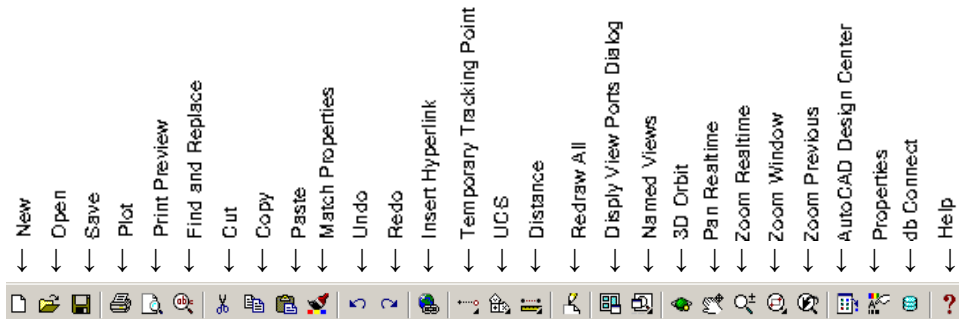


Figure 11.9: Standard Toolbar

In Figure 11.10 and Figure 11.11, draw and modify toolbars are presented respectively. The various icons of the draw and modify commands have been introduced in these figures. One can choose the required command say, line, construction line, erase, copy etc. from these toolbars directly. It becomes time saving and efficient to invoke the command from these toolbars.

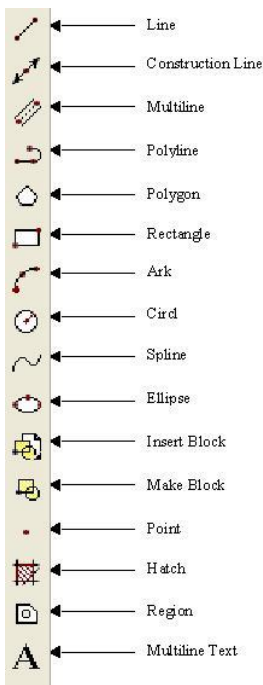


Figure 11.10: Draw Toolbar

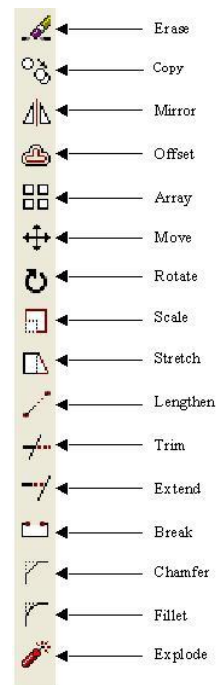


Figure 11.11: Modify Toolbar

In order to turn a Toolbars on and off one has to click on View title in the menu bar, a view menu will appear (Figure 11.12). Now clicking on Toolbars in view menu a Toolbars dialog box will be displayed (Figure 11.13). There are many toolbars in the list of Toolbars dialog box. When the crosscheck box beside the desired toolbar in the toolbars list is selected, that toolbar appears in the

AutoCAD window. On the other hand if the crosscheck box is deselected, the toolbar disappears from the window.



Figure 11.12: View Menu

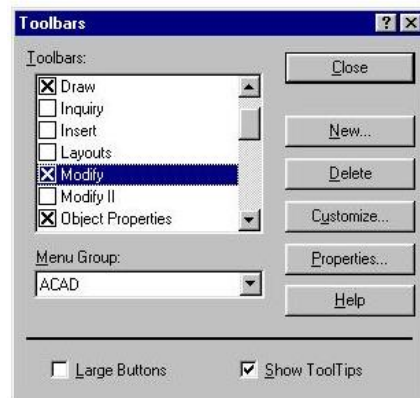


Figure 11.13: Toolbars Dialog Box

The view of AutoCAD window for every user may not look the same. Some toolbars (Draw and Modify toolbars) as shown in Figure 11.5 may be present; on the other hand some other toolbars may be absent, which are not shown in the figure. One can select the dimension toolbar and insert toolbar etc. from the Toolbars dialog box, if necessary.

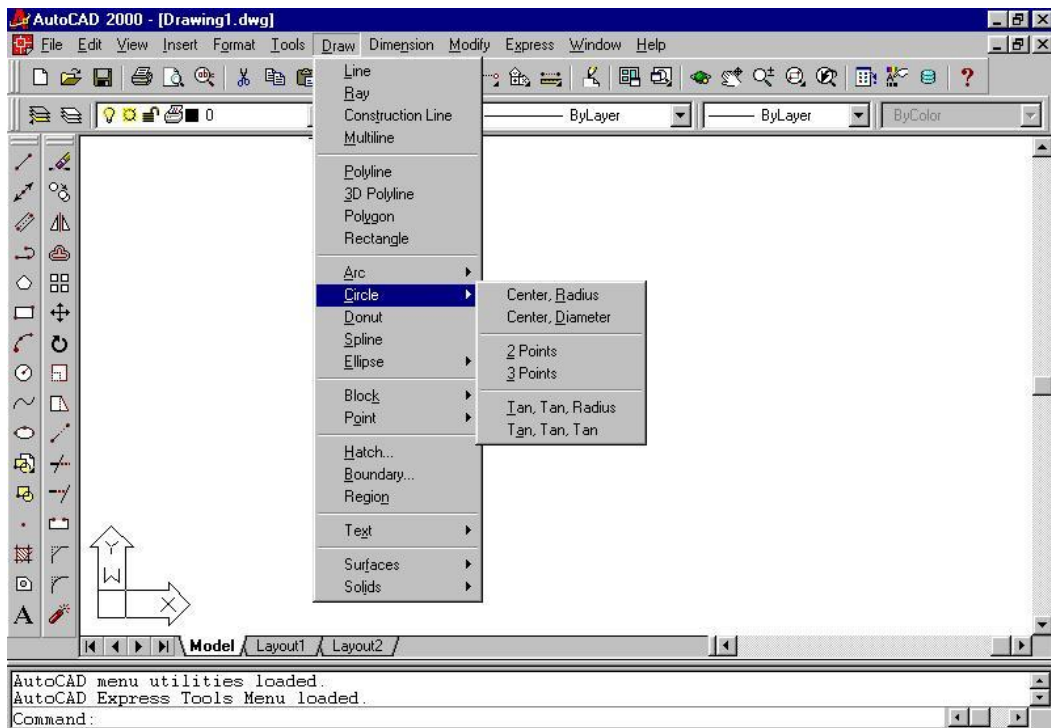


Figure 11.14: Draw Commands from Menu

Commands of Draw Toolbar and Modify Toolbar can also be picked up from the menu bar. In Figure 11.14 the Draw commands in the draw menu is shown. One can find the Modify commands from the modify menu by selecting Modify title in the menu bar. Pull down menus offers more variations and detail in some menus.

11.1.7 Command Area

For the beginners it is suggested always to keep an eye on the command area because once an operation is in process, AutoCAD prompts the operator and mentions in the command area what the next step would be. For example, when the user performs an operation to draw a line by clicking the Line command either from the toolbar or from the menu bar, AutoCAD prompts the command in the command area as shown in Figure 11.15. Any command can also be initiated by inserting it directly in the command area.

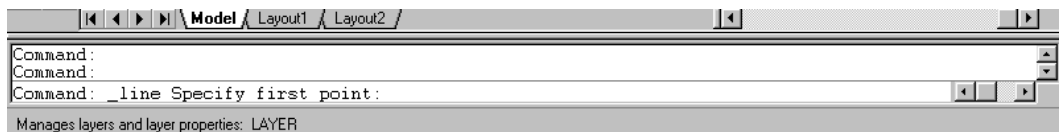


Figure 11.15: Command Area of Auto CAD Window

11.1.8 Options Dialogue Box

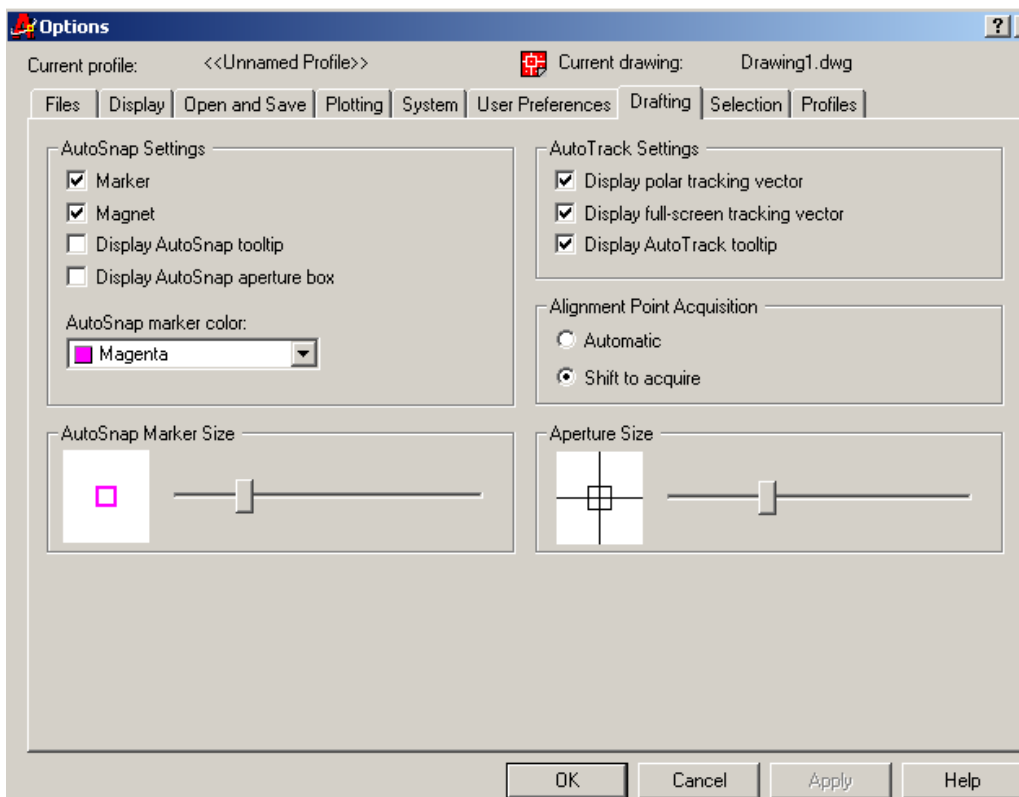


Figure 11.16: Options Dialogue Box

Menu: **Tools→ Options**

Command: **OPTIONS**

Options dialog box can be invoked from the pull-down menu clicking **Options** command. By default Drafting tab is set as shown in Figure 11.16. One can change the AutoSnap Settings using this dialogue box. AutoSnap marker color can be changed from the drop down list displayed by clicking down arrow button. AutoSnap marker size can be adjusted by moving the slider bar horizontally in the AutoSnap marker size area, which is displayed in the image box at the left of the AutoSnap marker size area. The aperture size can also be adjusted by moving the slider bar horizontally in the aperture size area. The aperture size is displayed at the left image box in the aperture size area.

11.1.9 Saving File

Standard Toolbar: 

Menu: **File → Save** or **Save As**

Command: **SAVE** or **SAVEAS**

One can invoke the **Save** command either from the toolbar directly or from the menu. It can also be initiated by inserting in the command area. To create a duplicate file one has to click on **Save As** command. To save in a new file or a duplicate file, a Save Drawing As dialogue box (Figure 11.17) will be displayed, where the new file name is to be inserted. Then click on the **Save** button, the file will be saved in the new file name.

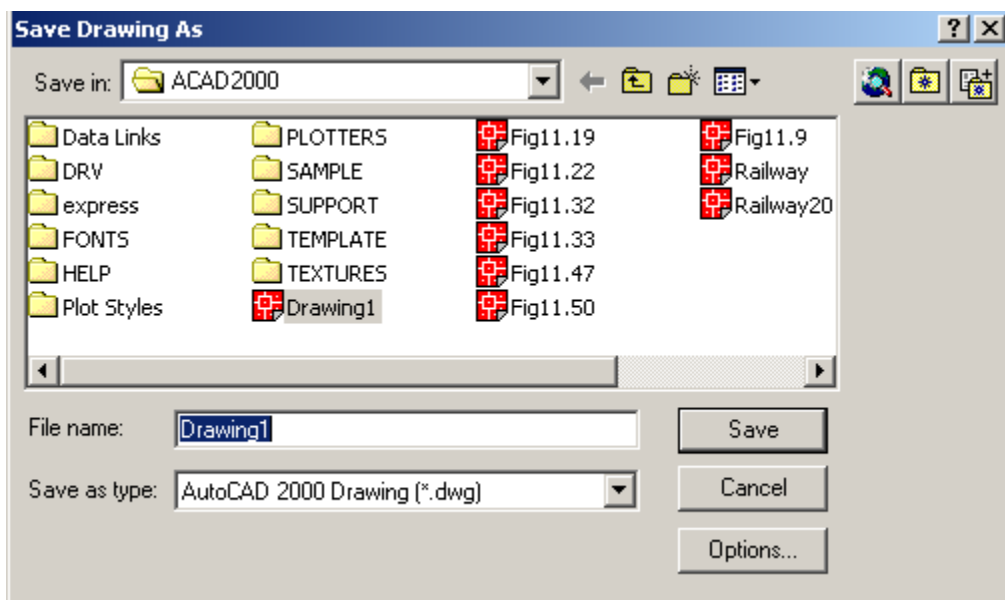


Figure 11.17: Save Drawing As Dialog Box

Save as type list box (Figure 11.18) is used to specify the drawing format in which it is needed to save the file. To save the file as AutoCAD 2000 drawing file, select AutoCAD 2000 drawing from the drop-down list.

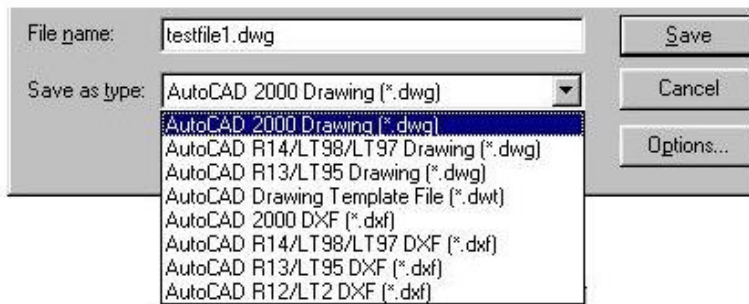


Figure 11.18: Save as Type List Box

11.1.10 *Opening Existing File*

Standard Toolbar: 

Menu: **File** → **open**

Command: **OPEN**

One can open the existing file either from the menu or the standard toolbar. It can also be initiated by inserting the **Open** command in the command area. A Select File dialogue box (Figure 11.19) will be displayed. Now the required file can be selected from the list of the files. When selected the preview of the drawing is displayed in the preview area as shown in the figure. The required files may be in the different locations, e.g. Desktop, and My Documents etc., which can be obtained from the drop down list by clicking the down arrow button of Look in area.

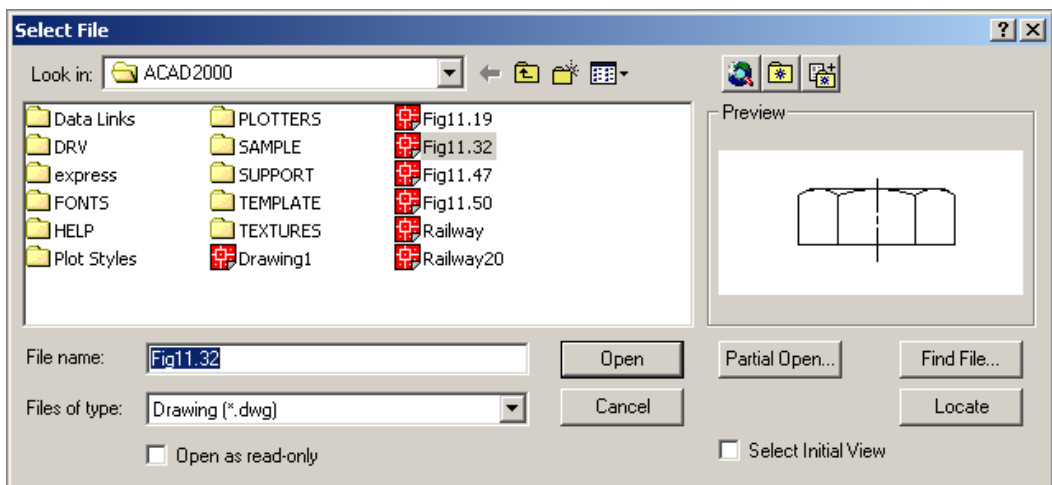


Figure 11.19: Select File Dialogue Box

11.1.11 *Starting New Drawing*

Standard Toolbar: 

Menu: **File → New**

Command: **NEW**

For starting a new drawing, one can either use the standard toolbar or the menu. It can also be initiated by entering the **New** command in the command prompt area. A Create New Drawing dialogue box will appear, which is identical to the Start Up dialogue box (Figure 11.2). Now to accept the Start from Scratch option after selection of the required units, when the **OK** button is clicked, the new drawing area is displayed, where the necessary drawing can be performed.

11.1.12 Closing Drawing

Menu: **File → Close**

Menu: **File → Exit**

Command: **CLOSE** or **EXIT**

The current drawing file can be closed from the menu or one can close it by clicking the close window button  located at the top right corner of the drawing area. It can also be initiated by entering the **Close** command in the command area. If the drawing has not been saved in case of a new drawing or changes in the existing drawing, an AutoCAD dialogue box will appear showing warning “**Save changes to**”. To save the drawing click on the **Yes** button, for no save click on the **No** button and for cancel click on the **Cancel** button. Now the user may open another drawing file to work. To exit from the AutoCAD, one can click the close upper button  located at the top right corner of the screen or it can be done from the menu or inserting **Exit** command in the command area.

11.2 Drawing Aids

This section contains some basic drawing aids, which are necessary to perform drawing precisely and quickly. Setting units, setting limits and drafting settings are described in short in this section. Various snap modes are taken into consideration for discussion in a nutshell.

11.2.1 Setting Units

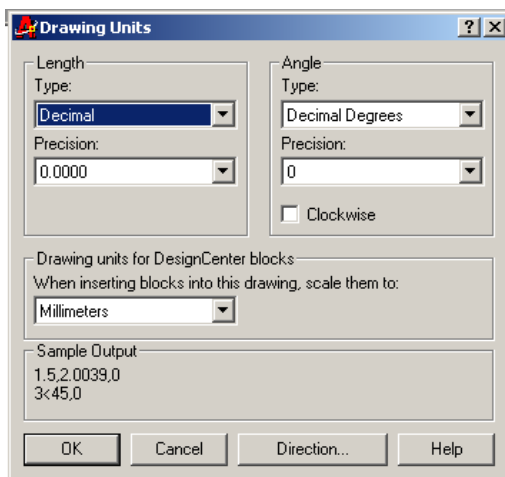


Figure 11.20: Drawing Units Dialog Box

Menu: **Format → Units**

Command: **UNITS**

When the **Units** command is invoked from the menu, a Drawing Units dialogue box (Figure 11.20) is displayed. The user can use this dialogue box to set the desired units. The desired units, length, angle can be selected from the drop-down list by clicking the down arrow button. The precision for length and angle can also be selected from the drop down list by clicking the down arrow button.

11.2.2 *Setting Limits*

Menu: **Format → Limits**

Command: **LIMITS**

To perform the drawing in AutoCAD it is necessary to set limits of the drawing area. Depending on the space required for the different views, dimensions, distance between views, border, title block and notes etc. the limits of the drawing area are calculated. Then the final limits are specified matching with the standard sheet size so that the calculated limits can be accommodated in the standard sheet, as a result during printing in the standard sheet no problem arises. The **Limits** command can be invoked from the menu or it can be initiated inserting in the command area. In the prompt sequence in the command area the values of the co-ordinates of the lower left corner and the upper right corner of the drawing area have to be inserted. For example, the limits of the A3 sheet are **0,0** for the lower left corner and **420,297** for the upper right corner.

11.2.3 *Status Bar*

To make the drawing easier and convenient, AutoCAD has provided some drawing aids. Various snap modes on status bar are shown in Figure 11.21. These snap modes such as, SNAP, GRID, ORTHO (orthogonal mode), POLAR (polar mode) and OSNAP (object snap mode) etc. are useful drawing aids to perform the drawing smoothly. One can easily toggle any desired mode on and off by clicking on it in the status bar.

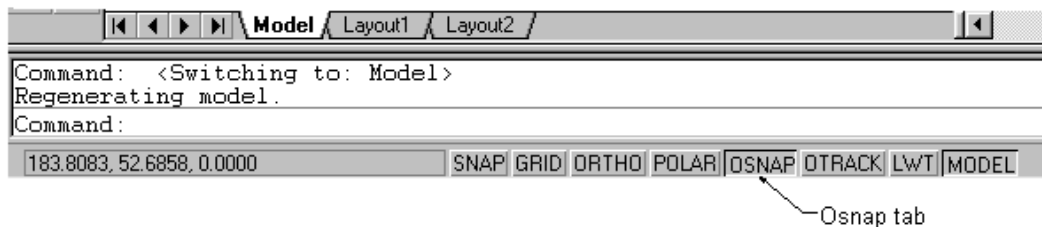


Figure 11.21: Various Snap Modes on Status Bar

11.2.4 *Drafting Settings*

Menu: **Tools → Drafting Settings**

Command: **DSETTINGS**

The Drafting Settings dialog box can be invoked from the menu. A Drafting Settings dialog box for snap and grid mode is shown in Figure 11.22. To display it **Snap and Grid** button of the Drafting Settings dialog box is selected. Clicking right mouse button on any tab except ORTHO and MODEL tabs in the status bar and then clicking settings, one can also display Drafting

Settings dialog box to select the necessary option. The different snap modes in the status bar can be customized and controlled with the help of the Drafting Settings dialog box.

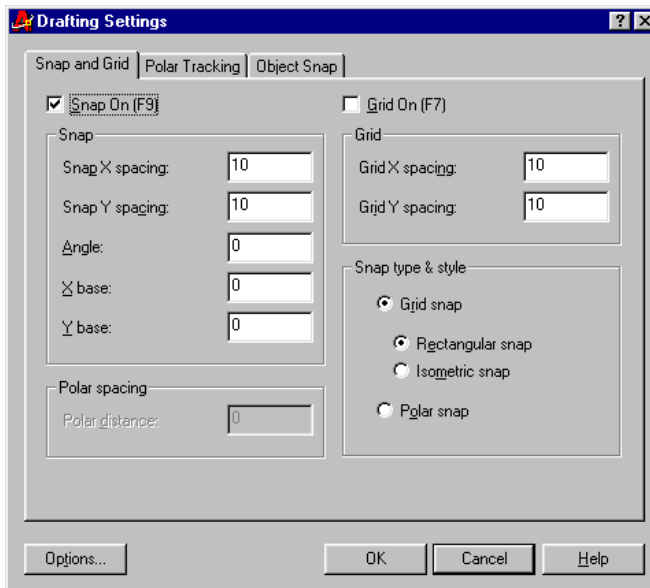


Figure 11.22: Drafting Settings Dialog Box (For Snap And Grid)

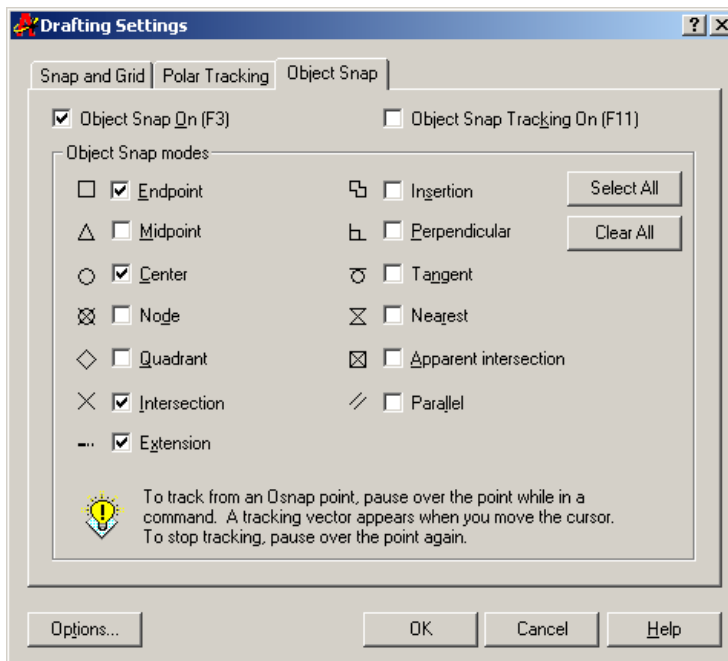


Figure 11.23: Drafting Settings Dialog Box (For Object Snap)

In Figure 11.23 a Drafting Settings dialog box for object snap mode is shown. It appears when the **Object Snap** button is selected in the dialog box. There are thirteen different snap modes in this dialog box. One can easily enable or disable any particular snap mode in this dialog box by selecting or deselecting the check box against each snap mode.

11.2.5 Object Snap Mode

Status Bar: **OSNAP**

Menu: **Tools → Drafting Settings**

Command: **OSNAP**

The OSNAP (object snap) is one of the useful drawing aids, which is used in conjunction with other commands to perform the drawing accurately. One can pick up a specific point using the OSNAP mode. The usage of OSNAP for location of the center point of a circle (Figure 11.24a), the end point of a line (Figure 11.24b), the mid-point of a line (Figure 11.24c) and intersecting point of two lines (Figure 11.24d) are shown. There are many other options as well. It can be noticed that when the cursor is brought near to the point to be picked up, the OSNAP marker is highlighted and locates that desired point accurately. Each OSNAP has different shapes of marker e.g. circle, square, triangle, cross etc. (Figure 11.24).

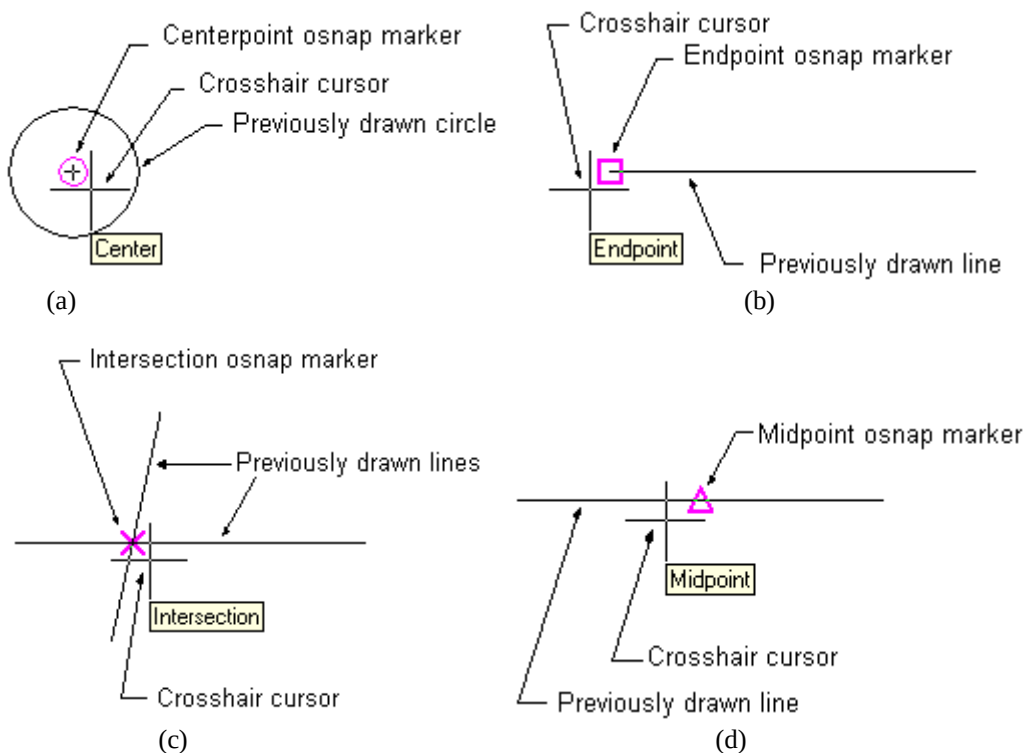


Figure 11.24: Use of OSNAP Mode

11.2.6 *Orthogonal Mode*

Status Bar: **ORTHO**

Command: **ORTHO**

When the ORTHO (orthogonal mode) is made on, one can only draw orthogonal lines. Moreover, with orthogonal mode on, one can only move the objects vertically or horizontally. Click ORTHO on the status bar to toggle orthogonal mode on and off. It is not allowed to have polar tracking on, at the same time when orthogonal mode is on. Picking a point with object snap or by entering the values of coordinate will override ORTHO mode activity.

11.2.7 *Drawing Grid*

Status Bar: **GRID**

Menu: **Tools → Drafting Settings**

Command: **GRID**

The drawing grid (Figure 11.25) is a regular pattern of dots displayed on the screen, which acts as a visual aid; it is tantamount to making a drawing on the graph paper placed on the drawing board. Drawing grid can be turned on or off by clicking on **GRID** tab in the status bar. It can also be selected in the Drafting Settings dialog box. Selecting the grid mode on, a drawing grid appears in the drawing area. The Grid X spacing and Grid Y spacing can be inserted in the dialog box (Figure 11.22). The grid spacing is 10 units in Figure 11.25a, while it is 5 units in Figure 11.25b. The grid is available in orthogonal mode (Figure 11.25c) and in isometric mode (Figure 11.25d). The grid gives the idea about the size of the drawing objects.

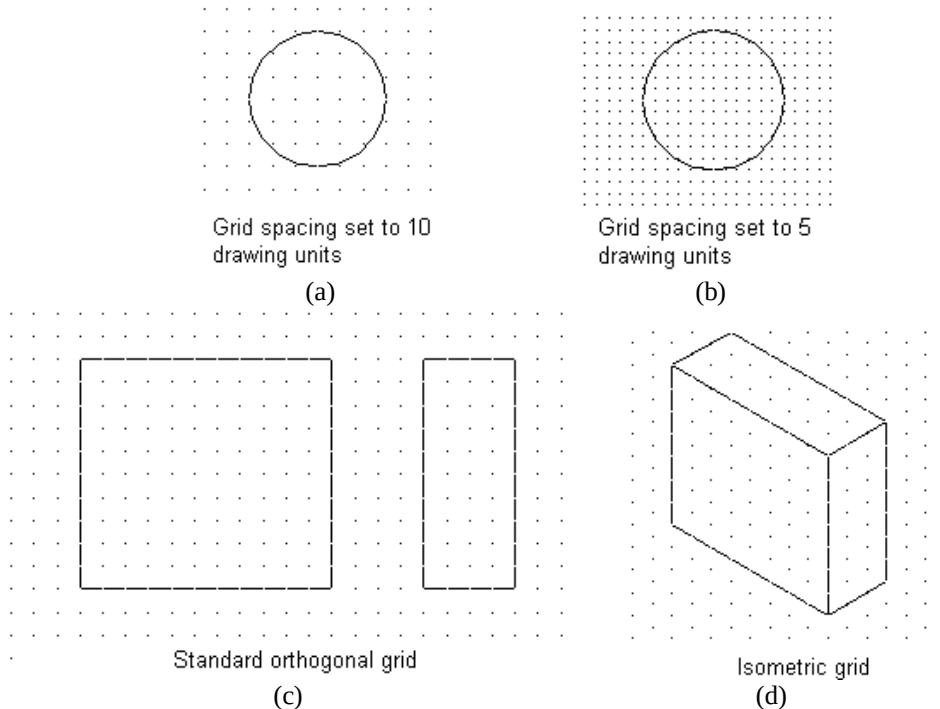


Figure 11.25: Drawing Grid

11.2.8 Snap Mode

Status Bar: **SNAP**

Menu: **Tools → Drafting Settings**

Command: **SNAP**

Snap mode can be made on and off by clicking SNAP button in the status bar. It can be selected in the Drafting Settings dialog box also. The values of Snap X spacing and Snap Y spacing can be inserted in the dialog box (Figure 11.22). The snap mode is used to set increments for the cursor movement. The snap spacing is independent of the grid spacing; as a result they can have same or different values. Usually the snap and grid increments are set to the same value. With the snap mode turned on, AutoCAD only allows to pick points, which lie on a regular grid.

11.3 Use of Draw Command

In AutoCAD, the Draw commands can be invoked in a number of ways. They can be initiated entering the required command (e.g. line, circle etc.) in the command prompt area with the help of the keyboard. Often it is necessary to insert any value in the command area as well. After entering any command or inserting any value in the command area, **press enter** every time. Draw commands can be invoked from the Draw toolbar (Figure 11.10) directly or they can be obtained clicking on Draw title in the menu bar (Figure 11.14).

11.3.1 Line

Draw Toolbar:  (Figure 11.10)

Menu: **Draw → Line** (Figure 11.14)

Command: **LINE** or **L**

Using the Line command, a line can be drawn between any two points within the drawing area.

Click on the **line** icon  at Draw toolbar **or** click **Draw→Line** in the menu. It can also be initiated by entering the line command in the command prompt area. Now click on any point at P₁ in the drawing area and move the cross hair to the next point at P₂ and click, P₁P₂ will turn into a line. Next move the cross hair to the point P₃ as in Figure 11.26, the dashed line along P₂P₃ is known as the rubber band line. After clicking at P₃, P₂P₃ will also turn into a line.

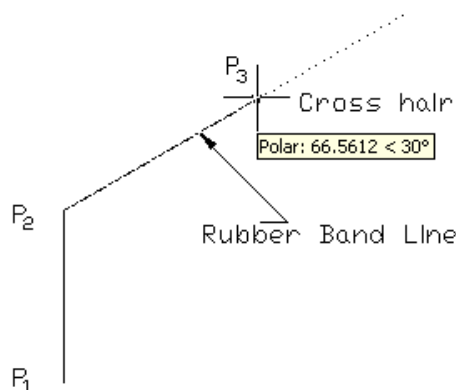


Figure 11.26: Rubber Band Lines

Drawing lines by Cartesian co-ordinates

In two dimensional Cartesian co-ordinate system, one can draw lines (Figure 11.27) specifying x- and y-coordinates in the command area following the sequence as mentioned below (example).

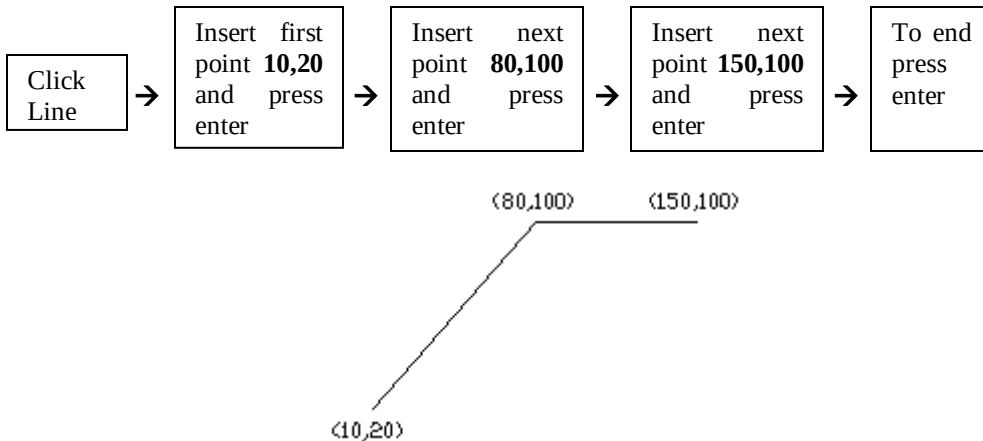


Figure 11.27: Lines by Cartesian Co-ordinates

Drawing Line by Direct Distance Entry

Using direct distance entry one can draw lines very efficiently. It is the most useful and timesaving approach for drawing lines. Following the sequence as mentioned below (example) lines are drawn (Figure 11.28) in ORTHO mode (located at the status bar). In the ORTHO mode lines are drawn vertically and horizontally. Line at an angle can also be drawn using direct distance entry. To draw lines at an angle the cross hair has to be moved at a desired angle and the similar sequence as shown below has to be followed.

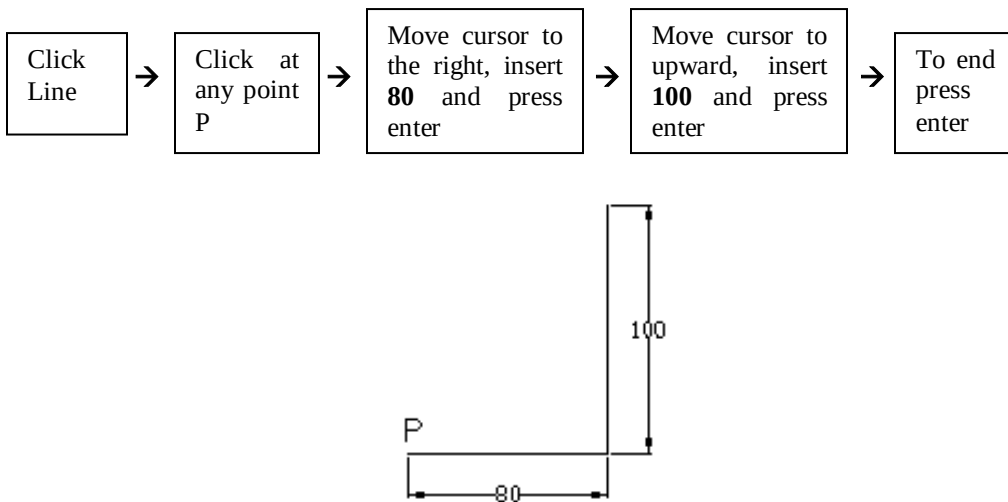


Figure 11.28: Lines by Direct Distance Entry

Drawing Lines by Relative Co-ordinates

In this system after specifying the first point, the next point is measured with respect to the previous point. In AutoCAD the relative co-ordinate system is identified by the symbol **@**. As an example, the steps as shown below are followed to draw lines by the relative co-ordinate system (Figure 11.29).

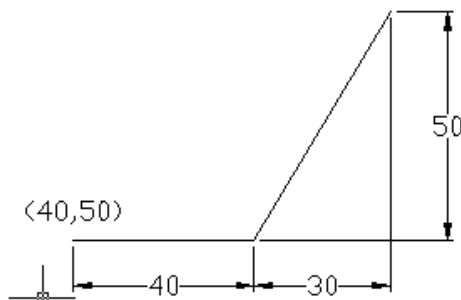
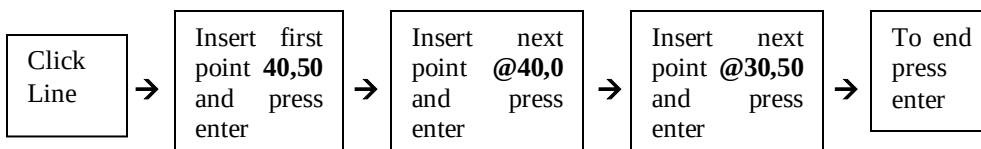


Figure 11.29: Lines by Relative Co-ordinates

Drawing Lines by Polar Co-ordinates

In the Polar co-ordinate system, one can draw a line at an angle. After specifying the first point, the next point is specified at a distance and an angle with the x-axis. The distance is measured in reference to the first point (Figure 11.30). The sequence of drawing a line in polar co-ordinate is shown below as an example.

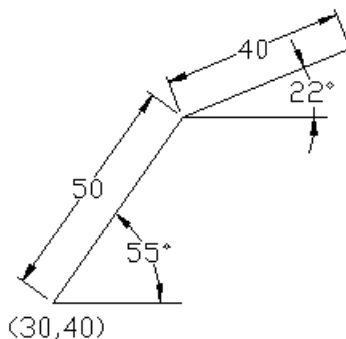
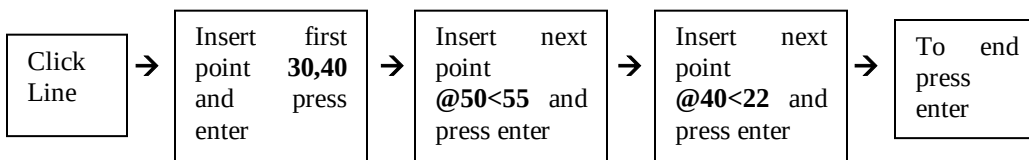


Figure 11.30: Lines by Polar Co-ordinates

There is very efficient and useful method of drawing lines at an angle using Polar Tracking. On menu bar click on **Tools** title, a Tools menu (Figure 11.7) will appear, then click on **Drafting Settings** on Tools menu, a Drafting Settings dialogue box for polar tracking (Figure 11.31) will be displayed after invoking **Polar Tracking** tab on it.

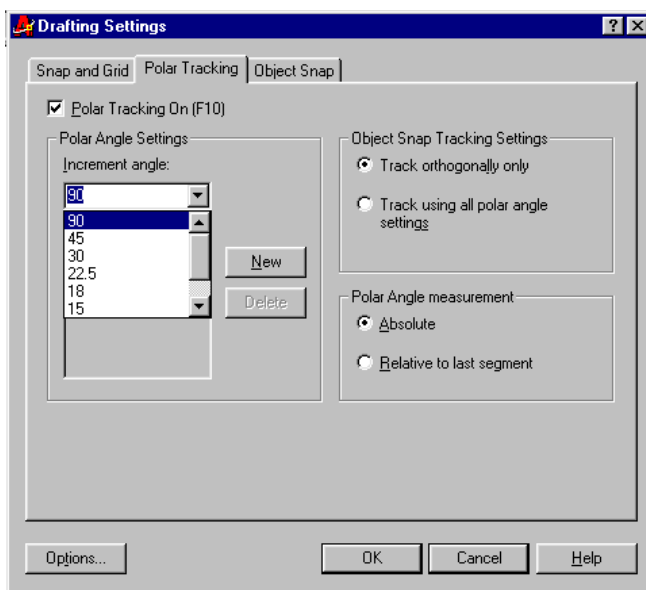


Figure 11.31: Drafting Settings Dialogue Box For Polar Tracking

On this Drafting Settings dialogue box there is a drop down list named Increment Angle showing several increment angles available by default. Any desired value of increment angle (say 15, 30 etc.) can be selected from the drop down list. Once the desired increment angle is set, the line at an angle (Figure 11.32) can be drawn easily from the following steps as shown below.

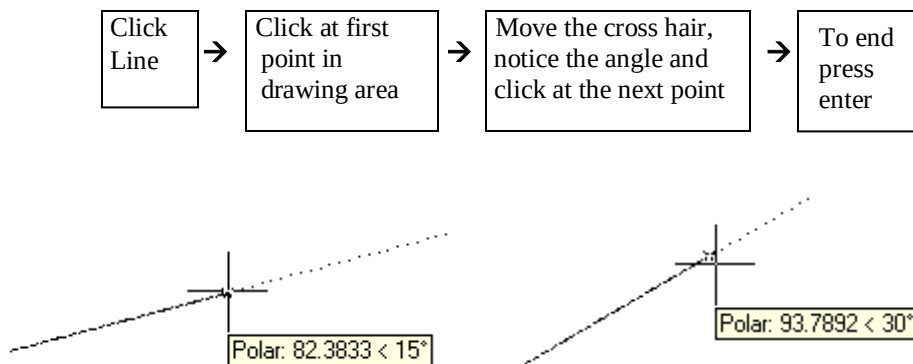


Figure 11.32: Drawing Lines at an Angle

Setting the increment angle at 15° a line at an angle of 15° , 30° , 45° etc. can be drawn with a step of 15° . In the drawing area the angle will be indicated showing the value of the angle (Figure 11.32). A rubber band line with the dotted tracking line also will guide to draw the line at an angle. The length of the line can be inserted in the command prompt area.

11.3.2 Polyline

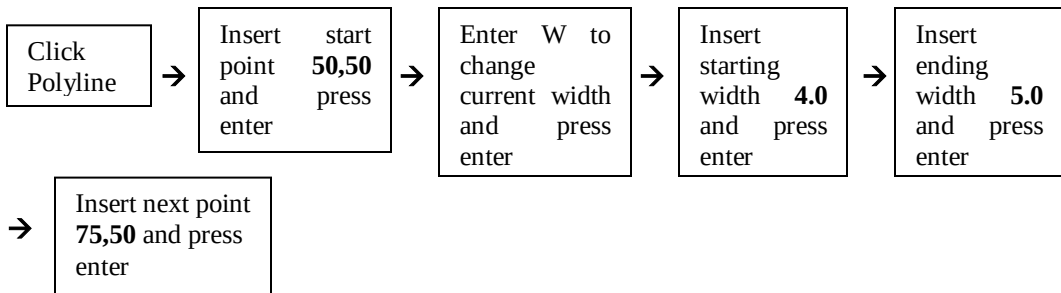
Draw Toolbar: 

Menu: **Draw** → **Polyline**

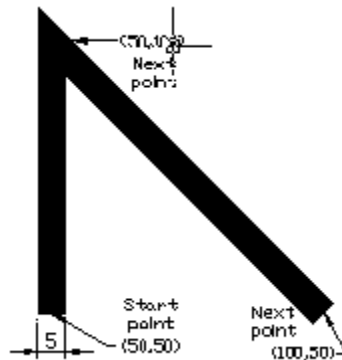
Command: **PLINE** or **PL**

Polyline command can be used to draw different types of features. Drawing can be done using **Polyline** command in several options. (Poly means many).

Width Option: In this option a polyline can be drawn with the desired thickness of the line. The sequence to draw a polyline (Figure 11.33a) is shown below as an example. Clicking at the points by the cross hair one can also specify the start point and end point. Another example of a polyline is shown in Figure 11.33b.



(a)



(b)

Figure 11.33: Polylines Using Width Option

Half Width Option: This option is similar to the width option. In place of the full width, the half width is inserted and W for width option is replaced by H for half width option.

Arc Option: One can draw a polyarc using arc option. There are many ways to draw an arc. Here only one way is taken into consideration. The sequence to draw an arc (Figure 11.34) is given below as an example. The values of the co-ordinates can be inserted for the start and end points but here the start and end points have been selected by clicking with the cross hair.

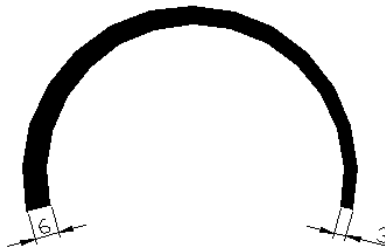
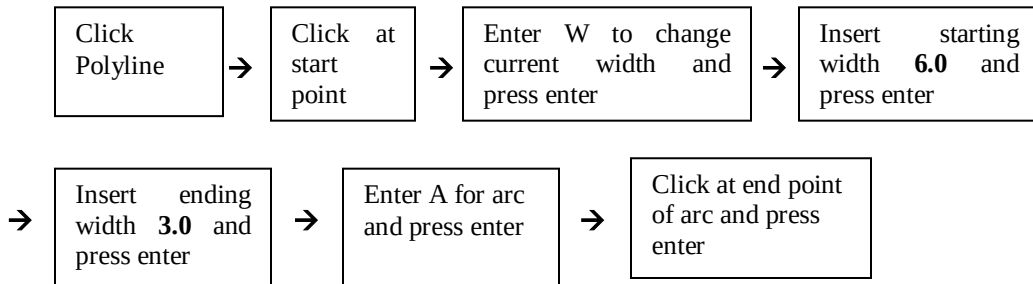


Figure 11.34: Polyarc Using Arc Option

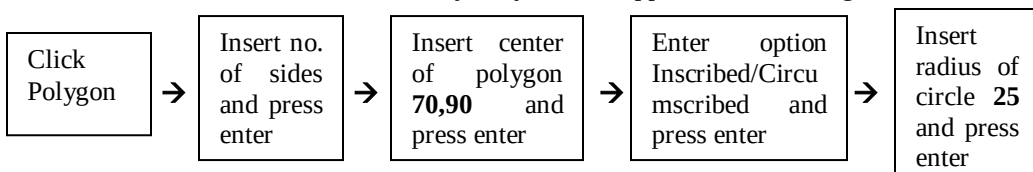
11.3.3 Polygon

Draw Toolbar: 

Menu: **Draw → Polygon**

Command: **POLYGON**

To draw a regular two-dimensional polygon, **Polygon** command is used. The regular polygon has the equal sides and equal angles. The center of the polygon may be specified either by clicking only at the center point by the cross hair or by inserting values of co-ordinates in the command area. The polygon may be drawn with the inscribed or circumscribed option. In the inscribed option (insert **I** for inscribed), the polygon remains within the imaginary circle and each corner of the polygon touches the circumference of the circle. While in the circumscribed option (insert **C** for circumscribed), the polygon remains outside the imaginary circle and each side of the polygon touches the circumference of the circle. The steps of drawing a polygon are given below. Examples of drawing polygons in inscribed and circumscribed options are given in Figure 11.35. The circles shown are for clarification only; they will not appear in the drawing.



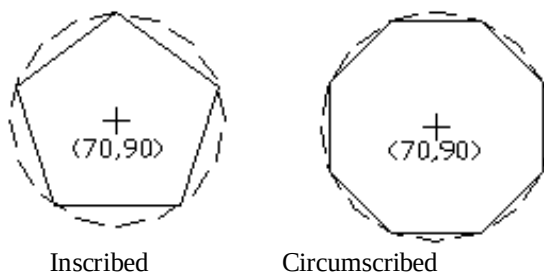


Figure 11.35: Regular Polygon

11.3.4 Rectangle

Draw Toolbar: 

Menu: **Draw** → **Rectangle**

Command: **RECTANG**

Using **Rectangle** command one can draw a rectangle. The two corner points at the opposite ends of the diagonal of the rectangle are to be specified either by clicking only at the points by the cross hair or inserting the values of co-ordinates in the command area. The steps to draw a rectangle with fillet are given below. Examples of drawing rectangles are shown in Figure 11.36. The chamfer, fillet, width etc. can be set inserting the desired values in the command area. By default they have zero values. In case of sharp corner the second and third steps have to be omitted in the following sequence.

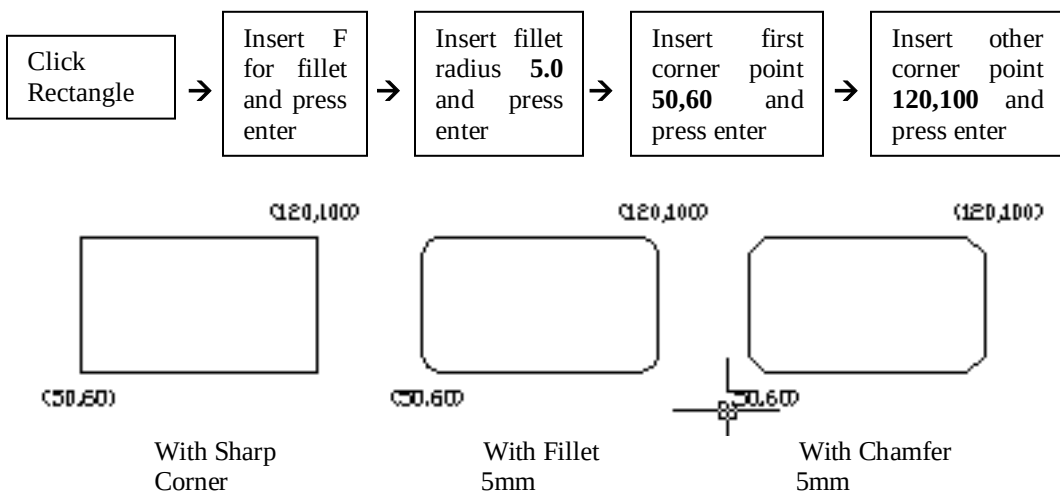


Figure 11.36: Rectangle

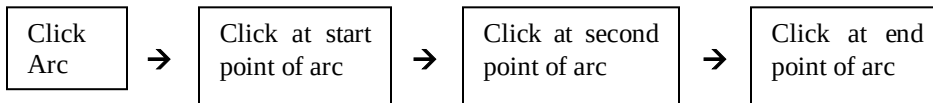
11.3.5 Arc

Draw Toolbar: 

Menu: **Draw** → **Arc**

Command: **ARC** or **A**

An arc is a part of a circle. It can be drawn using eleven different options. Of them the most useful one is the three points option. One can obtain the various options from the menu by clicking on **Draw → Arc**. To draw the arc by the three points option, one can specify the points either by clicking only with cross hair at the points or by inserting the values of co-ordinates in the command area. An arc by the **Three Points Option** (Figure 11.37) can be drawn following the steps as shown below.



An arc by **Start, Center, and Angle Option** can be drawn (Figure 11.37) in accordance with the following steps.

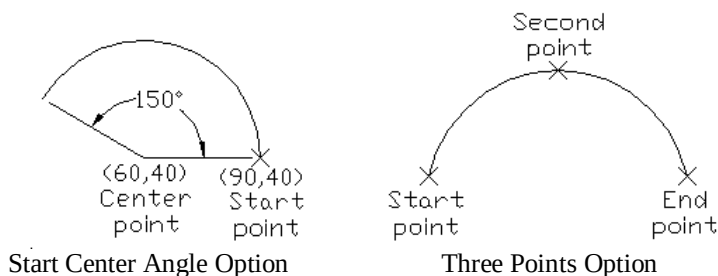
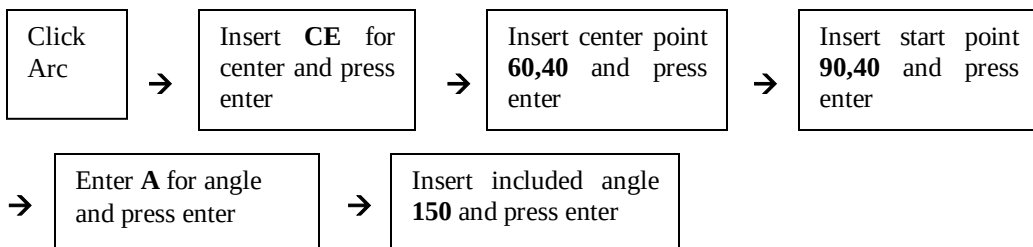


Figure 11.37: Arcs

11.3.6 Circle

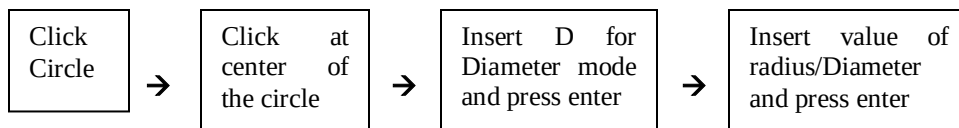
Draw Toolbar:

Menu: **Draw → Circle**

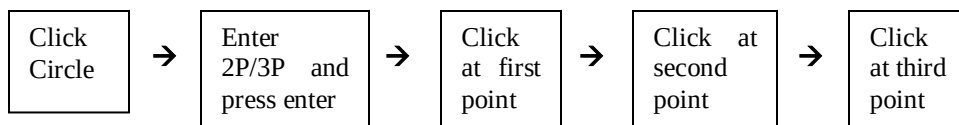
Command: **CIRCLE** or **C**

In AutoCAD a circle can be drawn by several options.

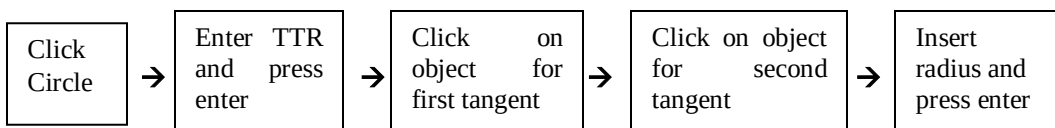
Center and Radius/Diameter Option: In this option the center of the circle is to be specified. The center may be specified either by clicking only with the cross hair at the center point or inserting the values of co-ordinates in the command area. By default the radius mode appears. For diameter the mode has to be changed inserting D. To draw the circle (Figure 11.38) the sequence as mentioned below is to be followed. For radius mode the third step in the following sequence is to be omitted.



Two/Three Points (2P/3P) Option: In this option circles are drawn (Figure 11.38) specifying two points for two points (2P) option and three points for three points (3P) option on the circumference of the circle. The points may be specified either by clicking only with the cross hair at the points or inserting the values of co-ordinates in the command area. The steps to be followed to draw the circle are provided below. For the two points option the last step has to be omitted.



Tangent, Tangent Radius Option: In this option one can draw a circle (Figure 11.38) by clicking on two objects at any point on them which will be the tangents to the circle and inserting the value of the radius of the circle. The steps that are followed to draw the circle in this option are given below.



Tangent, Tangent, Tangent Option: In this option a circle (Figure 11.38) can be drawn by clicking on three objects at any point on them which will be the tangents to the circle. It has to be invoked from the menu clicking **Draw** → **Circle** → **Tan, Tan, Tan**. The steps as mentioned below are followed to draw the circle in this option.

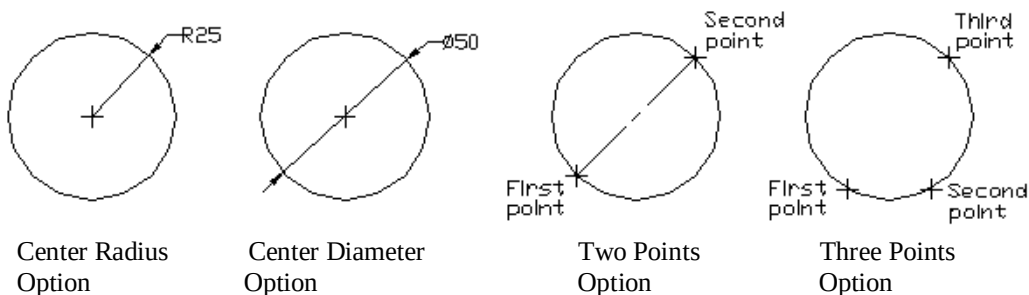
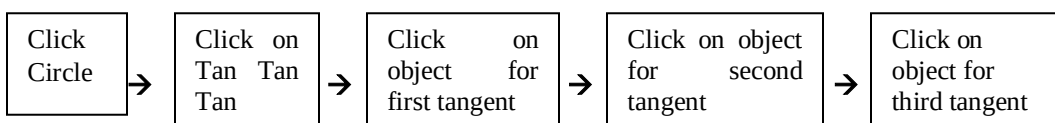


Figure 11.38: Circles (contd.)

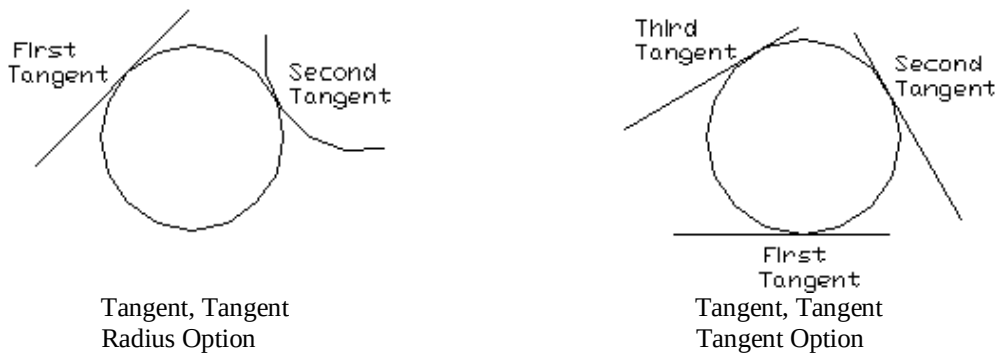


Figure 11.38: Circles (contd.)

11.3.7 Spline

Draw Toolbar: 

Menu: **Draw** → **Spline**

Command: **SPLINE** or **SPL**

Spline command can be used to draw a curved line of any desired shape. One can specify the points for drawing the curve by inserting the values of co-ordinates of the points in the command area or by clicking only at the points by the cross hair. A curved line is drawn (Figure 11.39) using **Spline** command according to the following sequence as mentioned below. Inserting the values of the co-ordinates in the command prompt area, a curve of any regular shape e.g. sinusoidal one may be drawn easily.

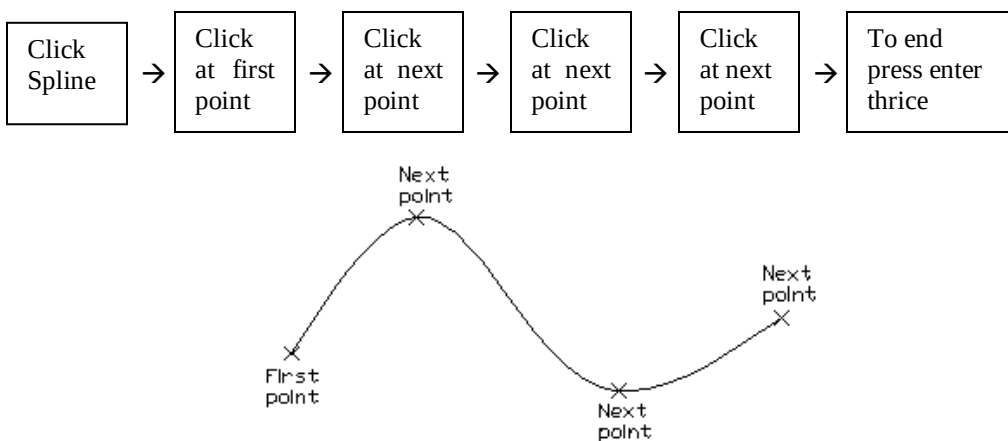


Figure 11.39: Curved Line

11.3.8 Ellipse

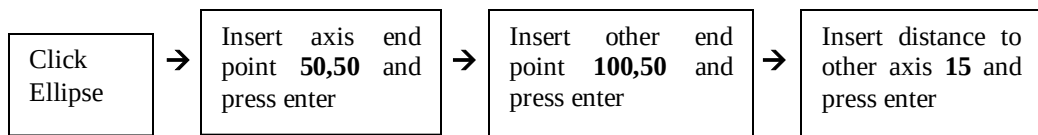
Draw Toolbar: 

Menu: **Draw** → **Ellipse**

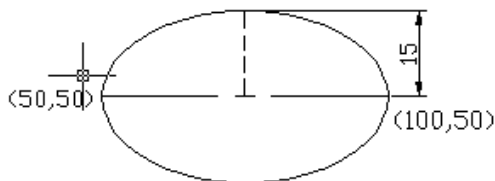
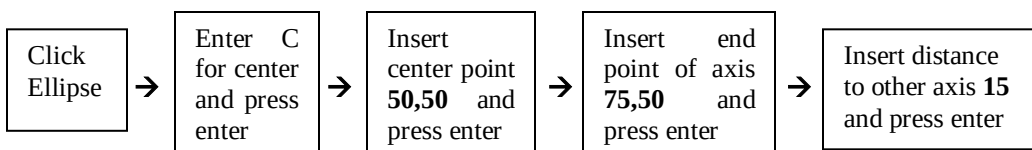
Command: **ELLIPSE** or **EL**

An ellipse is the most common geometric feature. It can be drawn using **Ellipse** command. Some options are there to draw ellipses. The most common two options will be presented here only.

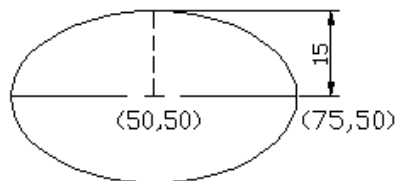
Axis and Eccentricity Option: In this option an ellipse is drawn inserting the values of the co-ordinates at the two end points of an axis of the ellipse and the distance to the other axis (Figure 11.40). By clicking at the points with the cross hair only the ellipse can be drawn. The steps are shown below to draw the ellipse by inserting values in command area.



Center and Two Axes options: In this option to draw an ellipse (Figure 11.40) the values of the co-ordinates of the center point and end point of an axis and the distance to the other axis have to be inserted. By clicking at the points with the cross hair only the ellipse can be drawn. The sequence of the steps to draw the ellipse is shown below.



Axis and Eccentricity Option



Center and Two Axes Option

Figure 11.40: Ellipse

11.3.9 Make Block

Draw Toolbar: 

Menu: **Draw → Block**

Command: **BLOCK** or **B**

When a part of a drawing, the whole one or any symbol is necessary to be redrawn time and again in the different location in the same drawing or another one, **Make Block** command can be used to store that under a block name. All the objects within the block will be treated as the single object. The entire block can be selected by picking a point on it. The block can be placed in a drawing in the desired orientation and with the desired scale factor. It can be moved, erased or listed as the single object. Invoking **Make Block** command, a Block Definition dialogue box (Figure 11.41) will be displayed. The sequence of the steps for making a block (Figure 11.42) using **Make Block** command is given below.

In the Block Definition dialogue box in the base point area clicking on **Pick point** button insertion base point can be specified. This point is used as the reference point to insert the block. Usually the lower left corner or the center point of the objects to be converted to block is chosen as the insertion base point. When the block is inserted, it is observed that the block appears at the insertion point. Any suitable unit can also be chosen from the drop down list in the dialogue box.

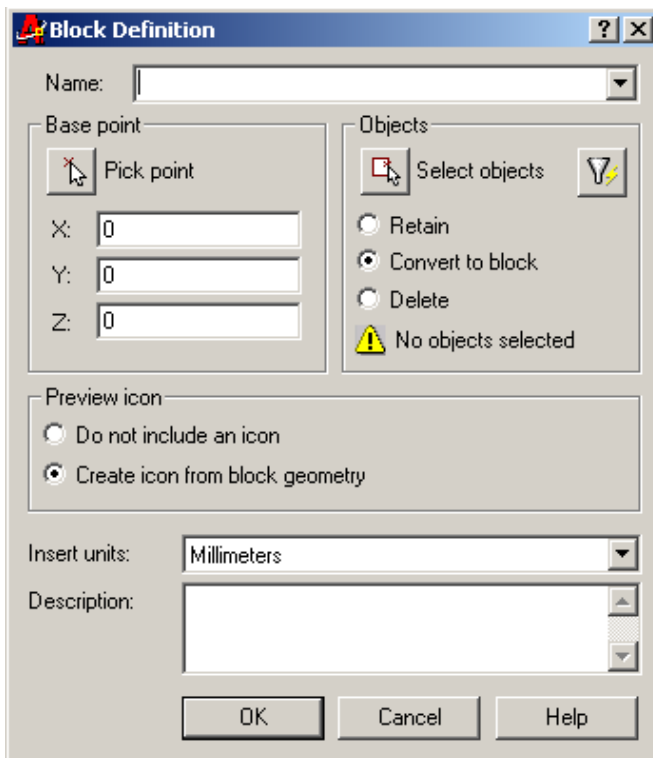
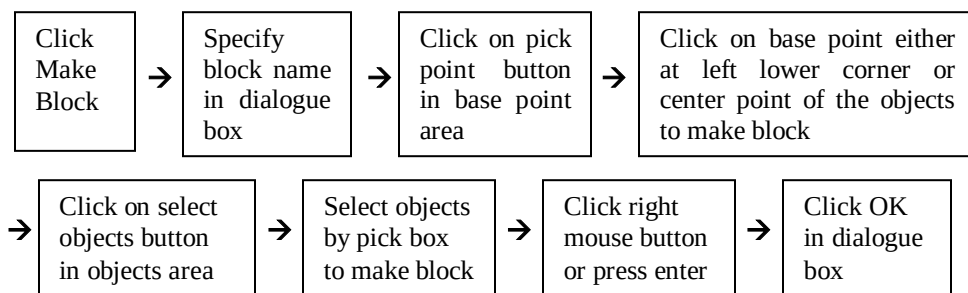


Figure 11.41: Block Definition Dialogue Box

If a block name is given, which already exists, an AutoCAD dialogue box will appear showing warning. It occurs during closing the Block Definition dialogue box clicking **OK** button. One can redefine block name clicking on the **Yes** button or exit clicking on the **No** button.



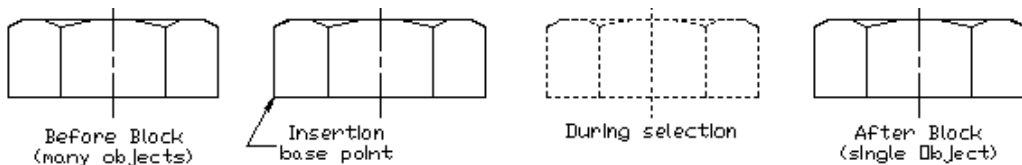


Figure 11.42: Making a Block

11.3.10 *Insert Block*

Draw Toolbar: 

Command: **INSERT**

With the help of **Insert Block** command one can insert the block in any desired location of the drawing. Invoking the **Insert Block** command, Insert dialogue box (Figure 11.43) will be displayed. Now in the dialogue box one can put the values of the co-ordinates of the insertion point, scale and rotation angle directly or they can be specified on the screen by selecting **Specify On-screen** with tick. For the insertion point by default **Specify On-screen** is selected. There is option in the dialogue box to insert different scales in the x-, y- and z- directions separately. Uniform scale may also be chosen. In Figure 11.44 the uses of scale and rotation are shown to insert block. The steps as mentioned below are to be followed to insert any block in the drawing.

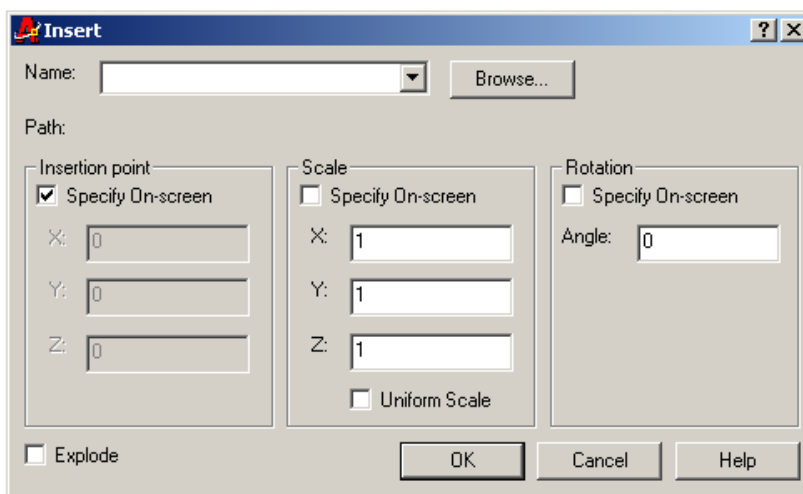
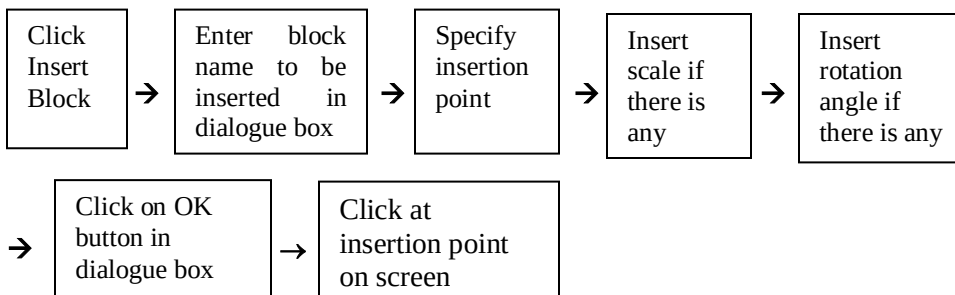


Figure 11.43: Insert Dialogue Box

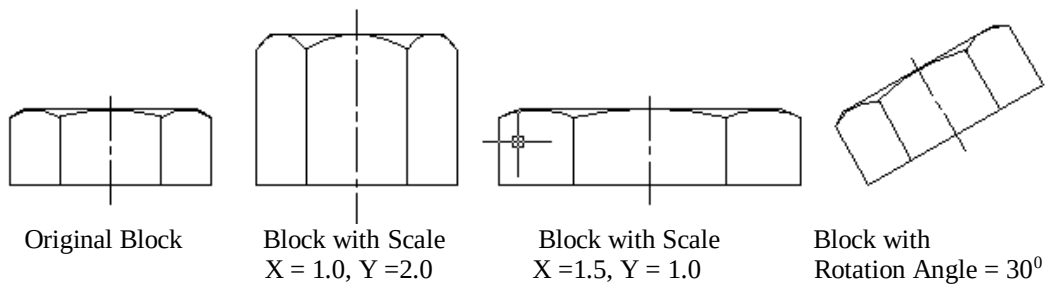


Figure 11.44: Block Inserted with Scale and Rotation

11.3.11 **Point**

Draw Toolbar: 

Menu: **Draw** → **Point**

Command: **POINT** or **PO**

The point is the basic drawing object. It can be drawn on the screen by using **Point** command. If the **Point** command is invoked from the menu by clicking Draw title, single or multiple point option can be selected. But when the **Point** command is invoked from the Draw toolbar only the multiple point option is selected. Using single point option one point can be drawn only while using multiple point option many points can be drawn. In case of the multiple point option one can exit by pressing **ESC** button while for the single point option after drawing the single point, it comes to exit automatically.

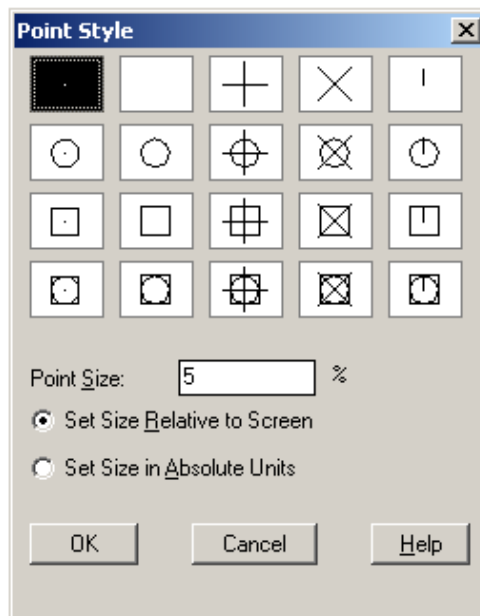


Figure 11.45: Point Style Dialogue Box

The type of point to be drawn can be changed from the Point Style dialogue box. Point Style dialogue box (Figure 11.45) can be displayed from the menu clicking **Format → Point Style**. By default the dot point is selected. Any type of desired point can be selected from the Point Style dialogue box and the size of it can also be changed according to the requirement. Then clicking **OK** button in the Point Style dialogue box the type of point is set for drawing.

11.3.12 Hatch

Draw Toolbar: 

Menu: **Draw → Hatch**

Command: **HATCH** or **H**

Hatching is very important to represent section in mechanical drawing. This hatching can be done using the **Hatch** command. AutoCAD provides with a number of hatch patterns for some common materials. When the **Hatch** command is invoked, a Boundary Hatch dialogue box (Figure 11.46) appears.

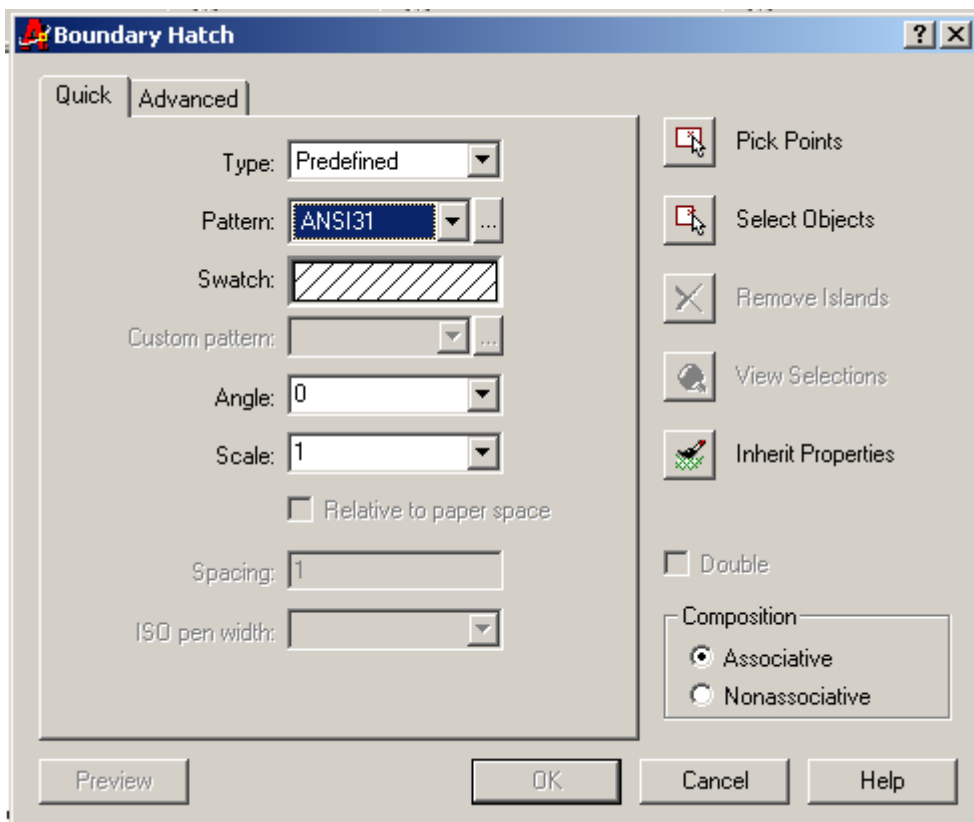


Figure 11.46: Boundary Hatch Dialogue Box

The steps as mentioned below are followed to give hatch line in an area (Figure 11.47). The objects to be selected for hatching may be done by clicking on the **Pick Points** tab or the **Select**

Objects tab in the dialogue box. In case of the **Pick Points** tab, the area is selected for hatching by picking a point within the area while for the **Select Objects** tab; the area to be selected is kept within the window made by the pick box. Pattern, angle and scale may be chosen from the drop down list displayed by clicking on the down arrow button of the respective area in the Boundary Hatch dialogue box. Different patterns of the section can also be made visible for selection by clicking on the tab indicated by three dots, which is located at the right side of Pattern area in the dialogue box.

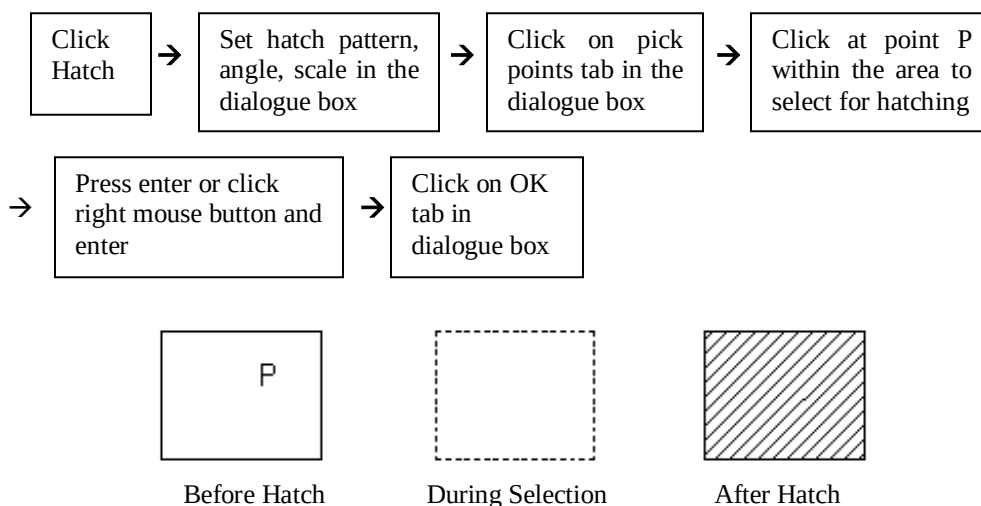


Figure 11.47: Hatching an Area

11.4 Use of Modify Command

Sometimes it is necessary to modify the basic drawing object e.g. line, circle, rectangle, polygon etc. in order to obtain the final object. AutoCAD provides a number of modify commands such as, **Erase, Copy, Mirror, Offset, Array, Move** and **Rotate** etc., the uses of them are described below in a nutshell. One can invoke the modify command either from the modify toolbar (Figure 11.11) directly or from the menu (Figure 11.48). It can also be invoked entering the desired command in the command prompt area. Sometimes it may be necessary to insert any value in the command area. After entering any command or inserting any value in the command area, **press enter** every time.

When any modify command will be invoked, the cross-hair cursor will turn into the shape of a pick box (Figure 11.49a) with some exception. In case of the offset command, the shape of a cross hair will appear in place of a pick box. In order to select the object to be modified, it is brought within the window, which is done by picking at point P₁ and then at point P₂ by using the pick box. The whole objects will be selected (Figure 11.49c). An object is selected by picking on it directly with the pick box (Figure 11.49d).

The selection of the object can be done before invoking any modify command. One can do this bringing the whole objects within the window with the help of the cross-hair cursor picking at point P₁ and then at point P₂, the whole objects will be selected (Figure 11.50b). An object may

also be selected by picking on it directly with the cross-hair cursor (Figure 11.50c). The selected lines turn into the dashed lines.



Figure 11.11: Modify Toolbar

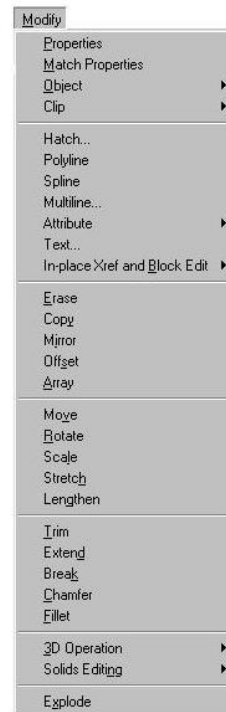


Figure 11.48: Modify Commands From Menu

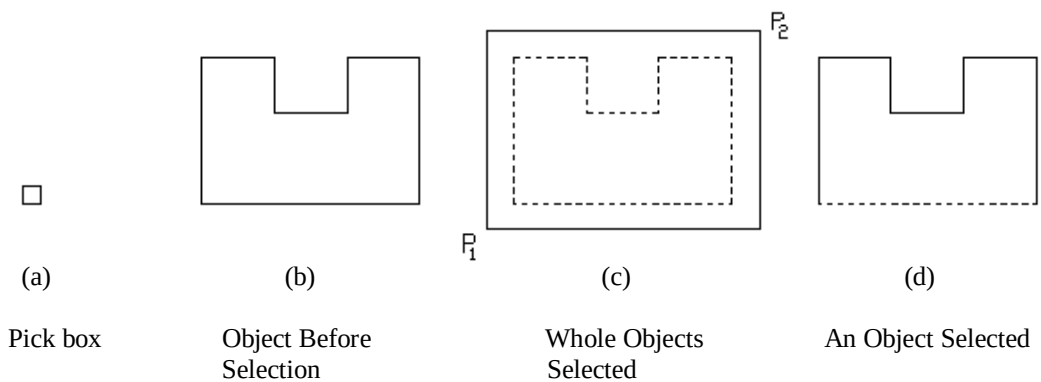


Figure 11.49: Selection of Object with Pick Box

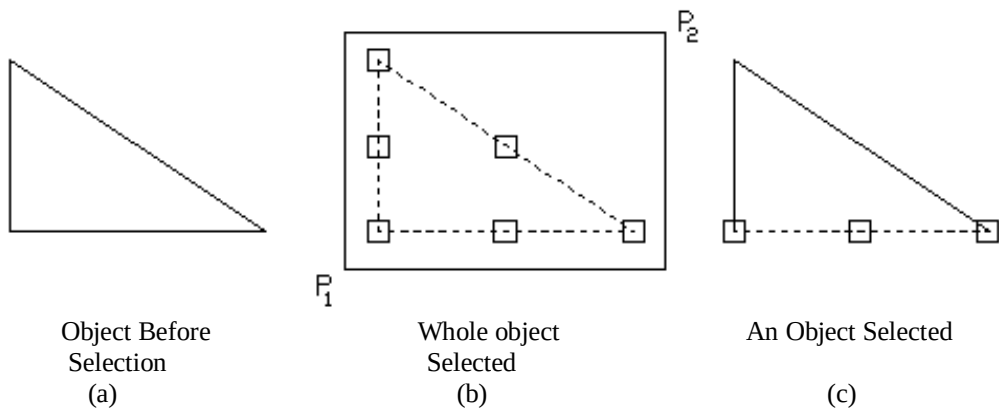


Figure 11.50: Selection of Object with Cross hair Cursor

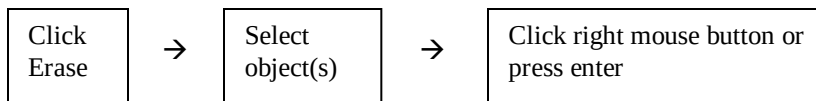
11.4.1 Erase

Modify Toolbar:  (Figure 11.11)

Menu: **Modify** → **Erase** (Figure 11. 48)

Command: **ERASE** or **E**

The **Erase** command is one of the simplest and most useful AutoCAD commands. The command erases (deletes) any selected object(s) from the drawing. The delete key of keyboard can also be used to erase a selected object by cross-hair cursor in AutoCAD. The following sequence may be followed to erase any object(s).



11.4.2 Copy

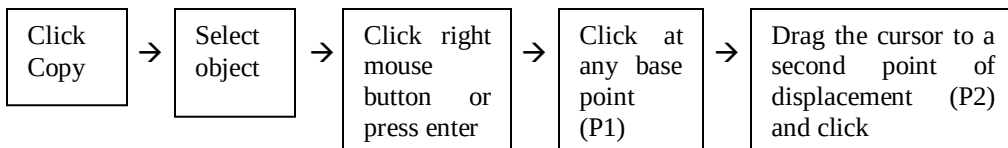
Modify Toolbar: 

Menu: **Modify** → **Copy**

Command: **COPY** or **CP**

The **Copy** command can be used to create one or more duplicates of any object that was previously created. Copy is a very useful and timesaving command because if any complex drawing is needed to be redrawn, it can be done easily using copy command as many times as are required (Figure 11.51).

The steps as mentioned below are followed to make the copy of any object(s).



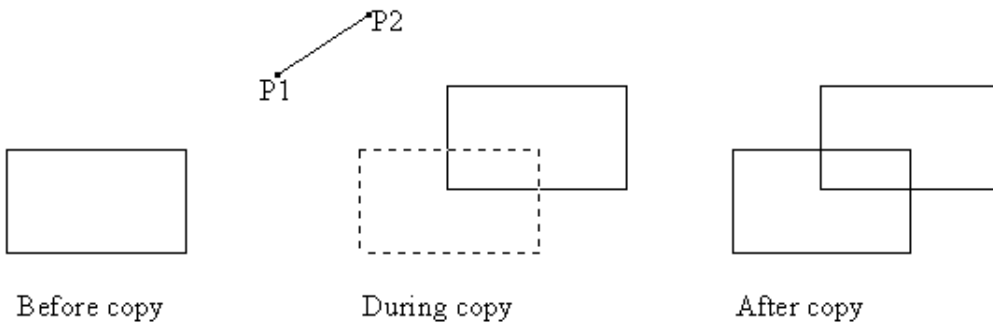


Figure 11.51: Use of Copy Command

11.4.3 *Mirror*

Modify Toolbar: 

Menu: **Modify** → **Mirror**

Command: **MIRROR** or **MI**

When it is necessary to draw the image of an object the **Mirror** command is used. It makes the mirror copy of the selected object(s) (Figure 11.52). The mirror copy can also be made at any angle. The sequence as shown below is followed to make the mirror of any object(s).

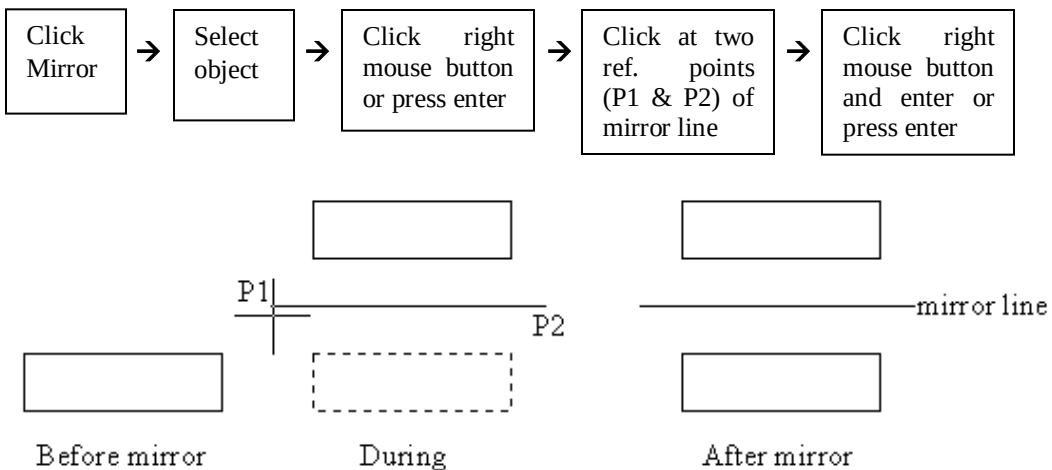


Figure 11.52: Use of Mirror Command

11.4.4 *Offset*

Modify Toolbar: 

Menu: **Modify** → **Offset**

Command: **OFFSET** or **O**

To draw the parallel lines, polylines, concentric circles, ellipses, arcs, curves, etc., **Offset** command may be used. An object similar to the selected object is created using offset command

(Figure 11.53). To offset an object the offset distance is to be inserted in the command prompt area. The steps as mentioned below are followed to make the offset of any object(s).

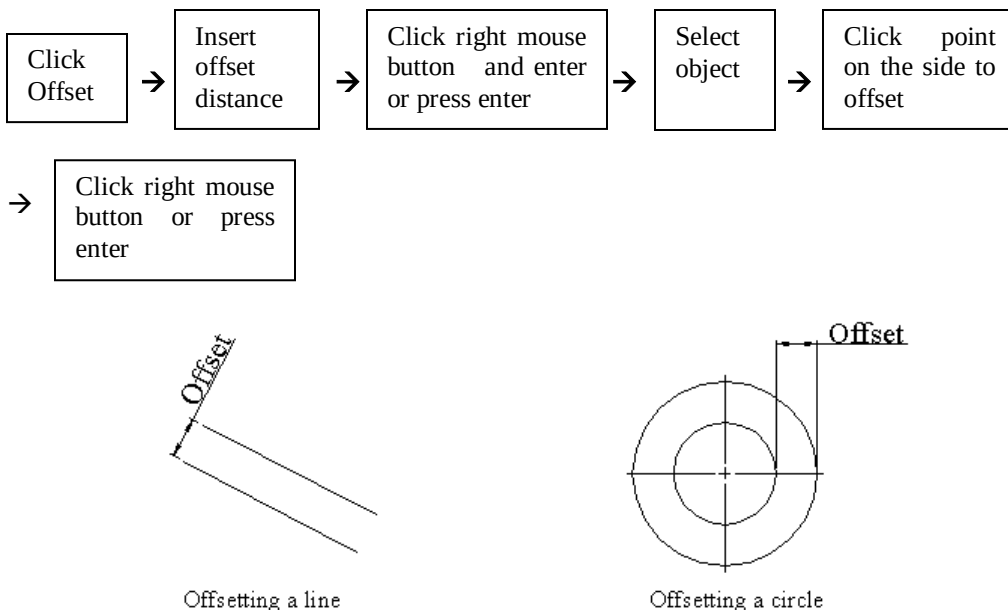


Figure 11.53: Use of Offset Command

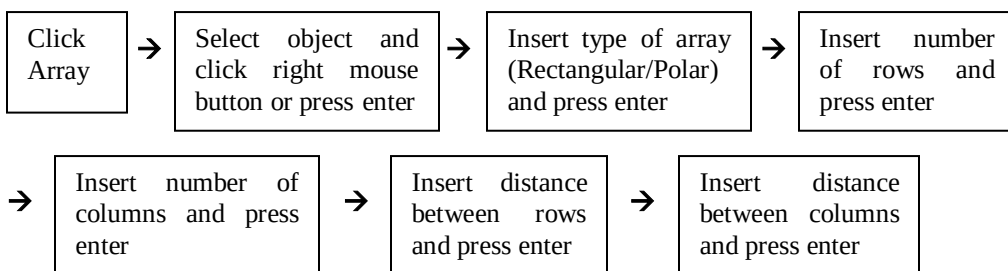
11.4.5 Array

Modify Toolbar: 

Menu: **Modify → Array**

Command: **ARRAY** or **AR**

The **Array** command makes multiple copies of selected object(s) either in a rectangular matrix (columns and rows) (Figure 11.54) or in a polar (circular) pattern (Figure 11.55). The necessary data is to be inserted in the command prompt area. The steps of rectangular array are as follows. In AutoCAD 2004 the array command is activated in a different way.



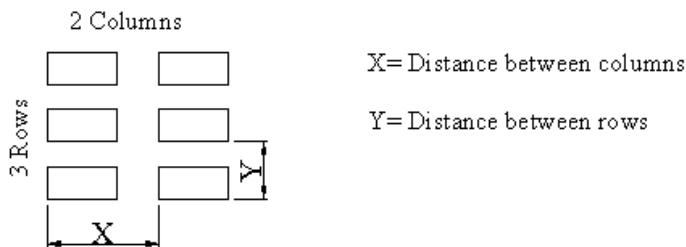


Figure 11.54: Use of Rectangular Array Command

The steps to make the polar array of any object(s) are shown below.

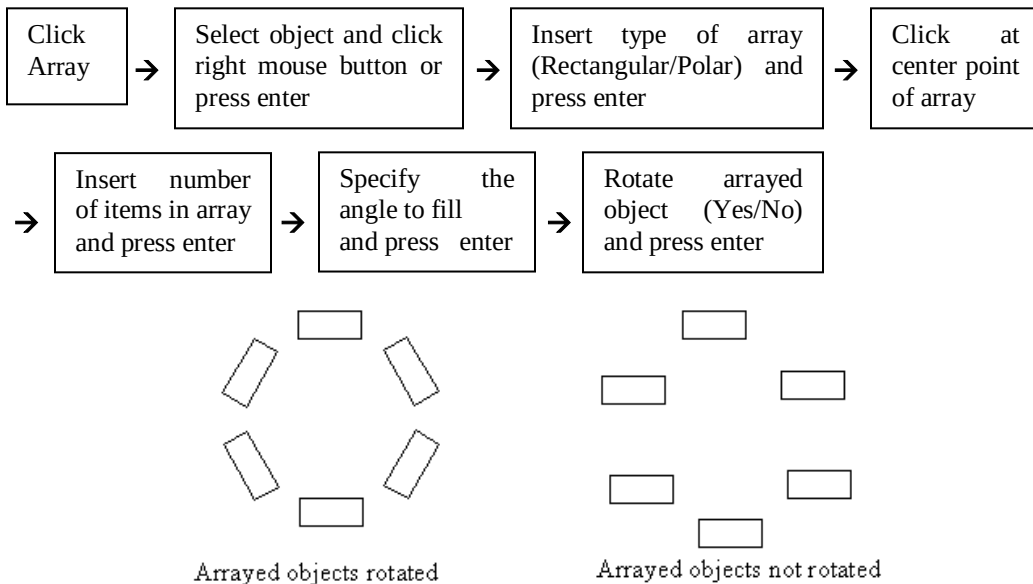


Figure 11.55: Use of Polar Array Command

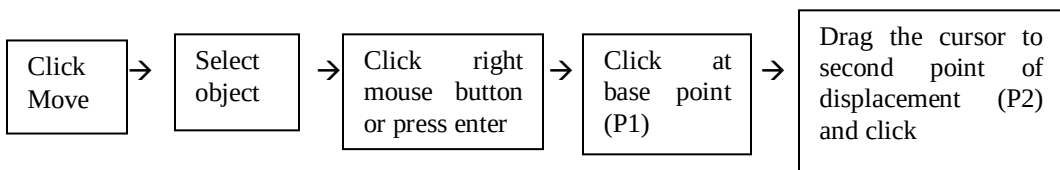
11.4.6 Move

Modify Toolbar:

Menu: **Modify → Move**

Command: **MOVE** or **M**

Sometimes it is necessary to move object from one place to another (Figure 11.56), which can be achieved by using **Move** command following the steps as shown below.



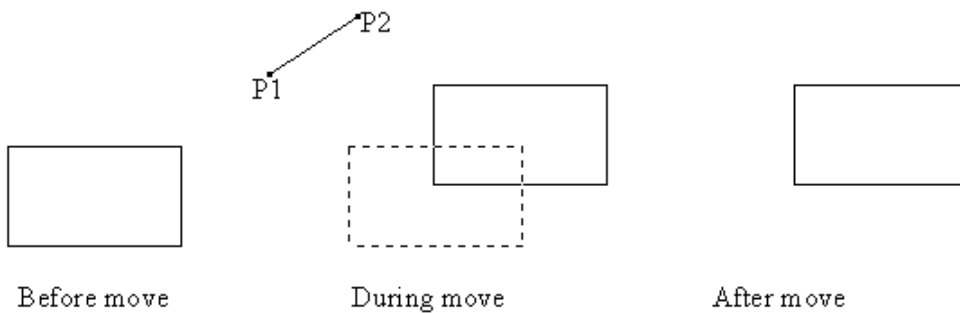


Figure 11.56: Use of Move Command

11.4.7 Rotate

Modify Toolbar: 

Menu: **Modify → Rotate**

Command: **ROTATE** or **RO**

One can rotate an object at any angle either in anticlockwise or clockwise direction about any specified point (base point) by using the **Rotate** command (Figure 11.57). For anticlockwise rotation insert positive angle and for clockwise rotation insert negative angle in degree in the command prompt area. The steps to rotate any object(s) are as follows.

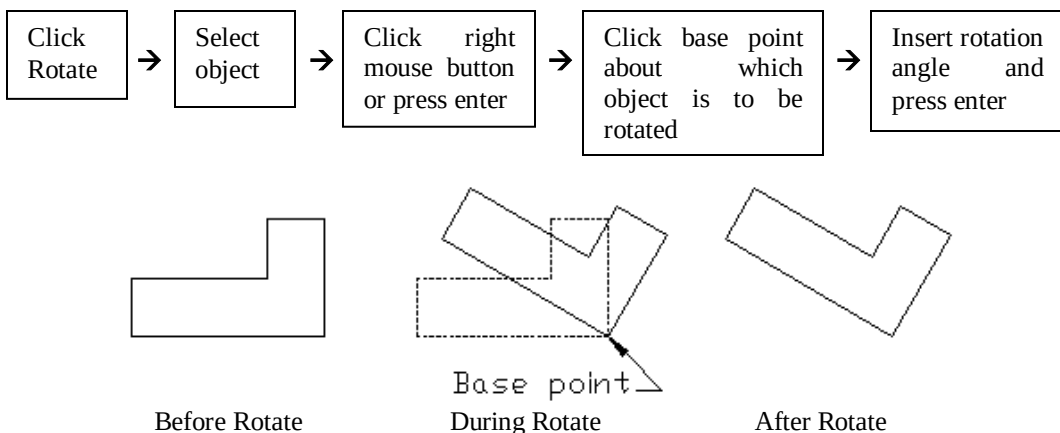


Figure 11.57: Use of Rotate Command

11.4.8 Scale

Modify Toolbar: 

Menu: **Modify → Scale**

Command: **SCALE** or **SC**

The **Scale** command is used to enlarge or shorten the size of an object in accordance with the desired scale. By doing this, the shape of the object is not changed at all. It is a timesaving command to edit. If the scale of the object is changed, the dimensions of the object are also changed according to the scale (Figure 11.58).

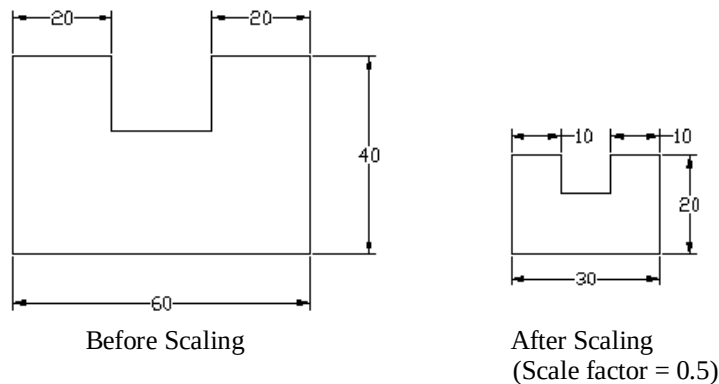
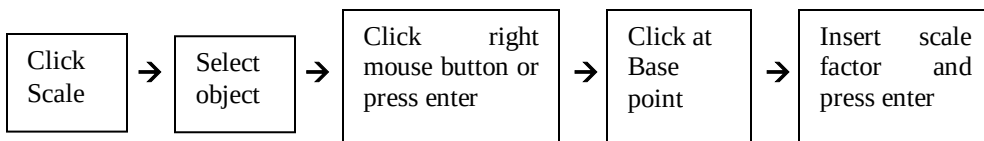


Figure 11.58: Use of Scale Command

The base point is that point in respect of which, the scaling is done. The base point may be at any point on the corner, inside or outside of the object. The value of the desired scale factor is to be inserted in the command prompt area. The sequence to scale any object(s) is as follows.



11.4.9 Stretch

Modify Toolbar:

Menu: **Modify** → **Stretch**

Command: **STRETCH**

The **Stretch** command can be used to stretch the object(s) (Figure 11.59). One can either lengthen or shorten the objects using this command.

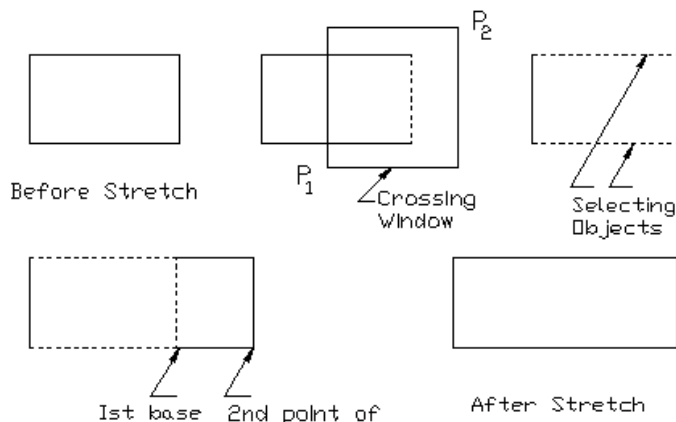
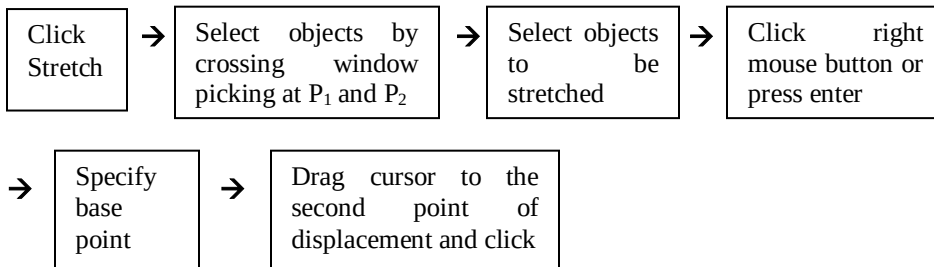


Figure 11.59: Use of Stretch Command

The steps to stretch any object(s) are as follows.



11.4.10 **Lengthen**

Modify Toolbar: 

Menu: **Modify → lengthen**

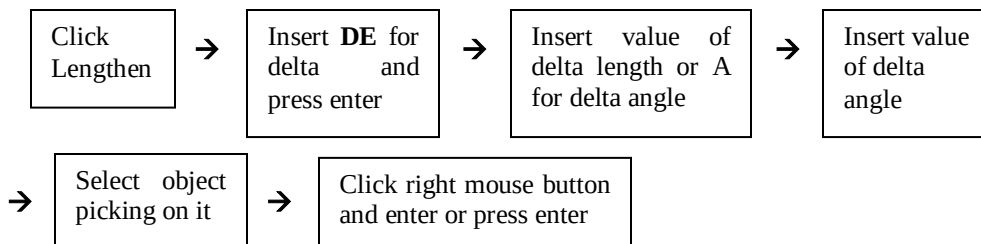
Command: **LENGTHEN**

One can use the **Lengthen** command to extend or shorten lines, polylines, arcs and elliptical arcs. The Lengthen command has several options to change the length of the object. The side at which it is necessary to extend or shorten, picking has to be done on that side to select the object. In case of the closed object like circle lengthen command has no effect. While the lengthen command is invoked, the following prompt sequence is displayed.

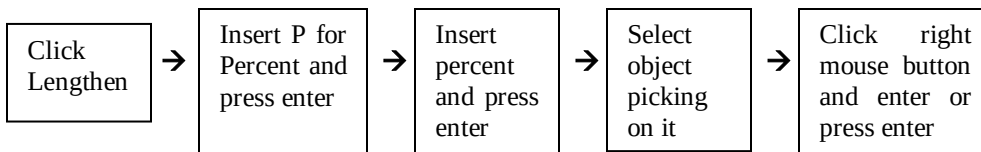
_Lengthen

Select an object or [Delta/Percent/Total/Dynamic]:

Delta Option: In this option the length of the angle of an object is either increased or decreased by Delta length or Delta angle (Figure 11.60). The Delta value is inserted either numerically or specifying two points. The positive value will increase and the negative value will decrease the length of the selected object. If the value of the delta length is inserted in the third step in place of inserting A for delta angle, the fourth step is to be omitted in the following sequence.



Percent Option: In this option the percentage of the original line is to be specified in respect of which the line will be extended or shortened (Figure 11.60). For percent of 125 the original length will be increased by 25 percent while for percent of 75 the original length will be decreased by 25 percent. The steps are as follows.



Total Option: In this option the value of the total length is specified, as a result the original length of the line is either increased or decreased to make it equal to the total length (Figure 11.60). If the inserted value of the total length is 95, the original length will either increase or decrease to make its length equal to 95. The value can be inserted either numerically or specifying two points. The steps are as follows.

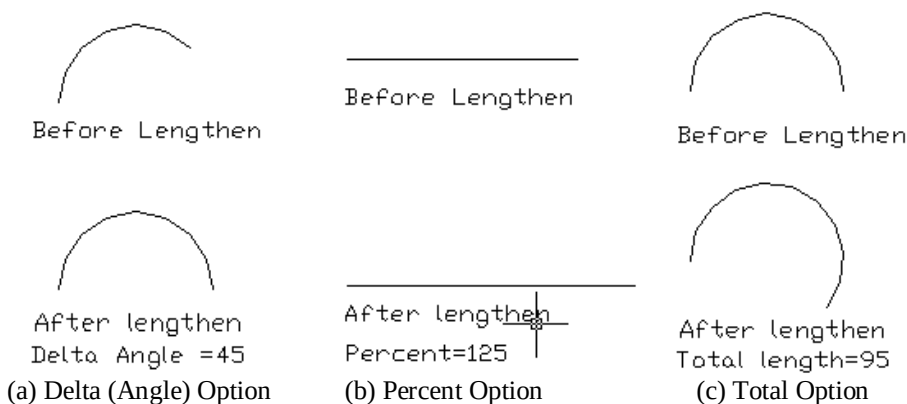
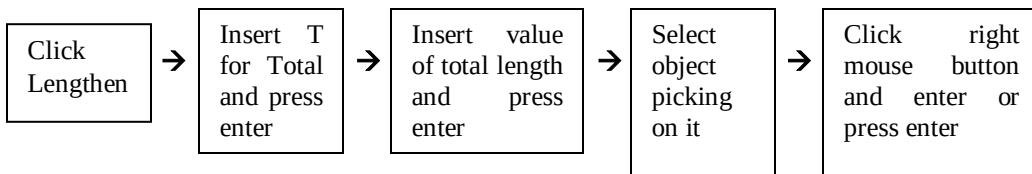


Figure 11.60: Use of Lengthen Command

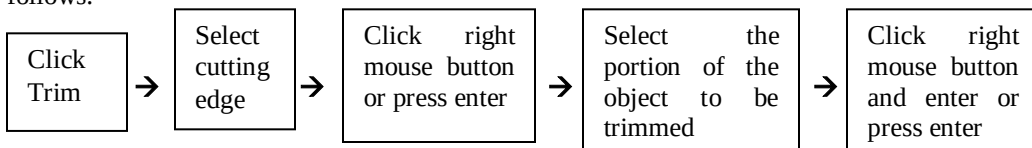
11.4.11 Trim

Modify Toolbar: 

Menu: **Modify** → **Trim**

Command: **TRIM** or **TR**

The **Trim** command is used in order to trim a part of an object. An object is trimmed with respect to another object known as the cutting edge (Figure 11.61). The sequence to trim any object is as follows.



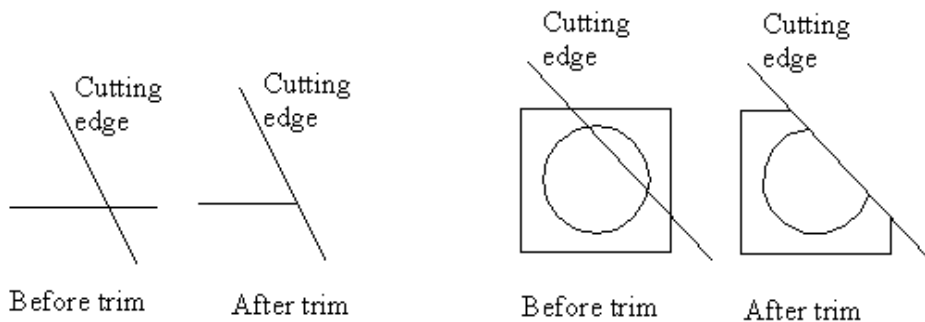


Figure 11.61: Use of Trim Command

11.4.12 *Extend*

Modify Toolbar: 

Menu: **Modify** → **Extend**

Command: **EXTEND** or **EX**

Using the **Extend** command one can extend a line, polyline or arc up to another object known as the boundary edge (Figure 11.62). This command may be considered as opposite to the Trim command. The steps to extend any object are shown below.

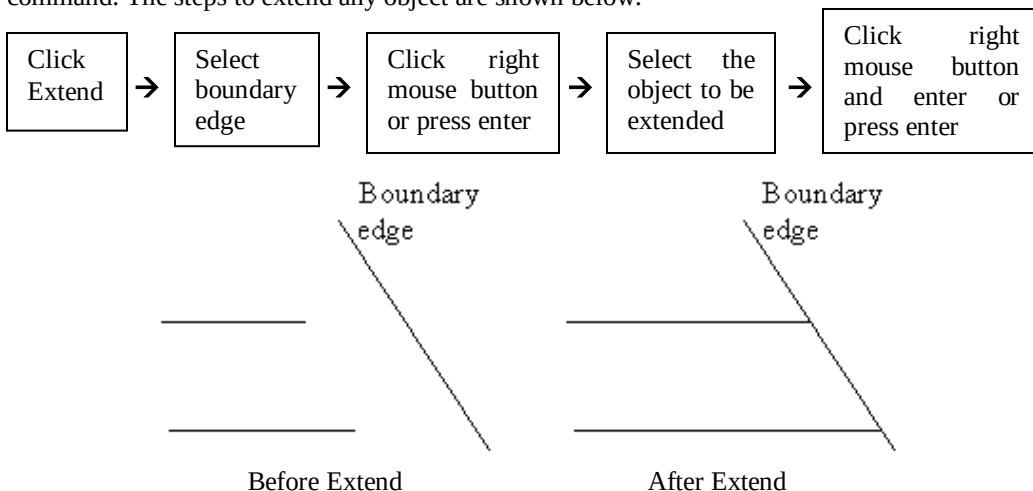


Figure 11.62: Use of Extend Command

11.4.13 *Break*

Modify Toolbar: 

Menu: **Modify** → **Break**

Command: **BREAK** or **BR**

The **Break** command can be used to break or remove a part of the object, e.g. line, polyline, arc, circle, ellipse, etc. The object is thus divided into two parts (Figure 11.63). For the circle with position of the first and second break points as shown, the feature of Figure 11.50b will appear and

interchanging the break points the feature of Figure 11.50c will appear. The steps to break any object are as follows.

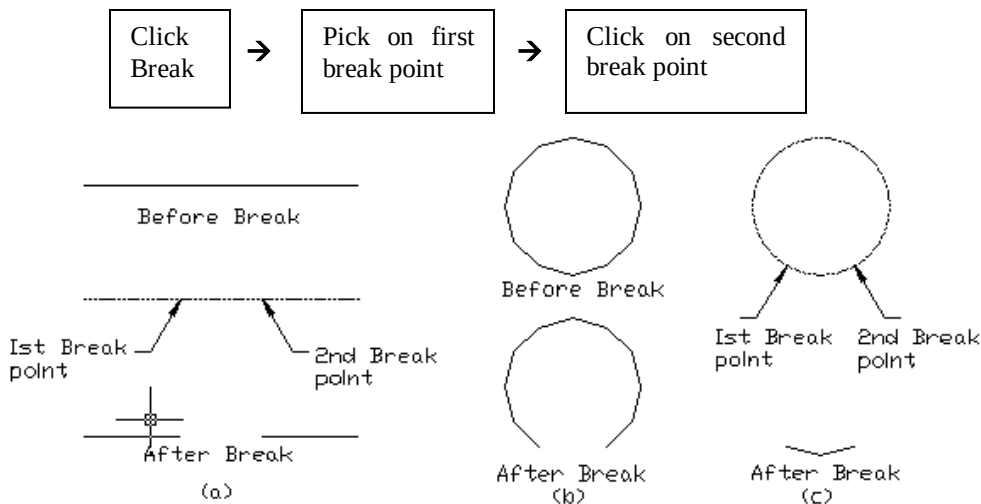


Figure 11.63: Use of Break Command

11.4.14 Chamfer

Modify Toolbar: 

Menu: **Modify** → **Chamfer**

Command: **CHAMFER**

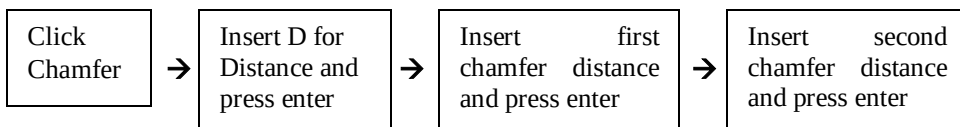
With the help of the **Chamfer** command the sharp edge of an object can be chamfered according to the requirement. By invoking chamfer command the following prompt sequence will be displayed.

(TRIM mode) Current chamfer distance Dist1 = 10.0000, Dist2 = 10.0000

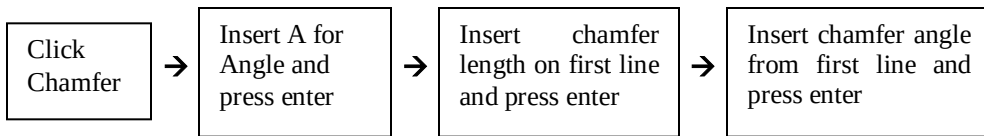
Select first line or [Polyline/Distance/Angle/Trim/Method]:

Necessary option as required by the user can be set from the prompt sequence before chamfering the object.

Distance Option: In this option the values of the distances are inserted, if the current chamfer distances do not match with the requirement of the user. The steps to make chamfer by distance option are as follows.



Angle Option: In this option the values of distance on one side and angle are inserted. The value of the angle has to be inserted in degree. The steps in angle option are as follows.



Trim Option: One can set Trim or No Trim mode in this option by entering T for Trim mode and No for No Trim mode. With Trim mode the extended part beyond the chamfer will be removed while with No Trim mode, it will be retained (Figure 11.64). To make the desired chamfer, if the current Trim mode matches with the requirement of the user, the second and third steps should be avoided from the following sequence.

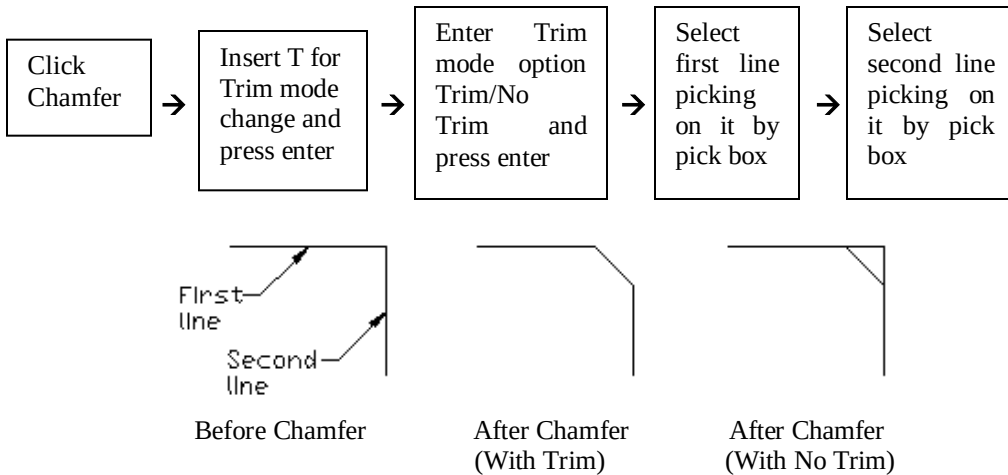


Figure 11.64: Use of Chamfer Command

11.4.15 Fillet

Modify Toolbar: 

Menu: **Modify → Fillet**

Command: **FILLET**

The **Fillet** command is used to make the fillet (round arc) or round connecting two objects. The fillet is the inside round corner and the round is the outside one. By invoking Fillet command the following prompt sequence will be displayed.

Current Settings: Mode = TRIM, Radius = 10.000

Select first object or [Polyline/Radius/Trim]:

Radius Option: If the required fillet radius of the user does not match with the current fillet radius, one has to change the current fillet radius in this option. To obtain the required fillet radius the steps as mentioned below are followed.



Trim Option: One can set Trim or No Trim mode in this option by entering T for Trim mode and No for No Trim mode. With Trim mode the extended part beyond the fillet or round will be removed while with No Trim mode, it will be retained (Figure 11.65). To make the fillet, if the current Trim mode matches with the requirement of the user, the second and third steps should be avoided from the following sequence. The fillet command can be used between two parallel lines, where the fillet will turn into the shape of a semicircle, the radius of which is equal to half the distance between the parallel lines.

The sequence to make fillet by trim option is as follows.

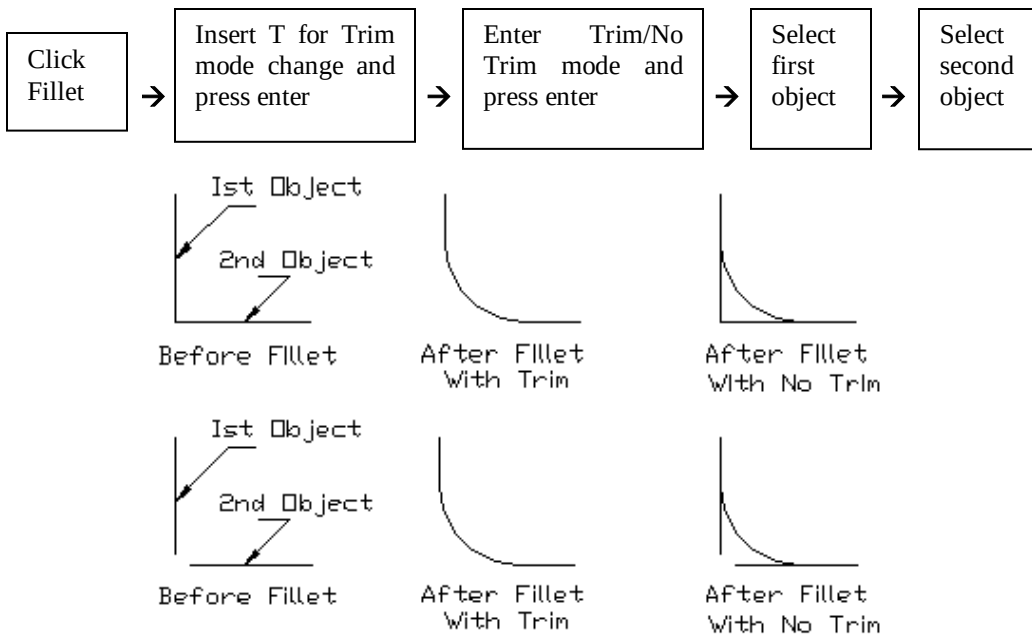


Figure 11.65: Use of Fillet Command

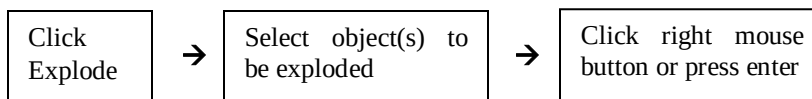
11.4.16 *Explode*

Modify Toolbar: 

Menu: **Modify → Explode**

Command: **EXPLODE**

Using the **Explode** command, any dimension or block can be broken into separate objects (Figure 11.66). In case of a dimension line, it is separated into the text, arrowhead and line objects. Then each object can be selected and edited individually. When a block is exploded, it turns into a group of objects that can be edited separately. To perform the explode command; the steps as mentioned below are to be followed.



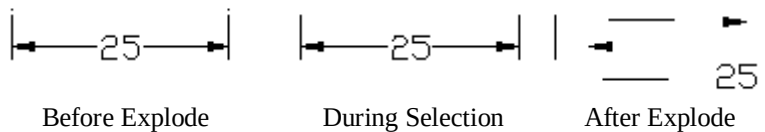


Figure 11.66: Use of Explode Command

11.5 Object Properties Toolbar

In Figure 11.67 an object properties toolbar is shown. It gives the provision to change the color, line weight, and line types of the object(s). This toolbar also contains another important layers option.

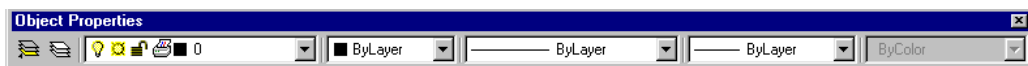


Figure 11.67: Object Properties Toolbar

11.5.1 Color

A drop down list of various available colors is shown in Figure 11.68. Any one color may be chosen from this list according to the requirement. This drop down list is obtained by clicking the down arrow button just above the list. If one selects the option **Other** in the list, a Select Color dialog box will be displayed from where the desired color can also be chosen. The Select Color dialog box can also be displayed inserting COLOR in the command area.

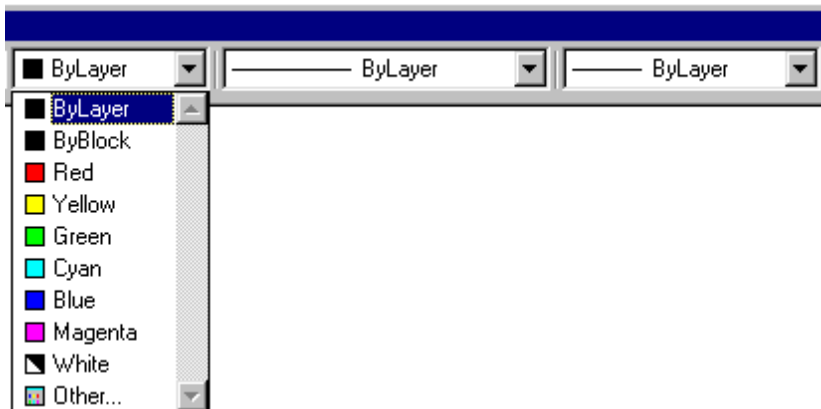


Figure 11.68: Drop Down List Showing Various Colors

11.5.2 Line Weights

The thickness of the various lines (e.g. continuous line, center line and hidden line etc.) in the drawing can be changed by selecting the required thickness from the drop down list as shown in Figure 11.69. The drop down list is obtained by clicking the down arrow button just above the drop down list.

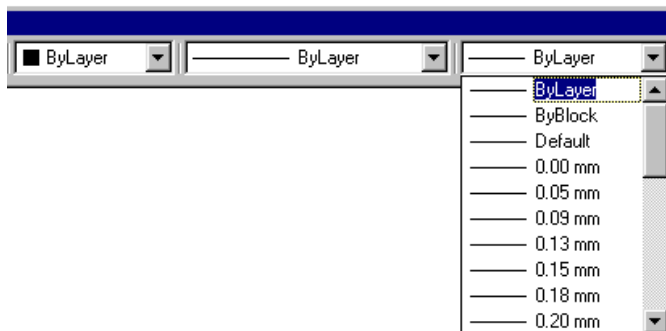


Figure 11.69: Drop Down List Showing Various Line Thickness

11.5.3 Line Types

From the drop down list as shown in Figure 11.70, one can select any desired line type. When the option **Other** is selected from the list, a Linetype Manager dialogue box as shown in Figure 11.71 will be displayed. Now clicking on the **Load** tab located at top right side of this dialogue box, another Load or Reload Linetypes dialogue box will also appear as shown in Figure 11.72.

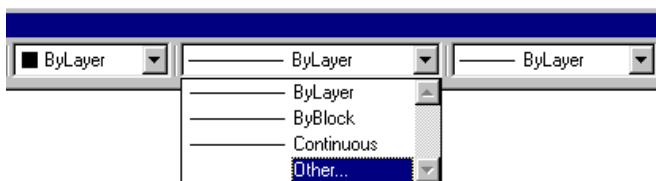


Figure 11.70: Drop Down List Showing Line Types

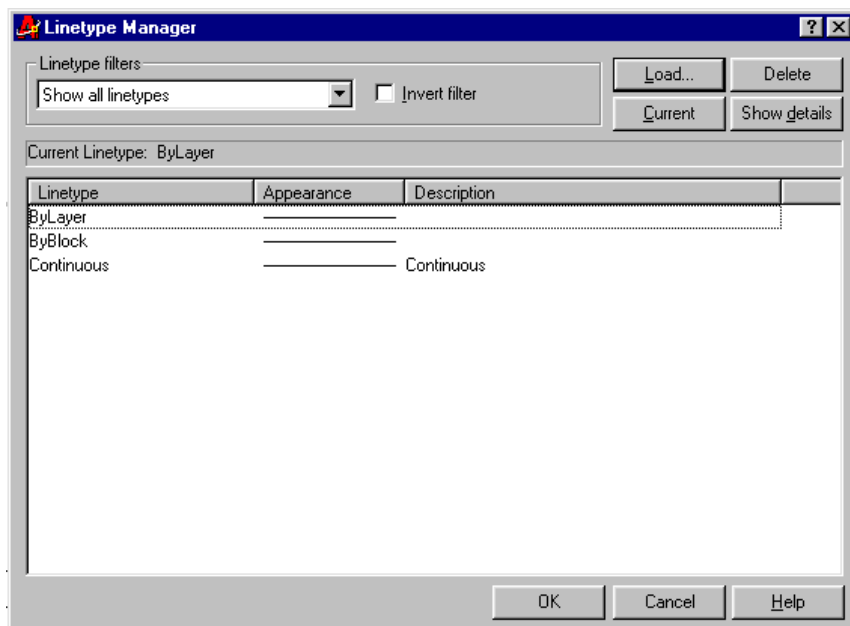


Figure 11.71: Linetype Manager Dialog Box

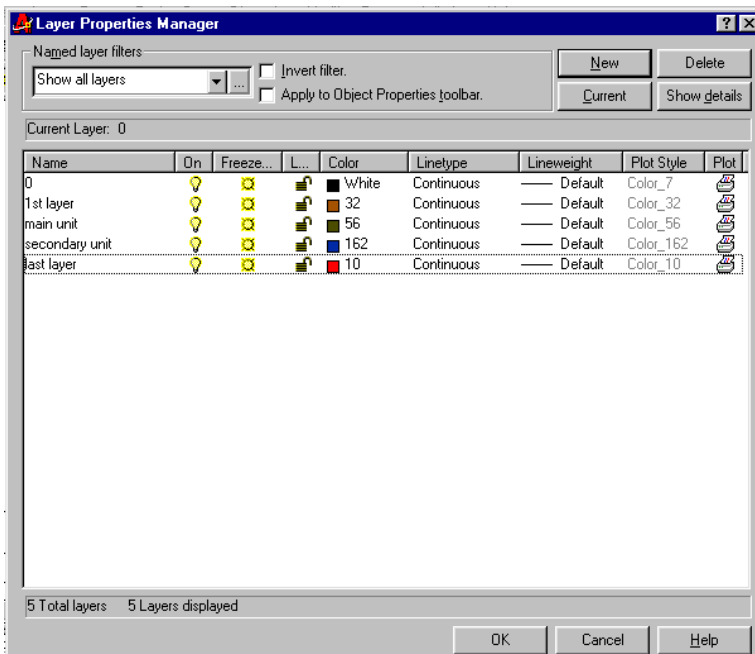


Figure 11.74: Layer Properties Manager Dialog Box

11.5.5 Properties Dialog Box

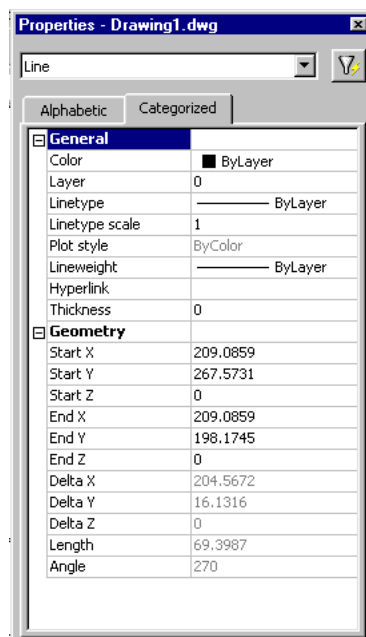


Figure 11.75: Properties Dialogue Box

The properties (thickness, scale, layer, color etc.) of the different objects such as, line, circle, arc etc. can be modified using the Properties dialogue box as shown in Figure 11.75. This dialog box can be displayed by clicking **Tools → Properties** from the menu. One can find this dialog box by clicking **Properties** in the short cut menu as well. When the object will be selected for changing its properties the name of it (e.g. Line, Circle, Arc etc.) will be displayed at the top of this dialog box.

11.6 *Drawing Display*

In AutoCAD some commands such as, Redraw, Zoom are necessary to perform the drawing precisely and smoothly. These commands are very useful; as a result they have been taken into consideration for discussion in short in this section.

11.6.1 *Redraw*

Standard Toolbar: 

Menu: **View → Redraw**

Command: **REDRAW** or **R**

During drawing in AutoCAD it is necessary to pick at different points and erase different objects. When picking is made at different points, small cross marks may appear. These small marks are called blip marks or **blips**. These blip marks are not the part of the drawing. After deleting an object from the drawing, some spots may remain there in the drawing as well. In order to clean them from the drawing **Redraw** command is used. This command may be either invoked from the standard toolbar or from the menu. When **Redraw** command is clicked the whole drawing is cleaned eliminating these undesired marks. While clicking on the command no prompt sequence is displayed in the command prompt area. In AutoCAD for some commands redraw action takes place automatically.

11.6.2 *Zoom*

Sometimes it happens that it is necessary to draw a small object or modify it in a drawing. It is difficult to accomplish that manually but in AutoCAD it can be done easily by making enlarged view of that portion of the drawing. In AutoCAD there is **Zoom** command, which can be used to enlarge or reduce size of the drawing, displayed on the screen. **Zoom** command has several options. Of them the most useful options are only taken into consideration here.

Zoom Window Option

Standard Toolbar: 

Menu: **View → Zoom → Window**

Command: **ZOOM** or **Z**

Zoom window option is the most useful one. The specific area, which is needed to be magnified, is brought within the rectangular window under this option. The window is created by two specified points at the opposite corners of the rectangular window. One may specify the two opposite corners of the window either clicking at the points with the cross hair or inserting values of the co-ordinates of the points in the command area. The center of the window becomes the center of the new display screen. The window option of **Zoom** command can be invoked either

from the standard toolbar or from the menu. It can also be initiated by entering the ZOOM in the command prompt area. The steps for zooming objects (Figure 11.76) are given below.

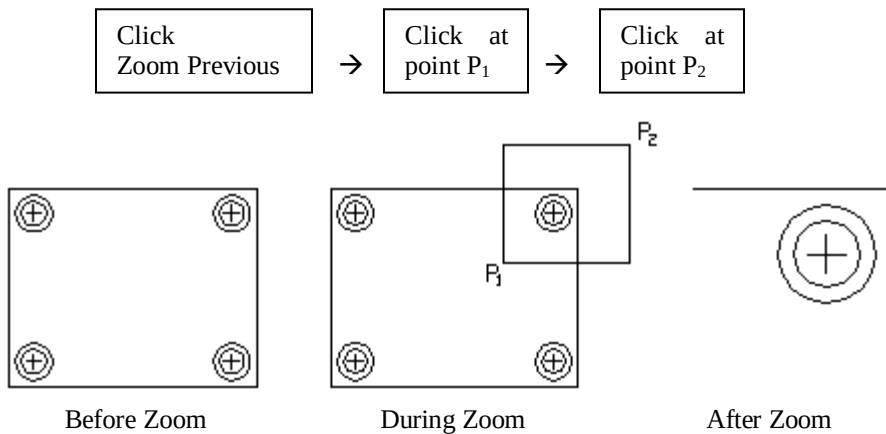


Figure 11.76: Zooming Object with Window Option

Zoom Previous Option

Standard Toolbar:

Menu: **View → Zoom → Previous**

Command: **ZOOM** or **Z**

To edit a portion of a drawing it may be required to magnify it by **Zoom** command. At the end of the edit it becomes necessary to go back to the previous view, which can be done by using the previous option of **Zoom** command. This command can be invoked either from the standard toolbar or from the menu. One can also initiate this **Zoom** command by entering ZOOM in the command prompt area. Ten previous views are stored in the view port; as such one may come back to the required view by using previous option of **Zoom** command successively until it appears on the screen.

Zoom Realtime Option

Standard Toolbar:

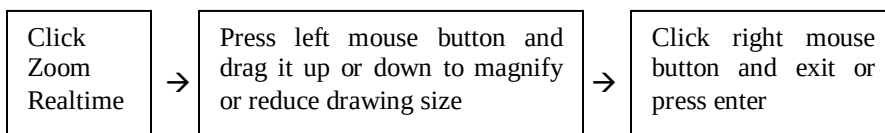
Menu: **View → Zoom → In/Out**

Command: **ZOOM** or **Z**

Realtime option of **Zoom** command is the most useful command for magnification or reduction of size of the drawing. During performing drawing, sometimes this command becomes necessary. This command can be invoked from the standard toolbar or from the menu or by entering ZOOM in the command area. When it is initiated from the standard tool bar, its positive sign indicates **Zoom in** while its negative sign indicates **Zoom out**. To activate the positive sign pressing the left mouse button drag the cursor to the upward direction, the drawing will be magnified. To further magnify bring back the cursor and drag it in the same way in the upward direction.

To reduce the size of the drawing the opposite action has to be done. That is, press the left mouse button and drag the cursor to the downward direction. To further reduce the size bring back the cursor and drag it in the same way in the downward direction. If the cursor is dragged vertically up

from the mid point of the screen to the top of it, the drawing will be magnified by 100% (Zoom in 2x magnification). On the other hand if the cursor is dragged vertically downward from the mid point of the screen to the bottom of it, the drawing will be reduced by 100% (Zoom out 0.5x magnification). If the command is invoked from the menu, for **In** option the drawing is magnified by 100% and for **Out** option it is reduced by 100%. While invoking the command from the toolbar the sequence as mentioned below has to be followed.

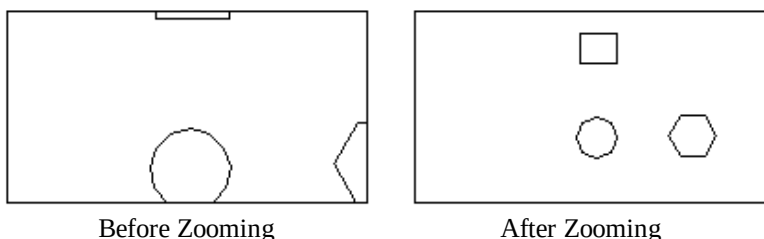


Zoom All Option

Menu: **View → Zoom → All**

Command: **ZOOM** or **Z**

All option of **Zoom** command can be invoked from the menu. It can also be initiated by entering ZOOM in the command area. It is the useful command for drawing. Sometimes some portion of a drawing in a file may remain outside the boundary limit of the screen, which is not displayed. Using this command all the objects of the file appear on the screen (Figure 11.77).



Before Zooming

After Zooming

Figure 11.77: Application of Zoom Command

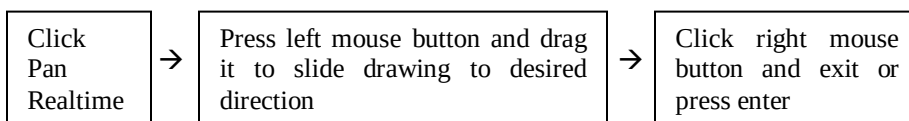
11.6.3 Realtime Pan

Standard Toolbar: 

Menu: **View → Pan → Realtime**

Command: **PAN** or **P**

With the help of Realtime option of **Pan** command one can slide the drawing to the left, right, up or down to bring the portion of the drawing inside the window for work. This command can be invoked either from the standard toolbar or from the menu. It can also be initiated by entering PAN in the command area. Pressing the left mouse button drag the cursor to the desired direction until the desired portion of the drawing appears in the window. The steps as mentioned below are followed to slide the drawing.



11.7 Dimensioning

Menu: **View → Toolbars → Dimension**

Menu: **Dimension**

Dimension indicates the measurement of the object, which is used for manufacturing. The Dimension commands can be invoked either from the Dimension toolbar or from the Dimension menu. Dimension toolbar (Figure 11.78) can be obtained selecting the crosscheck box beside Dimension in the Toolbars dialog box (Figure 11.13).

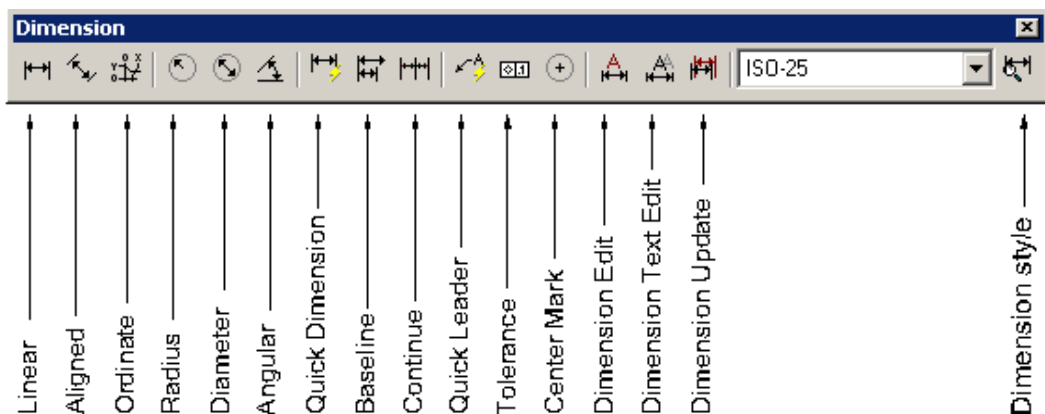


Figure 11.78: Dimension Toolbar

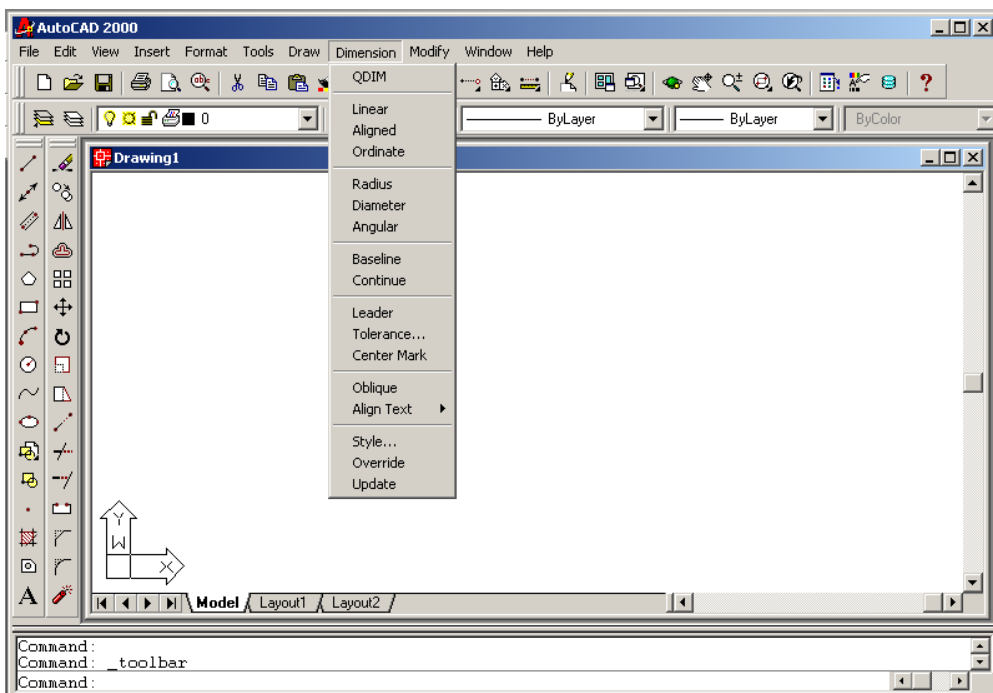


Figure 11.79: AutoCAD Window Showing Dimension Menu

On the other hand, Dimension menu (Figure 11.79) can be obtained selecting **Dimension** title in the menu bar. Dimension commands can also be initiated by entering command in the command prompt area.

11.7.1 Types of Dimensioning

As can be seen from the Dimension menu or Dimension toolbar that they consist of many dimension commands. But here only the most common types of dimension commands (e.g. Linear, Aligned, Baseline, Continue, Radius, Diameter, Angular, Center Mark and Leader) will be taken into consideration for discussion in short.

Linear

Dimension Toolbar: 

Menu: **Linear**

Command: **DIMLIN** or **DIMLINEAR**

To give the dimension between two points of a line either horizontally or vertically (Figure 11.80) **Linear** command is used. Object snap (OSNAP) mode is made on to locate the points accurately. The sequence to give linear dimension on an object, invoking command from the toolbar or menu is as follows.

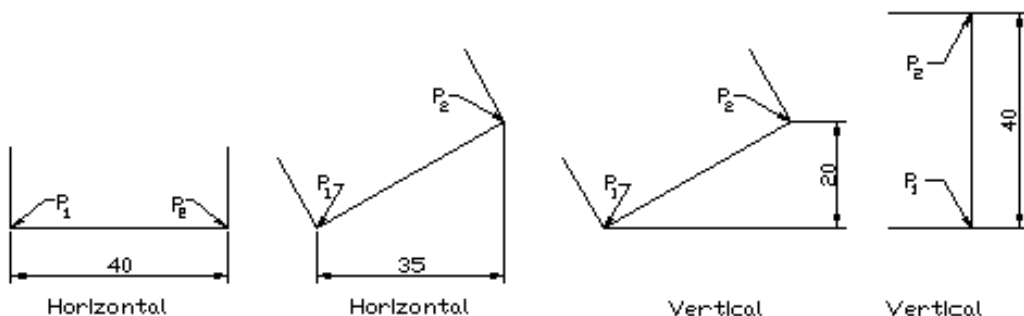
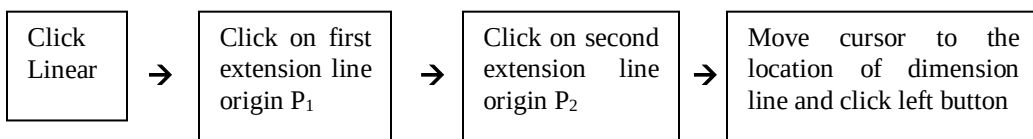


Figure 11.80: Linear Dimension

Aligned

Dimension Toolbar: 

Menu: **Aligned**

Command: **DIMALIGNED**

Using **Aligned** command, dimensioning between two end points of an inclined line can be given (Figure 11.81). With the help of this command linear dimensioning can also be given. The steps to give aligned dimension on an object, invoking command from the toolbar or menu are as follows.

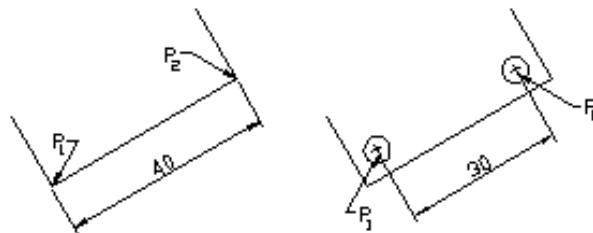
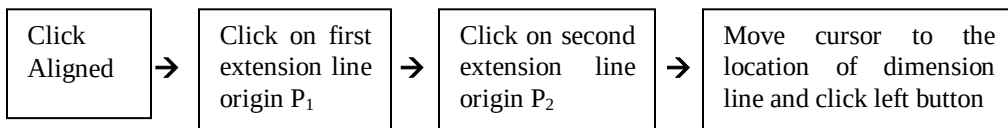


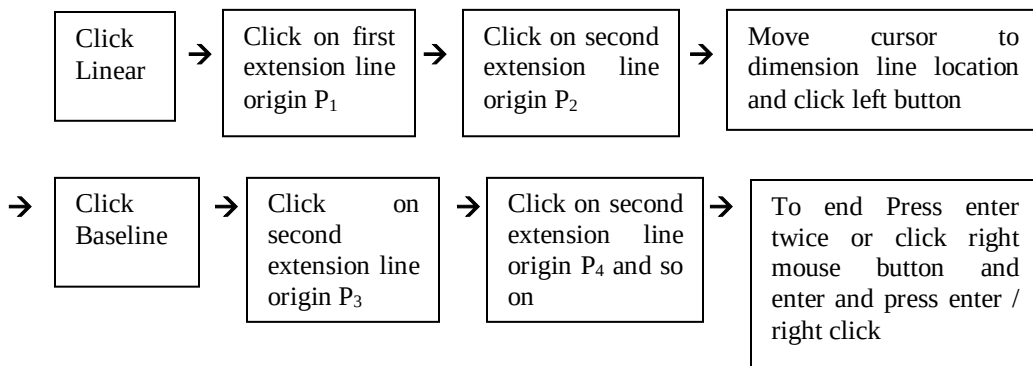
Figure 11.81: Aligned Dimension

Baseline

Dimension Toolbar:

Menu: **Baseline**Command: **DIMBASE** or **DIMBASELINE**

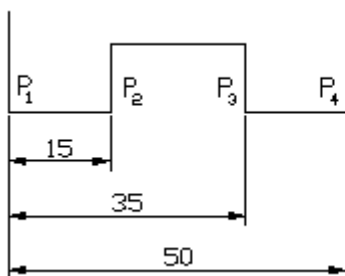
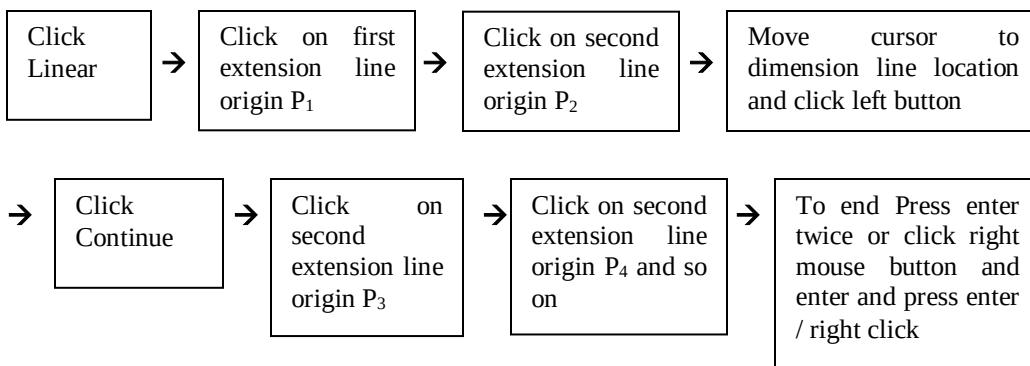
When dimensions (Figure 11.82a) are given in reference to a fixed point (base point or reference point), it is called baseline dimensioning. **Baseline** command is used for this dimension. The sequence for the baseline dimension invoking command from the toolbar or menu is mentioned below.

**Continue**

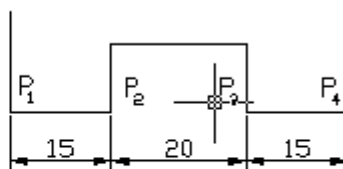
Dimension Toolbar:

Menu: **Continue**Command: **DIMCONT** or **DIMCONTINUE**

Using **Continue** command linear dimensioning can be continued from the second extension line of the previous dimension. This is a kind of chain dimensioning (Figure 11.82b). When the command is invoked from the toolbar or menu, the steps for the continue dimensioning become as follows.



(a)



(b)

Figure 11.82: Baseline and Continue Dimensions

Radius

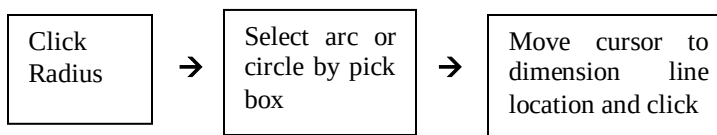


Dimension Toolbar:

Menu: **Radius**

Command: **DIMRAD** or **DIMRADIUS**

With the help of **Radius** command, the radius of an arc or the circle (Figure 11.83a) can be given. The steps to give the radius dimension invoking command from the toolbar or menu are as follows.



Diameter

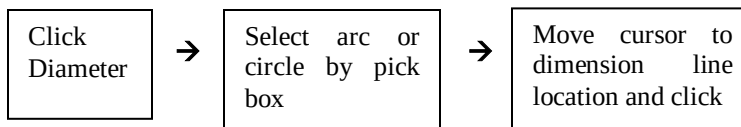


Dimension Toolbar:

Menu: **Diameter**

Command: **DIMDIA** or **DIMDIAMETER**

Using **Diameter** command, the diameter of an arc or the circle (Figure 11.83b) can be given. The steps to give the diameter dimension invoking command from the toolbar or menu are as follows



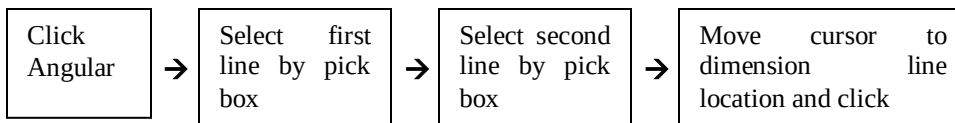
Angular

Dimension Toolbar: 

Menu: **Angular**

Command: **DIMANG** or **DIMANGULAR**

The angle between two intersecting lines can be given using **Angular** Command (Figure 11.83c). The sequence to insert the angular dimension invoking command from the toolbar or menu is shown below.



Center Mark

Dimension Toolbar: 

Menu: **Center Mark**

Command: **DIMCENTER**

Sometimes it is necessary to give the center mark of an arc or circle (Figure 11.83d), which can be done by using **Center Mark** Command. The sequence to provide center mark is shown below.

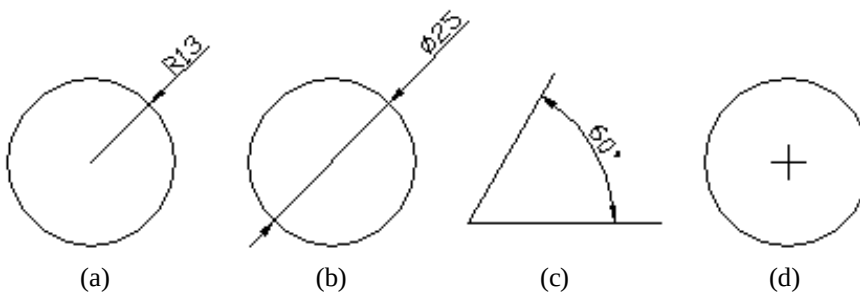
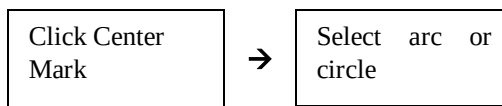


Figure 11.83: Radius, Diameter, Angular Dimensions and Center Mark

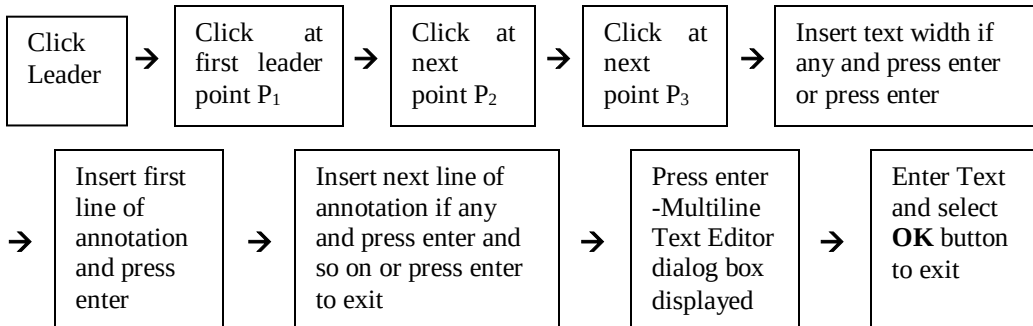
Leader

Dimension Toolbar: 

Menu: **Leader**

Command: **LEADER** or **QLEADER**

Often it becomes necessary to show dimensioning without using another dimension command or attach annotations to an object, which can be done using **Leader** command. The leader consists of an arrow and a leader line. The leader command offers several options as well. The sequence to draw a leader is shown below.



In option for no text (Figure 11.84a), omit 6th and 7th steps in the above sequence, and select **OK** button only in 9th step to exit. In option for annotation text (Figure 11.84b) omit the last two steps. In option for Multiline text (Figure 11.84c) omit 6th and 7th steps.

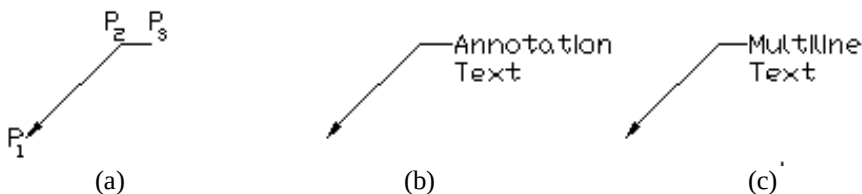


Figure 11.84: Leader in Different Options

11.7.2 Setting Dimension Properties

Dimension Toolbar: 

Menu: **Format** → **Dimension Style**
or **Dimension** → **Style**

Command: **DDIM** or **DIMSTYLE**

Invoking **Dimension Style** command from the dimension toolbar or menu a Dimension Style Manager dialog box as shown in Figure 11.85 will be displayed. It can also be initiated by entering DDIM/DIMSTYLE command in the command area.

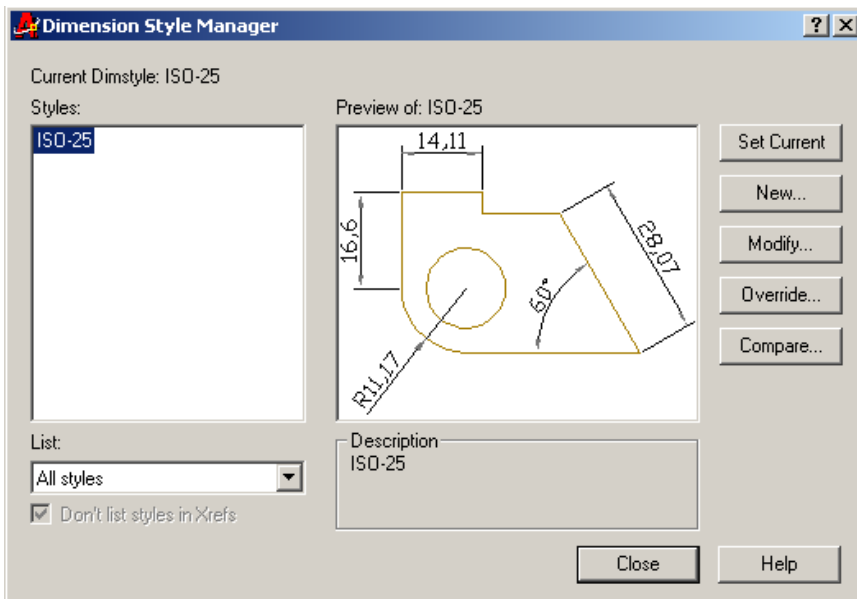


Figure 11.85: Dimension Style Manager Dialogue Box

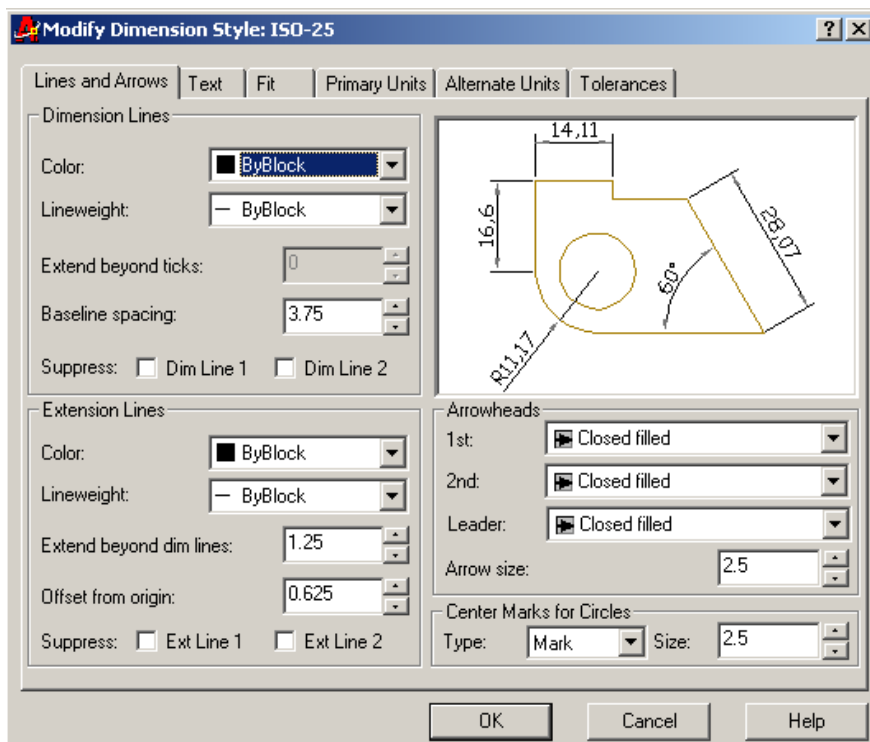


Figure 11.86: Modify Dimension Style Dialogue Box (For Lines and Arrows)

Now to set the different properties of the dimensions, click the **Modify** button in this dialog box. Then a Modify Dimension Style dialog box (For Lines and Arrows) will appear by default (Figure 11.86). This dialogue box consists of a number of tabs such as, Lines and Arrows, Text, Fit, Primary units, Alternate units, Tolerances from where any one can be selected according to the requirement. When the Text tab is selected a Modify Dimension Style dialog box (For Text) is displayed (Figure 11.87). The Modify Dimension Style dialog box (For Lines and Arrows) contains a number of areas such as, Dimension Lines, Extension Lines, Arrowheads and Center Marks for Circles. On the other hand the Modify Dimension Style dialog box (For Text) contains a number of areas such as, Text Appearance, Text Placement and Text Alignment.

According to the requirement of the user necessary settings can be made from here. When the Primary Units tab of the Modify Dimension Style dialog box will be selected that dialog box will provide different areas like Linear Dimensions, Measurement Scale, Angular Dimensions and ZeroSuppressions from where one can set the scale of drawing, precision and decimal separator etc. After necessary settings select the **OK** button in the Modify Dimension Style dialog box followed by **Close** button in the Dimension Style Manager dialog box.

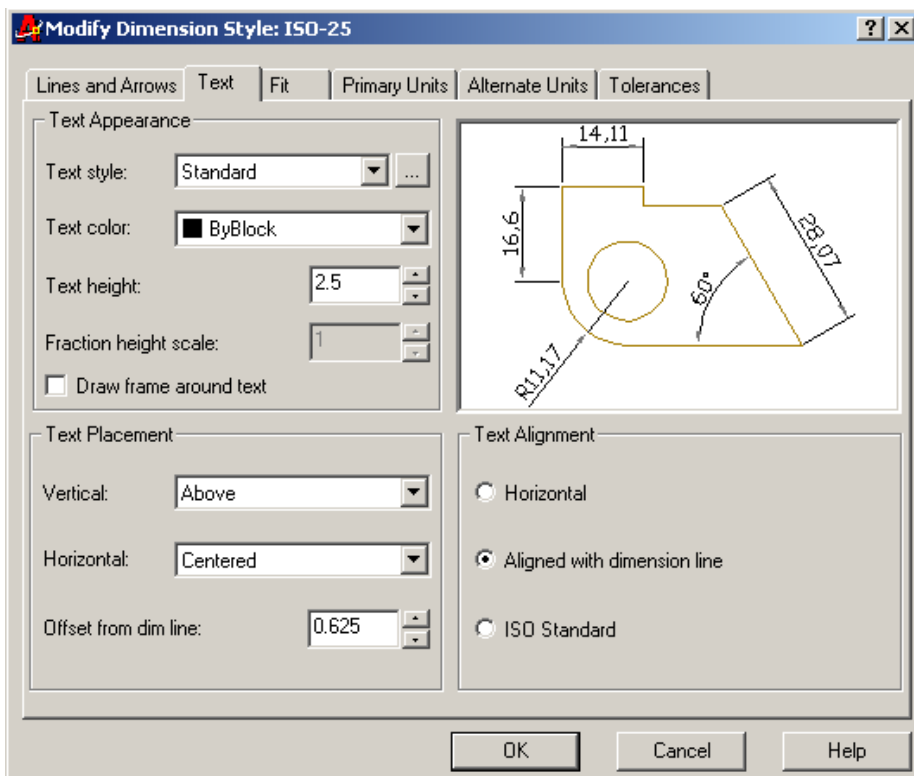


Figure 11.87: Modify Dimension Style dialogue Box (For Text)

11.7.3 Editing Dimensions

Menu: **Tools → Properties** or **Modify → Properties**

Command: **PROPERTIES**

The Properties window can be invoked from the menu or by entering **Properties** command in the command area. To change the properties of dimension, dimension text style, geometry, format etc. Properties window can be used. The Properties window for dimension (Figure 11.88a) consists of a number of sections such as, General, Misc, Primary Units, Fit, Lines & Arrows, Text, Tolerances and Tolerances & Text. While the Properties window for leader (Figure 11.88b) contains various sections like General, Geometry, Misc, Fit, Lines & Arrows and Text.

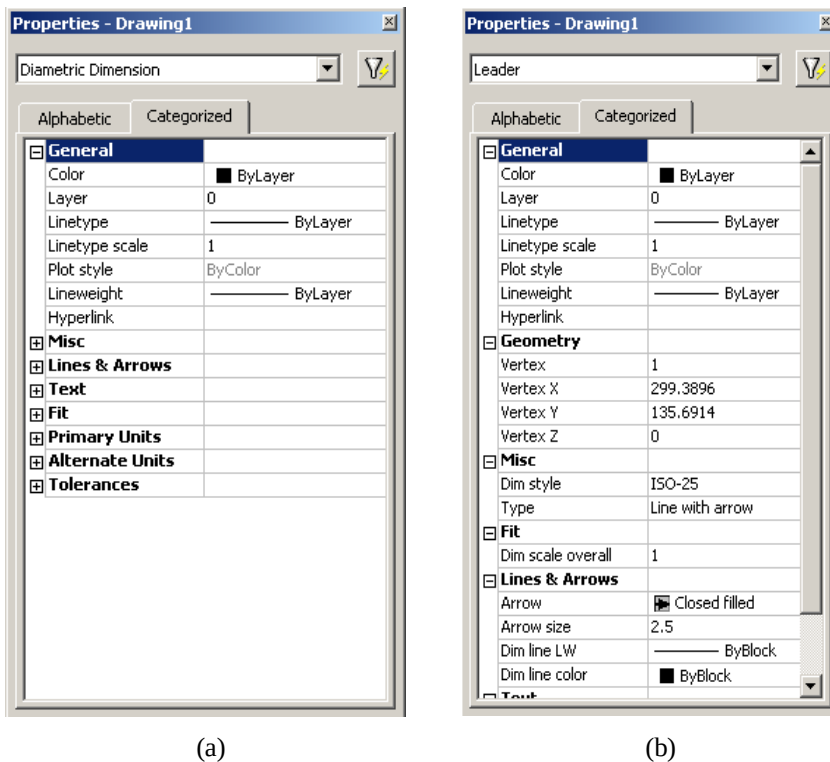


Figure 11.88: Properties Window

When any dimension is selected to edit, that dimension name appears at the top of the Properties window. In order to use any of them select it and perform the necessary change in the dimension.

11.8 Isometric Drawing

In this section a brief idea to make the isometric drawing will be provided. The dimensioning in the isometric drawing will also be taken into consideration for discussion.

11.8.1 Generating Isometric Drawing

Menu: **Tools → Drafting Settings**

Command: **DSETTINGS**

Drafting Settings dialogue box can be obtained from the menu or by entering command in the command prompt area. Drafting Settings dialog box can also be invoked from the status bar by right clicking on any of the button except the ORTHO and MODEL button, a short cut menu will

appear from where settings can be selected. Now clicking on **Snap and Grid** button at the top left corner of the Drafting Settings dialogue box, one finds the dialog box (Figure 11.22) where click on **Isometric snap** radio button in the Snap type & style area. Next click on **Polar Tracking** button, the dialogue box will turn into the feature as shown in Figure 11.89. Now in the area of Polar Angle Settings of this dialogue box, select the increment angle of 30 from the drop down list, which is found clicking on the down arrow button. In this condition the cross hair cursor is set for the isometric drawing.

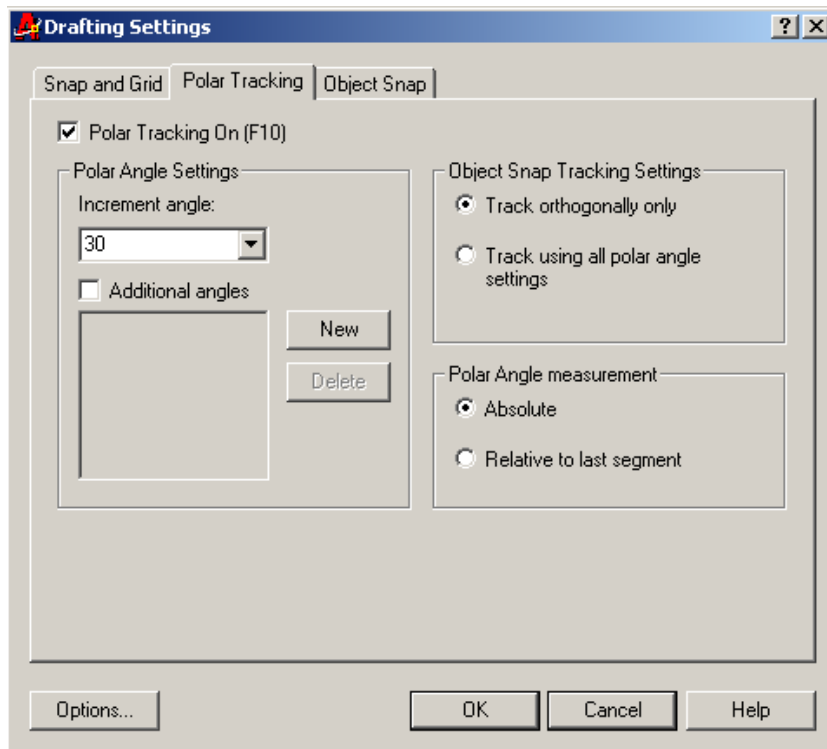


Figure 11.89: Drafting Settings Dialogue Box (For Polar Tracking)

The orientation of the crosshair for the isoplane left is shown in Figure 11.90.

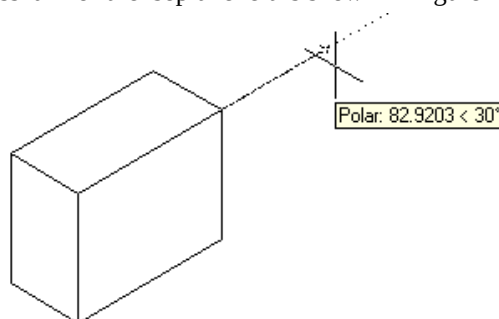


Figure 11.90: Crosshair Position For Isoplane Left

However, a user can easily change the orientation of the crosshair cursor for the isoplane right and isoplane top by pressing **Ctrl** and **E** keys simultaneously (**Ctrl** + **E**) or using the function key F5. The different isoplanes can also be obtained by entering **Isoplane** command in the command prompt area followed by L for left, R for right and T for top.

In the isometric view the circle in the different isoplanes turns into the shape of ellipse, which is called isocircle. To draw the isocircle the steps as mentioned below are followed. However, before doing that, the position of the crosshair has to be changed for the respective isoplane left, isoplane right and isoplane top as shown in Figure 11.91 from left to right.

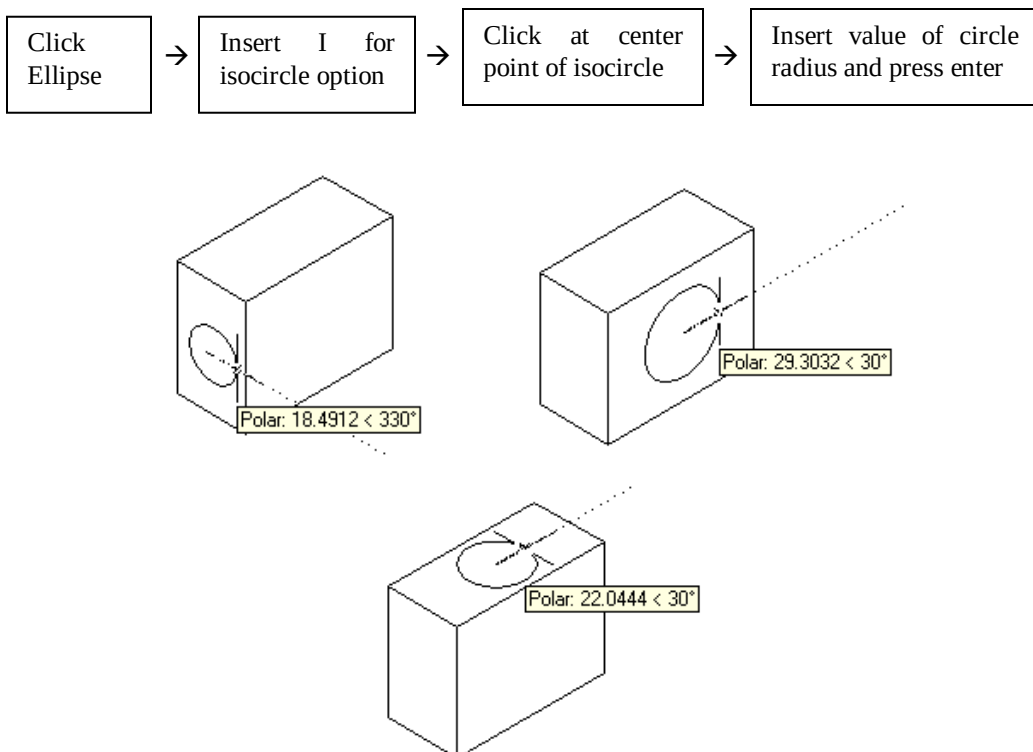


Figure 11.91: Isocircle in Different Isoplanes

11.8.2 Dimensioning Isometric Objects

In order to give the dimension in the isometric objects two steps are followed. In the first step using **Aligned** command (Section 11.7.1), the necessary dimensions are given (Figure 11.92a). Then in the second step these dimensions are edited with the help of **Oblique** command (Figure 11.92b). The sequence to edit the dimensions is as follows. In the third step of the sequence the value of oblique angle is to be inserted. The angle between the extension line of the isometric dimension and positive x-axis is called the oblique angle. The oblique angles 30° , 90° and 150° are shown in Figure 11.92b.

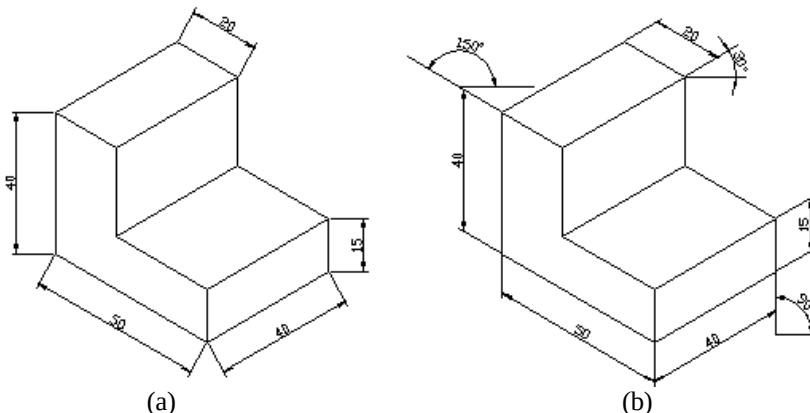
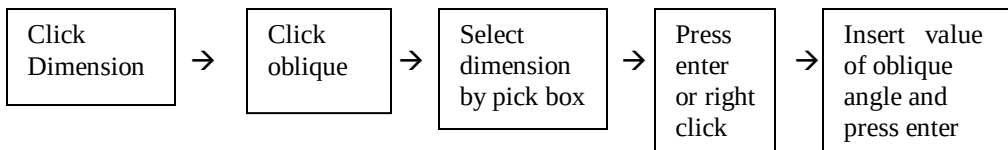


Figure 11.92: Dimensioning in Isometric Drawing

11.9 Solid Object Drawing

Menu: **View** → **Toolbars** → **Solids/Solids Editing**

Menu: **Draw** → **Solids**

Menu: **Modify** → **Solids Editing**

AutoCAD provides the scope to draw a great variety of solid objects. Some standard geometrical shapes are available in **Solids** toolbar. Solid drawings are edited with some tools that are available in **Solids Editing** toolbar. These toolbars can be obtained by selecting the crosscheck box beside **Solids** and **Solids Editing** in the Toolbars dialog box (Figure 11.13). Then the Solids and Solids Editing toolbars will be displayed, which are shown in Figure 11.93 and Figure 11.94 respectively. One can also obtain **Solids** commands from Draw menu (Figure 11.95) and **Solids Editing** commands (Figure 11.96) from Modify menu.

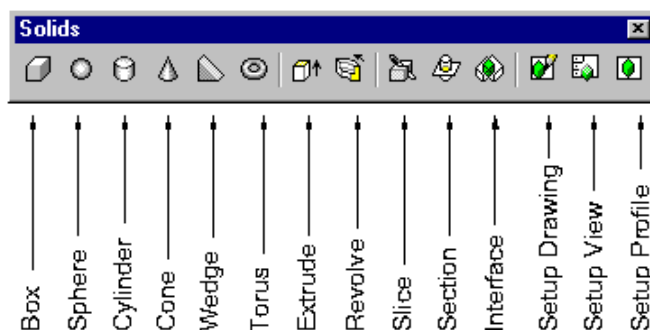


Figure 11.93: Solids Toolbars

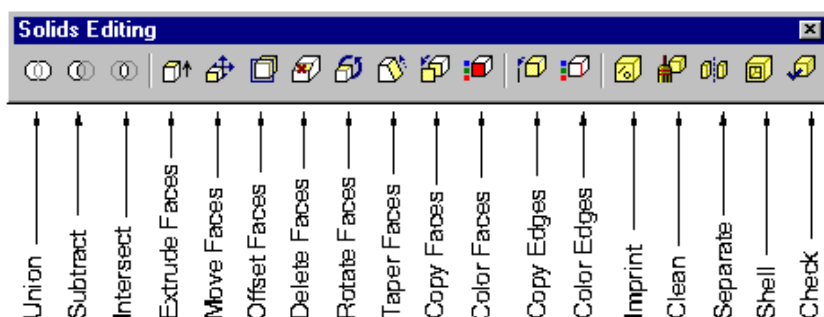


Figure 11.94: Solids Editing Toolbars

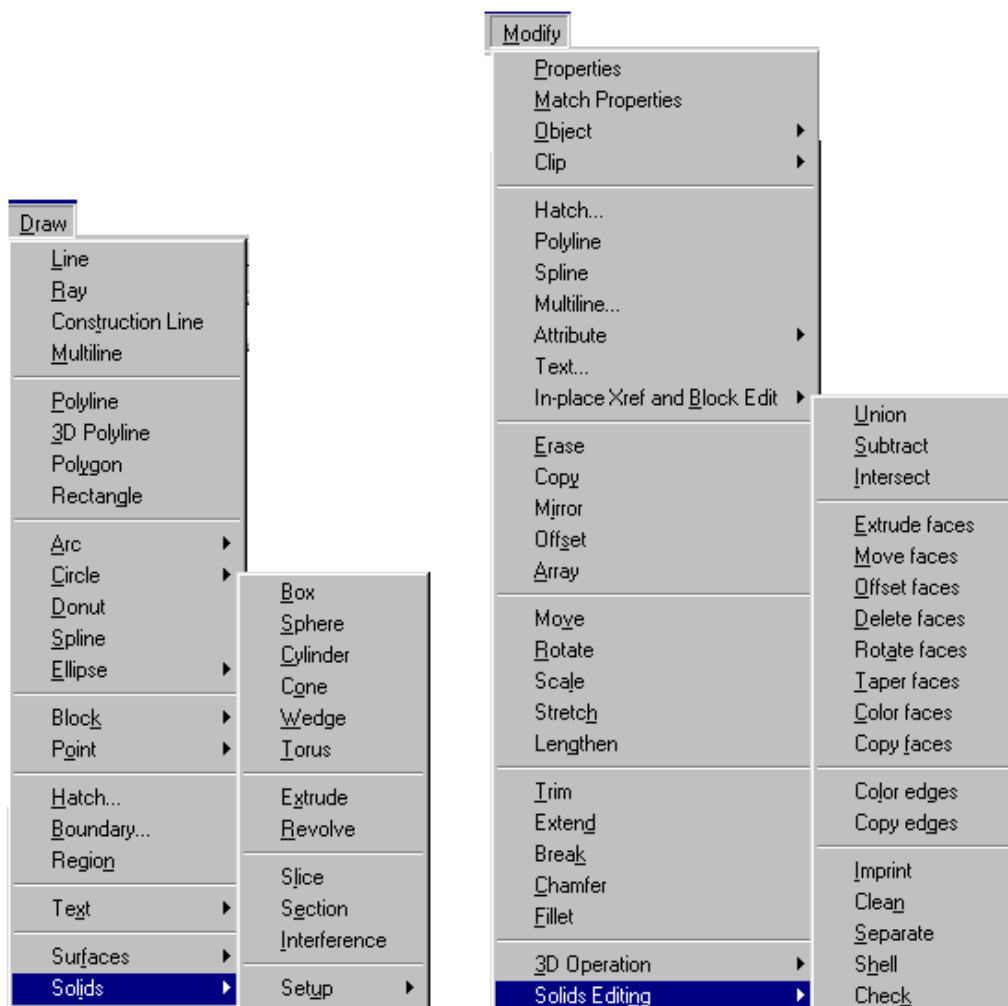


Figure 11.95: Solids in Draw Menu

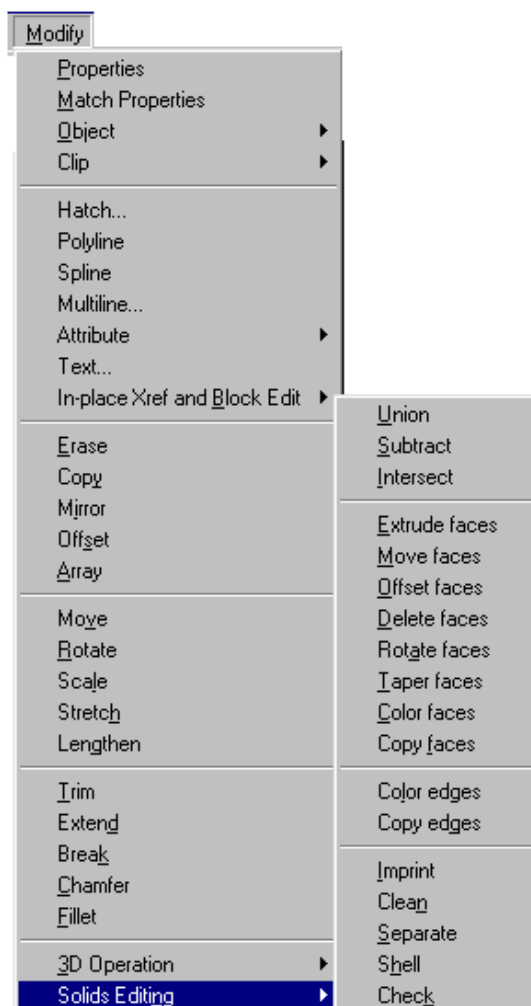
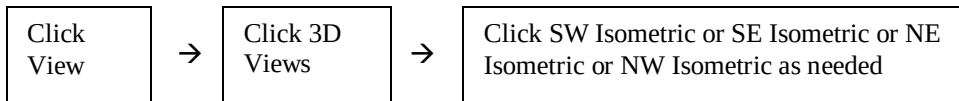


Figure 11.96: Solids Editing in Modify Menu

The most common solid objects available in the **Solids** toolbar are box, sphere, cylinder, cone, wedge and torus, which are shown from left to right in the **Solids** toolbar. Some additional solids drawing options are included in this toolbar amongst which extrude, revolve and slice may be mentioned. In the **Solids Editing** toolbar the first three tools are union, subtract and intersect. These three tools are of utmost importance while working with solid objects in AutoCAD.

To make the 3D drawing, the UCS orientation is changed beforehand. To do that steps as mentioned below are to be followed.



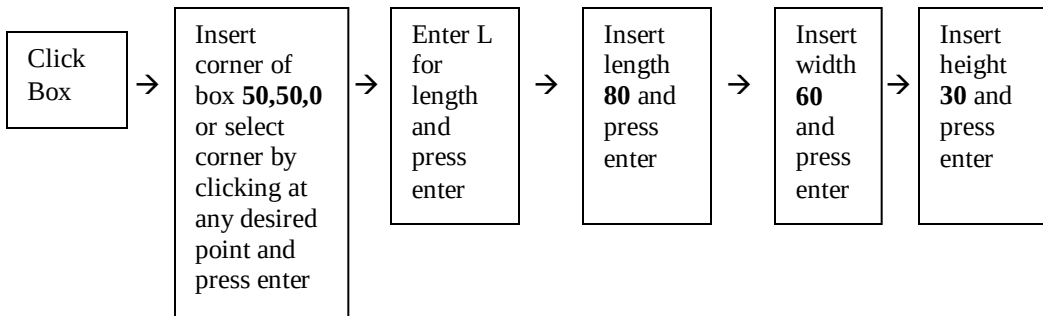
11.9.1 *Generating Solid Box*

Solids Toolbar: 

Menu: **Draw → Solids → Box**

Command: **BOX**

To generate a solid box (Figure 11.97a), **Box** command is used. This command has two options: **Corner** and **Center**. By default **Corner** option appears. The steps to generate a solid box by using corner option are given below as an example.



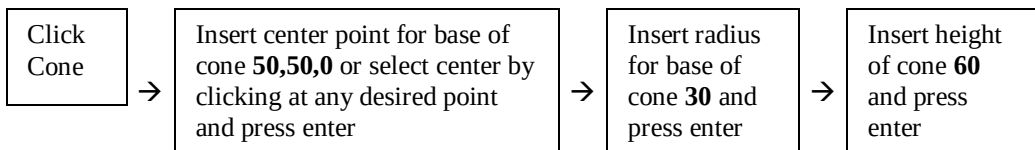
11.9.2 *Generating Solid Cone*

Solids Toolbar: 

Menu: **Draw → Solids → Cone**

Command: **CONE**

Using **Cone** command a solid cone is generated (Figure 11.97b). This command has two options: **Elliptical** and **Center point**. By default Center point option appears. The height of the cone occurs along the z-axis and the base in the xy-plane (current). However, UCS command option can be used to change the base plane. The sequence to generate a solid cone by using Center option is given below as an example.



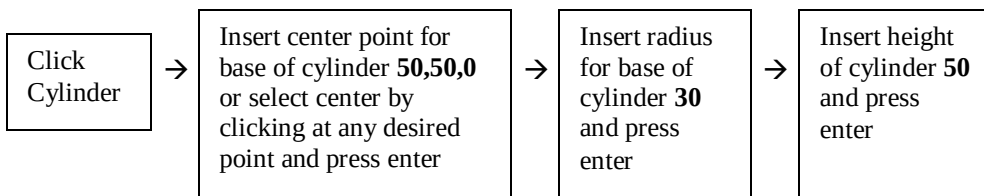
11.9.3 *Generating Solid Cylinder*

Solids Toolbar: 

Menu: **Draw** → **Solids** → **Cylinder**

Command: **CYLINDER**

With the help of **Cylinder** command, a solid cylinder is generated (Figure 11.97c). This command has two options: **Elliptical** and **Center point**. By default **Center point** option appears. The base of the cylinder occurs in the xy-plane and the height occurs in the z-direction (current). However, UCS command option may be used to change the base point. The steps as mentioned below are followed to generate a solid cylinder by using Center point option as an example.



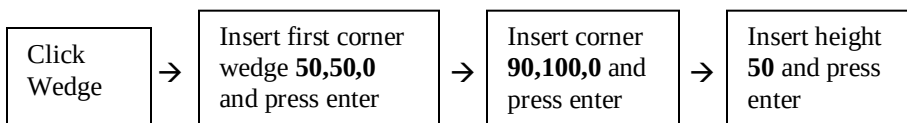
11.9.4 *Generating Solid Wedge*

Solids Toolbar: 

Menu: **Draw** → **Solids** → **Wedge**

Command: **WEDGE**

Wedge command is used to generate a solid wedge (Figure 11.97d). It has two options: **Corner** and **Center**. By default **Corner** option appears. The sequence to generate a solid wedge by using the Corner option is given below as an example. After making the solid objects some hidden lines remain behind them, which are removed by using the **Hide** command. This command can be invoked from the menu **View** → **Hide** or one can initiate it by entering HIDE command in the command area followed by press enter. Then the hidden lines will be removed (Figure 11.97e to Figure 11.97h).



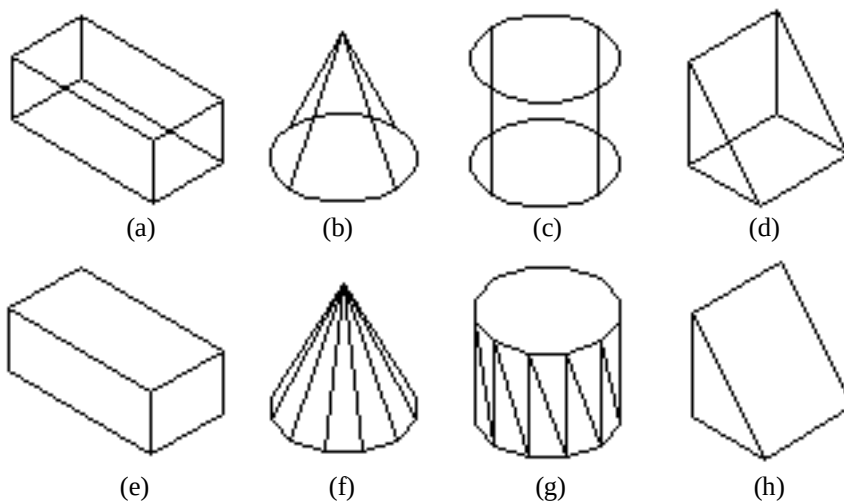


Figure 11.97: Solids Objects

11.9.5 *Revolve*

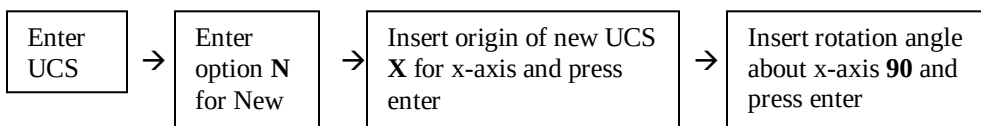
By using **Revolve** command a solid object is generated in accordance with the following steps provided as an example.

Step 1:

Menu: **Tools** → **New UCS**

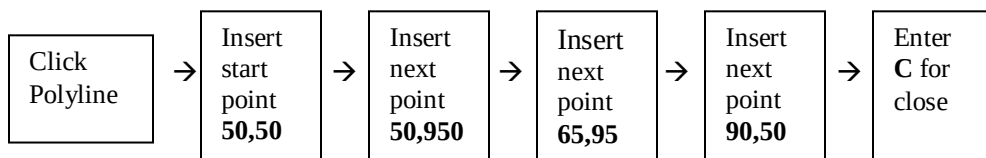
Command: **UCS**

In this step the orientation of the UCS (SE Isometric), which was created before (Figure 11.98a) is changed into the new orientation (Figure 11.98b) by entering UCS command in the command area from the following sequence. It can also be invoked from the menu by clicking on **New UCS** command.



Step 2:

In this step by using the **Polyline** command (section 11.3.2), the area is generated following the sequence as given below (Figure 11.98c).



Step 3:

Solids Toolbar: 

Menu: **Draw → Solids → Revolve**

Command: **REVOLVE**

By using the **Revolve** command, one can generate the solid object revolving the area (Figure 11.98c) in accordance with the following sequence. Now after using HIDE command, one finds the solid object as shown in Figure 11.98d.

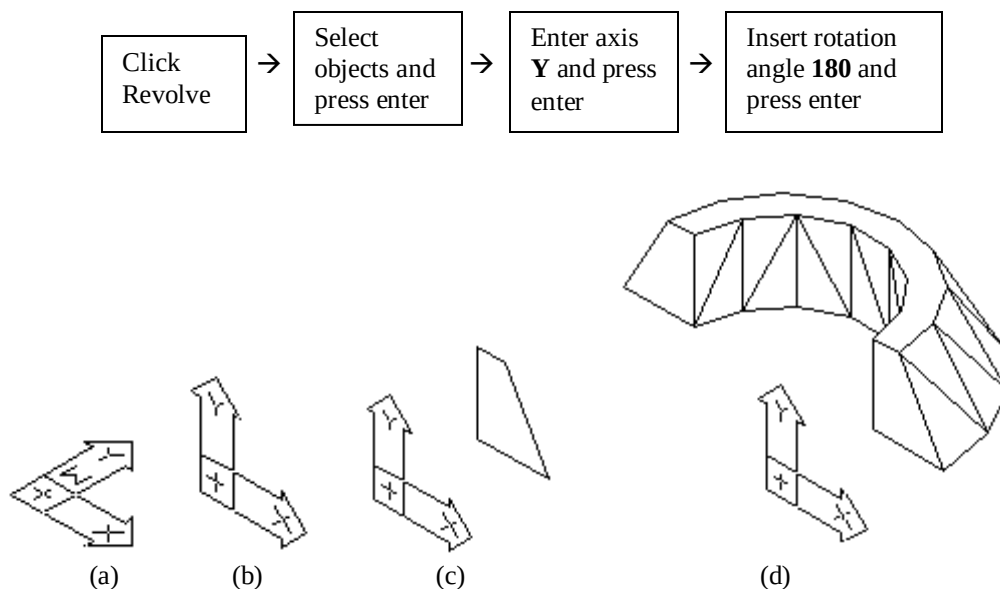


Figure 11.98: Use of Revolve Command

11.9.6 Union

Solids Editing Toolbar: 

Menu: **Modify → Solids Editing → Union**

Command: **UNION**

The **Union** command in the Solids Editing toolbar can be used to unite two solids. A union of a box and a cylinder is shown in Figure 11.99. The sequence as mentioned below is followed to make the union. Then the command DISPSILH is entered in the command prompt area followed by press enter. Next insert **1** for new value of DISPSILH. Now enter the command HIDE as a result the final solid object is obtained as shown in Figure 11.99.



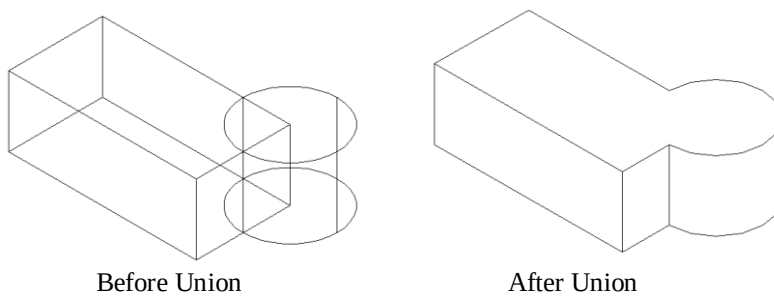


Figure 11.99: Use of Union Command

11.9.7 *Subtract*

Solids Editing Toolbar: 

Menu: **Modify** → **Solids Editing** → **Subtract**

Command: **SUBTRACT**

One solid object or a portion of it can be subtracted from another solid object by using the **Subtract** command from the **Solids Editing** toolbar. A subtraction of a portion of cylinder from a box is shown in Figure 11.100. The steps for subtraction are as follows. The object which is selected later, the shared or intersected part of that object is subtracted from the first object. Now using DISPSILH and HIDE command the final object as shown in Figure 11.100 is obtained.

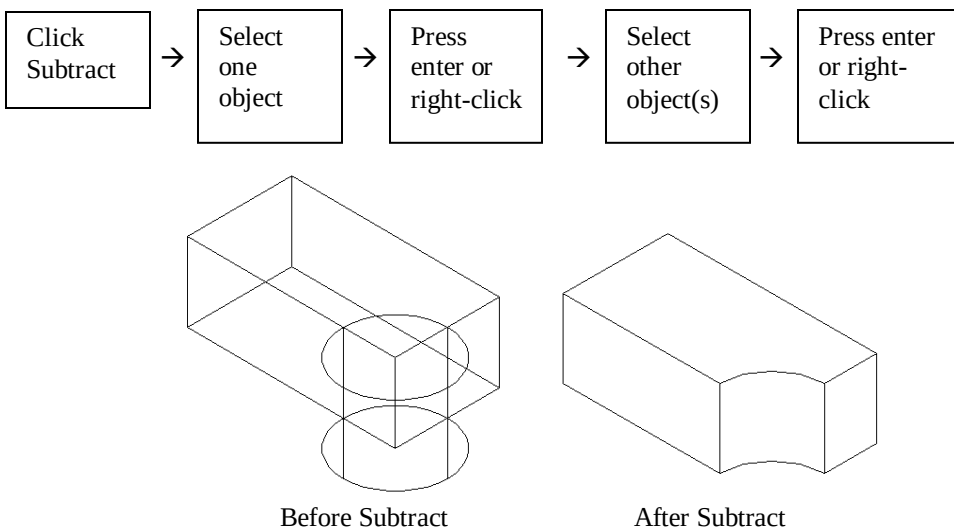


Figure 11.100: Use of Subtract Command

11.9.8 *Intersect*

Solids Editing Toolbar: 

Menu: **Modify** → **Solids Editing** → **Intersect**

Command: **INTERSECT**

When two solids are intersected or overlapped, the **Intersect** command results in keeping only the portion of solid they share with each other. Intersection of a box and a cylinder is shown in Figure 11.101. The sequence for intersection is mentioned below. Now using DISPSILH and HIDE command the final object as shown in Figure 11.101 is obtained.

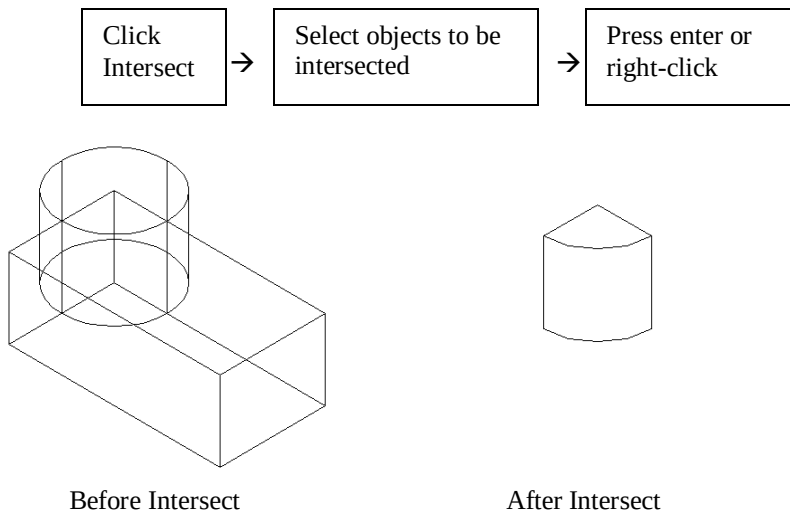


Figure 11.101: Use of Intersect Command

11.10 *Creating Text*

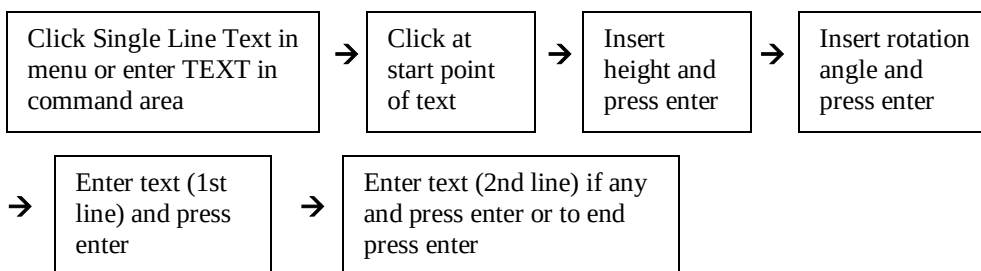
In AutoCAD a text can be written in two ways. One is the Single Line Text and the other is the Multiline Text. In this section both of them will be discussed in short. Some idea in regard to editing text and text style will also be taken into consideration in a nutshell.

11.10.1 *Single Line Text*

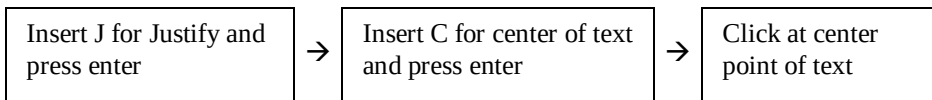
Menu: **Draw → Text → Single Line Text**

Command: **DTEXT or TEXT**

Single line text can be written by invoking **Text** command from the menu or it can be initiated by entering the TEXT command in the command area. There are many options in writing the text. Of them the most common ones will be considered here for discussion only. To write the single line text, the sequence as mentioned below has to be followed.



If the text is needed to be written in reference to the center of the text, the first and second steps of the above sequence are to be replaced by the following three steps. When J is inserted for justify, there appears 14 different options to place the text, C is one of them. Other options are TL (Top Left), TC (Top Center), TR (Top Right), ML (Middle Left), MC (Middle Center), MR (Middle Right), BL (Bottom Left), BC (Bottom Center) and BR (Bottom Right) etc. According to the requirement any one can be inserted.



11.10.2 Multiline Text

Draw Toolbar:

Menu: **Draw** → **Text** → **Multiline Text**

Multiline text can be written by invoking command from the toolbar or menu. The steps as shown below are followed to write Multiline text. When the window for the Multiline text is created in the drawing area, a Multiline Text Editor dialog box is displayed (Figure 11.102), where the text is to be written. After clicking on OK button in the dialog box, the text will appear within the window created previously.

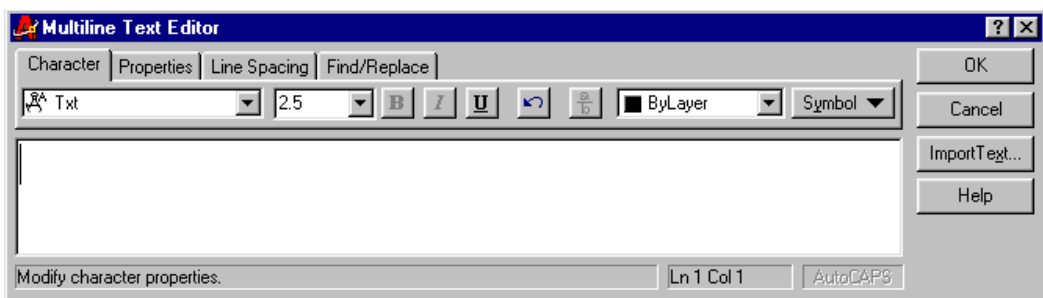
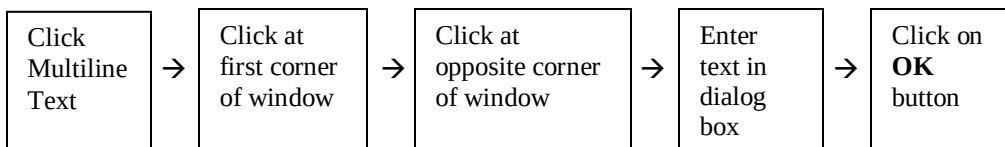


Figure 11.102: Multiline Text Editor Dialog Box

11.10.3 Editing Text

Modify II Toolbar:

Menu: **Modify** → **Text**

Command: **DDEDIT**

From the toolbars dialog box (Figure 11.13) the modify II Toolbar can be obtained by selecting crosscheck box beside Modify II. After selecting the text object to be edited, one can invoke the command from the toolbar or menu. It can also be initiated by entering DDEDIT command in the

command area. Then an Edit Text dialog box (Figure 11.103) will appear containing the selected text to edit. After required edit, click on the **OK** button in the dialog box, the necessary edit will be done. The text object can also be edited by using editing commands **Move**, **Erase**, **Rotate**, **Copy**, and **Mirror** etc. One can edit the text object by using the short-cut menu. As the text object is selected and right-clicked in the drawing area a short-cut menu appears where clicking **Text Edit**, the Edit Text dialog box (Figure 11.103) is displayed to edit the text.

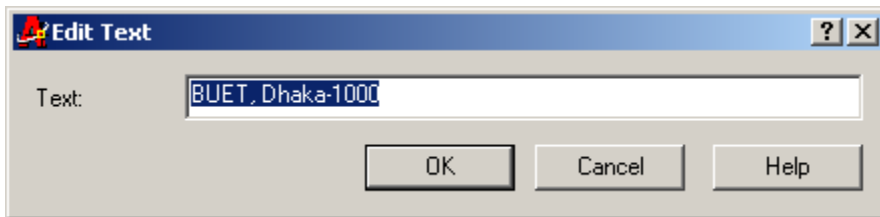


Figure 11.103: Edit Text Dialog Box

11.10.4 Text Style

Menu: **Format → Text Style**

Command: **STYLE** or **DDSTYLE**

Invoking command in the menu one can change the current text style or it can be initiated by entering **STYLE** command in the command area. Then a Text Style dialog box (Figure 11.104) will appear.

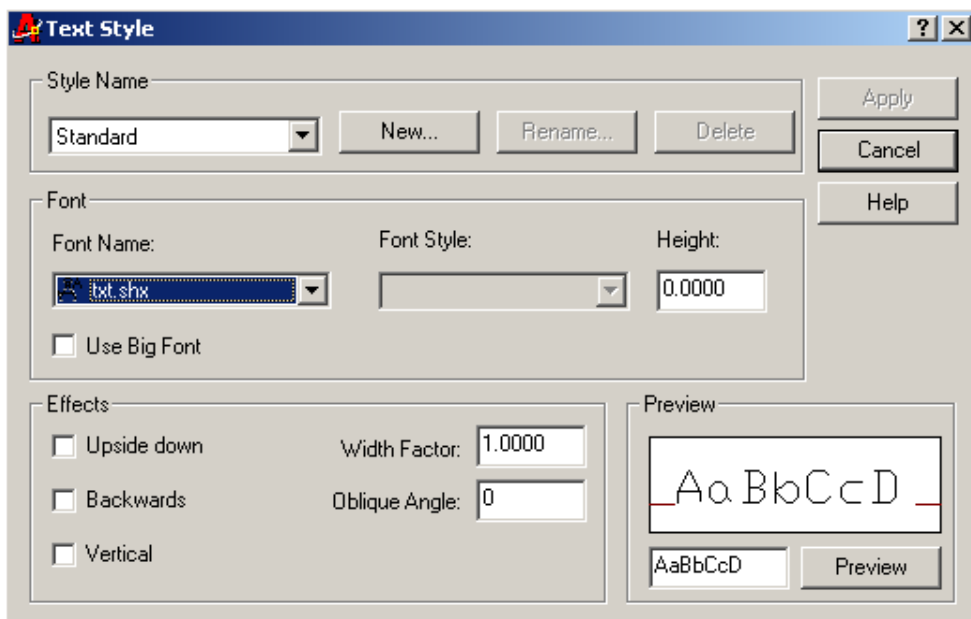


Figure 11.104: Text Style Dialog Box

In the Font area of this dialog box clicking the drop-down arrow, a drop down list consisting a number of Fonts will be displayed from where any desired Font can be selected. As the Font is selected, it is displayed in the Preview area of this dialog box. Once the selection is completed, click on the **Apply** button followed by **Close** button located at the top right corner of the dialog box. In this dialog box there are other options as well which can be selected according to the requirement.

11.11 Plotting Drawing

Standard Toolbar: 

Menu: **File** → **Plot**

Command: **PLOT** or **PRINT**

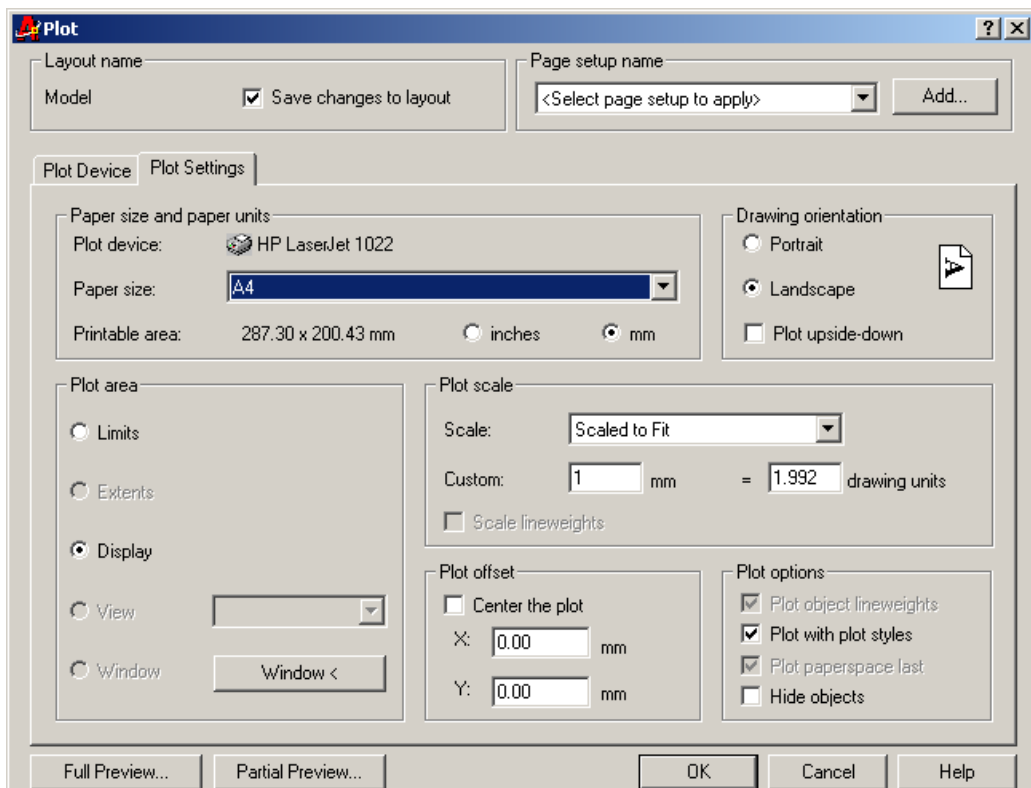


Figure 11.105: Plot Dialog Box (For Plot Settings)

Most drawing jobs remain incomplete until the final outcome is presented on paper. Many printers and plotters can handle a wide range of drawing sizes and paper types. When **Plot** command is invoked in the toolbar or menu, **Plot** dialogue box (Figure 11.105) is displayed containing many options for selection. It can also be initiated by entering PLOT command in the command area. In this dialog box the Plot Settings tab is selected. This dialog box contains different areas such as, Paper size and paper units area, Drawing orientation area, Plot area, Plot scale area, Plot offset and

Plot options area. Selecting Plot Device tab in the dialog box, one can select the appropriate Plot Device in the Plotter configuration area.

11.11.1 *Paper Size and Paper Units*

In Figure 11.106, a drop-down list consisting of various paper sizes is shown. It is obtained when the down arrow button in the Paper size and paper units area of the Plot dialog box is selected. Now the required paper size can be chosen from the drop down list. The measurements can be set either in inches or mm.

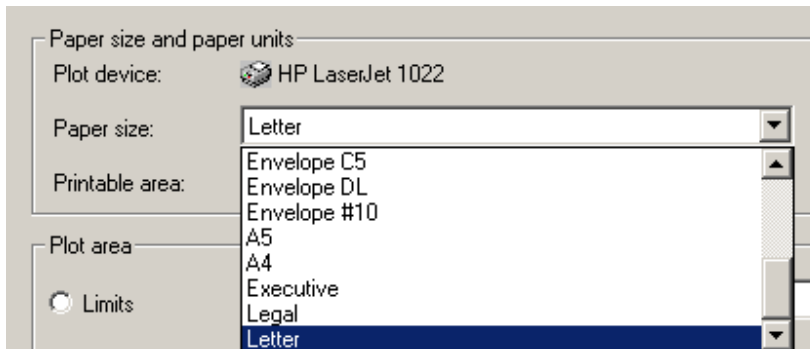


Figure 11.106: Paper Size in Drop Down List

11.11.2 *Plot Scale*

When the down arrow button in the Plot scale area of the Plot dialog box is selected a drop-down list containing a number of standard scale as shown in Figure 11.107 is displayed. Now the required scale can be chosen from this drop-down list.

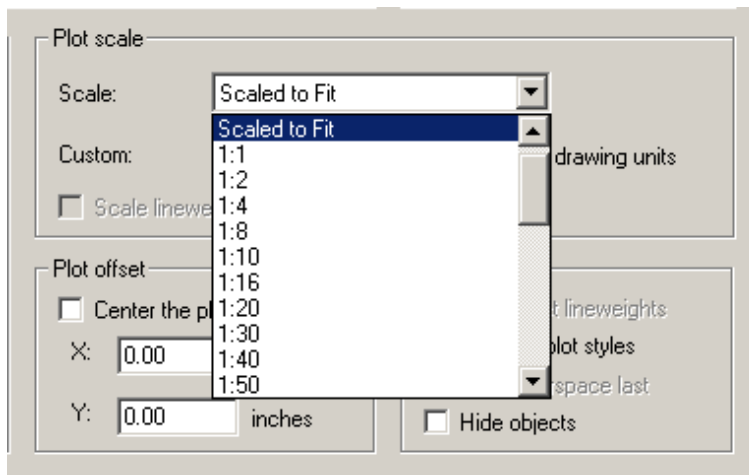


Figure 11.107: Drop Down List in Plot Scale

The scale is displayed in the custom edit box (Figure 11.105). The option **Scaled to Fit** is set by default, which will enable to plot the desired drawing accommodating in the paper selected previously. The scale factor can be changed manually in the custom edit box.

11.11.3 *Drawing Orientation*

In the drawing orientation area of the Plot dialog box (Figure 11.105) one can choose either the Portrait or Landscape option for the paper orientation by selecting the radio button. A plot upside-down option can also be chosen here if needed.

11.11.4 *Plot Area*

By default the plot is set to the layout. However, it can be set to Limits, Extents and Display in the **Plot** dialogue box by selecting the radio button in the Plot area of the Plot dialog box (Figure 11.105). By clicking on the **Window** tab the drawing will be displayed in the window from where a portion of the drawing may be selected within the window for printing.

11.11.5 *Plot offset*

In the Plot offset area (Figure 11.105) if **Center the plot** option is selected the plotting will be done in reference to the center of the paper by adjusting the values of X and Y. When the printing is not desired at the lower-left corner of the printable area on the paper the plot can be made offset using the X and Y text boxes. For example, if both X and Y are set to 2, the plot will be moved up and right by two units.

11.11.6 *Plot options*

In the Plot options area of the Plot dialog box (Figure 11.105), if the **Plot object line weights** is cleared, in the plotting line weights will not be shown in spite of using line weights in the drawing. When **Plot with Plot Styles** is cleared the plot styles will not be shown in the plotting in spite of using plot styles in the drawing. Clearing the **Plot paper space Last** option the objects are plotted on the paper space layout first. The hidden lines of 3D objects will not be shown if the checkbox of **Hide objects** is selected.

11.11.7 *Plot Preview*

The view of the drawing, which will be plotted on the specified paper size, can be seen before plotting by selecting the **Full Preview** or **Partial Preview** button located at the left bottom corner of the Plot dialog box (Figure 11.105).